

Weil classes and Hodge classes on abelian powers

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Abstract

We prove the rational Hodge conjecture in every codimension on every self-power of a complex abelian sixfold with an imaginary-quadratic action and a compatible polarization whose rational homological Hermitian form is hyperbolic of signature $(3, 3)$. We also prove it in every codimension on every self-power of a complex abelian variety of dimension at most five admitting an imaginary-quadratic action. Both results include nonsimple varieties and special periods with additional endomorphisms. The proof uses the companion theorem on the rational Hodge conjecture for CM abelian varieties.

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1 Introduction

For a smooth projective complex variety X , the rational Hodge conjecture asks whether every class in

$$H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$$

is the class of a rational algebraic cycle of codimension p . For an abelian variety, the question on every self-power is naturally a question about tensors in its first cohomology: classes can couple several copies even when the classes on the variety itself are already understood. This paper proves the conjecture on all self-powers in two families with an imaginary-quadratic action.

An action of an imaginary quadratic field E on an abelian variety A means a unital embedding $E \hookrightarrow \text{End}^0(A)$. Write $E = \mathbb{Q}(u)$, with $u^2 = -d$ and $d \in \mathbb{Z}_{>0}$. A rational Riemann form e is compatible with the action if $e(ax, y) = e(x, \bar{a}y)$ for every $a \in E$. Its associated homological Hermitian form is

$$h_E(x, y) = e(x, uy) + ue(x, y), \quad x, y \in H_1(A, \mathbb{Q}),$$

conjugate-linear in the first variable and linear in the second. A rank-six Hermitian form is hyperbolic if it contains a totally isotropic three-dimensional E -subspace. The form in the next theorem is the one coming from the actual polarization, not merely an abstract form of the same complex signature.

Theorem 1.1 (Split Weil sixfolds). *Let E be an imaginary quadratic field and let A be a complex abelian sixfold with an E -action. Suppose a compatible polarization has hyperbolic rational Hermitian form of signature $(3, 3)$. Then the rational Hodge conjecture holds in every codimension on A^m , for every $m \geq 1$.*

Theorem 1.2 (Quadratic actions in dimension at most five). *Let A be a complex abelian variety of dimension at most five admitting an action of an imaginary quadratic field. The rational Hodge conjecture holds in every codimension on A^m , for every $m \geq 1$.*

Both statements include nonsimple varieties and periods with additional endomorphisms. The quadratic action need not be the full endomorphism algebra or be central on a repeated simple factor. The second theorem requires no separately chosen compatible polarization: one is obtained by averaging any rational Riemann form, as shown in Section 3.

1.1 Weil classes and earlier results

Weil's determinant construction [41] provides a basic test of the Hodge conjecture beyond divisor classes. For an abelian variety of dimension $2n$ with a compatible imaginary-quadratic action of signature (n, n) , the two embedding determinants form a rational plane of classes of type (n, n) . For a general such variety with $n > 1$, these classes are not generated by divisors; see [40, Sections 4.9–4.11]. Early geometric constructions used automorphisms of curves and Prym varieties [33]. Schoen's addendum corrected the polarization hypothesis in the earlier fourfold statement and established algebraicity of the Weil plane for every fourfold with balanced action by $\mathbb{Q}(\sqrt{-3})$, without a discriminant restriction [34].

Markman's secant-sheaf construction proves algebraicity of the whole Weil plane for every split Weil sixfold. The preprint states the result for discriminant -1 , while the published account uses the equivalent split condition on the rational Hermitian form of the compatible polarization [25, Theorem 1.5.1] [26, Theorem 1.2]. The published account also proves the Hodge conjecture for every individual abelian variety of dimension at most five, with no quadratic-action hypothesis [26, Corollary 1.3]. These are all-period statements. The further question in our main theorems concerns

all Hodge classes on every self-power, including the extra tensors that can occur when the Hodge group drops at a special period.

The all-powers question also has substantial precedents. Moonen and Zarhin’s low-dimensional classification includes all-powers divisor-generation results, as well as a fourfold case whose square has exceptional classes [28, Theorems 0.1–0.2]. For every Weil fourfold of discriminant one, Floccari proves the Hodge conjecture on all powers through the Kuga–Satake correspondence [15, Theorems 1.3 and 5.12]. Floccari and Fu give another proof using singular varieties of O’Grady’s six-dimensional type [16, Theorem 1.2]. Our argument addresses the two stated families by combining the split-sixfold Weil plane with a determinant criterion for the actual Hodge group and the CM algebraicity input specified below.

The geometric proof is a detailed treatment of Markman’s construction, rather than an extension of its period range. It uses Mukai’s Fourier–Mukai duality and the algebraic spin interpretation of Golyshev–Lunts–Orlov [29, 17]. We work out the coherent seed, the fixed transform conventions, the full invariant derived obstruction space, and its passage through quotient descent and projective-bundle conversion. The full-group comparison and descent are already part of Markman’s method [25, Section 7.4.2 and Lemmas 9.3.2 and 9.3.11] [26, Sections 11.3–11.5]. The Hochschild–Atiyah comparison follows the character and evaluation identities of Căldăraru, Buchweitz–Flenner and Huang [7, 5, 21], with the abelian specialization of Todd-corrected compatibility [6]. The deformation argument reconstructs the smooth-projective-germ case of the coherent-sheaf argument in Buchweitz–Flenner’s Theorem 5.1 and its proof [4]. The needed injectivity is on the full obstruction group, not merely the image of Hochschild evaluation.

The separate passage from the Weil plane to all powers uses classical invariant theory to reduce tensors to pairings and determinants [23, 24, 35]. The two-weight restriction is the standard minuscule argument [38, Section 3.2, Lemmas 4–5]; comparison of product Hodge groups through graph links has a close predecessor in Moonen–Zarhin [28, Section 3.1 and Lemmas 3.3–3.4]. Adjoining auxiliary factors and extracting lower-dimensional Weil classes also appears in Schoen’s Proposition 10 [34]. Here CM elliptic factors supply smaller quadratic determinants, while the actual-group analysis identifies the products selected by the simultaneous balance equations. Those supply and balance steps, rather than the invariant spanning theorems alone, give the all-powers conclusions.

1.2 Proof strategy

Let $H = H^1(A, \mathbb{Q})$. A tensor in $H^{\otimes 2p}$ is balanced if it has type (p, p) . For A^m , antisymmetrization lifts a Hodge class to a balanced tensor in $(H^{\oplus m})^{\otimes 2p}$, with the copies labelled by the factors of the power. Diagonal pullback returns the exterior class. Section 2 gives a criterion for algebraicity of these tensors. The multilinear invariant theorems for symplectic, orthogonal, and special linear groups reduce the tensors to pairings and top exterior lines [23, 24, 35]. Pairings are divisor classes. The required products of the remaining lines are supplied by algebraic correspondences from the full cohomology of CM abelian varieties. A rational Hodge class in such a supplied subspace lifts to rational Hodge classes on the CM sources; Theorem 1.1 of *The rational Hodge conjecture for CM abelian varieties*, dated September 30, 2026, makes those lifts algebraic [30]. This is the precise CM input used in this paper.

Section 3 proves the two main theorems from the Weil-plane input, whose geometric proof follows in Sections 4–8. Adding CM elliptic curves supplies smaller quadratic determinants whose signature entries are at most three. The Albert decomposition, the two Hodge-circle weights, and the small minuscule representations determine the possible constituents of the actual derived Hodge group. A separate independence argument identifies their simultaneous product. The balance equations then cancel opposite determinants or combine them into a supplied quadratic determinant. This is how

the proof accommodates additional endomorphisms at special periods.

The cycle construction in Section 4 begins with a genus-three Jacobian X . Disjoint orbits of Abel curves give an ideal sheaf on X^2 ; a fixed Fourier–Mukai transform and duality produce a coherent perfect sheaf $\mathcal{E}$ on $X \times \hat{X}$, together with a coherent finite translation action twisted by line bundles. The normalized Chern character of this sheaf is the candidate cycle class. Two properties are needed: it must contain a nonzero Weil component and remain of Hodge type as the Weil period varies; and, after descent and a projective-bundle construction, an associated ordinary sheaf must deform so that the normalized character is algebraic near the initial point.

Section 5 proves the exact Hochschild–Atiyah trace identity and its transport for the fixed transform. A direct equivariant Ext calculation then shows that the evaluation image equals the entire invariant obstruction space, on which the trace map is injective. Section 6 computes the rational Betti character by a finite Clifford calculation. It identifies the actual ample polarization, proves that its homological Hermitian form is hyperbolic, and shows that the character is Hodge throughout the compatible marked domain.

Section 7 descends the sheaf through the finite quotient and converts the resulting twisted sheaf into an ordinary sheaf on a projective bundle without losing any obstruction class. Its semiregularity theorem produces the required local deformations. Section 8 extracts the normalized character by algebraic projective-bundle operations, spreads the resulting Weil plane through proper Hilbert parameter spaces, and clears rational lattice denominators by isogenies. This proves algebraicity of the whole rational Weil plane at every compatible hyperbolic period.

2 Algebraic supply and invariant tensors

The cohomology of an abelian variety is generated by its first cohomology. We will therefore prove algebraicity first for tensors in degree one. Classical invariant theory reduces these tensors to pairings and top exterior powers. Pairings are divisor classes. The remaining task in each application is to supply the required products of top exterior powers by algebraic correspondences from CM abelian varieties.

All varieties are complex, and cohomology is singular cohomology with the indicated coefficients. An algebraic correspondence has rational coefficients unless otherwise stated. We fix $k = \overline{\mathbb{Q}} \subset \mathbb{C}$. An automorphism of $k/\mathbb{Q}$ acts only on coefficients in the cohomology of the fixed complex variety; it does not replace that variety by a conjugate. For a pure Hodge structure V , our convention is $V(r)^{p,q} = V^{p+r,q+r}$, so the Tate twist $V(r)$ has weight $\text{wt}(V) - 2r$. A class is *algebraic over k* if it belongs to the k -linear span of rational algebraic cycle classes.

2.1 Correspondences from CM abelian varieties

For a smooth projective variety X , let

$$\mathcal{C}^d(X) = \sum_{B,j,\Gamma} \text{im}(\Gamma_* : H^{d-2j}(B, \mathbb{Q}) \longrightarrow H^d(X, \mathbb{Q})), \quad \Gamma \in \text{CH}^{\dim B+j}(B \times X)_{\mathbb{Q}},$$

where B ranges over CM abelian varieties, including a point, and j ranges over integers for which the displayed groups and cycles exist. Here and below a CM abelian variety means an abelian variety whose degree-one Hodge group is a torus. The displayed map is a morphism of weight- d Hodge structures from $H^{d-2j}(B, \mathbb{Q})(-j)$. We say that a subspace of $H^d(X, k)$ is *CM-supplied* if it is contained in $\mathcal{C}^d(X) \otimes_{\mathbb{Q}} k$. Thus supply concerns the full cohomology of the CM source, not only its Hodge classes.

We use the companion theorem that the rational Hodge conjecture holds for every complex CM abelian variety in every codimension [30, Theorem 1.1]. This is the only Hodge-conjecture input in the following lemma.

Lemma 2.1 (Operations and Hodge lifts). *CM supply is preserved by external products, permutations of factors, cup products, and rational algebraic correspondences. It is also preserved by k -linear combinations of such correspondences and by scalar conjugation. Every rational Hodge class in a CM-supplied subspace is algebraic.*

Proof. External products of correspondence cycles realize products of their maps, and a product of CM abelian varieties is CM. Composition is realized by pulling two cycles to the triple product, intersecting, and pushing forward. Graphs give pullback and pushforward. Diagonal pullback gives cup product; adding integration gives contraction. These operations are defined for the smooth projective varieties in question. Linearity proves the scalar assertion, and $\mathcal{C}^d(X) \otimes k$ is invariant under scalar conjugation because $\mathcal{C}^d(X)$ is rational.

Let $\alpha \in H^{2p}(X, \mathbb{Q})$ have type (p, p) and belong after scalar extension to $\mathcal{C}^{2p}(X) \otimes k$. Membership in a rational subspace descends under a field extension, so α already belongs to $\mathcal{C}^{2p}(X)$. Finitely many of the defining correspondence maps have an image containing α . They give a morphism

$$f : \bigoplus_{\nu=1}^s H^{2p-2j_\nu}(B_\nu, \mathbb{Q})(-j_\nu) \longrightarrow H^{2p}(X, \mathbb{Q})$$

of polarizable pure Hodge structures. This morphism splits onto its image as a rational Hodge morphism: the orthogonal complement of $\ker f$ for a polarization on the source is a Hodge substructure and maps isomorphically to $\operatorname{im} f$. Nondegeneracy on $\ker f$ follows from the positive Hermitian form obtained using the Weil operator. Hence α has a rational Hodge lift. Its ν th component before twisting has type $(p - j_\nu, p - j_\nu)$ on B_ν . The CM Hodge theorem makes this component algebraic. Applying the defining correspondence cycles and adding proves the assertion. $\square$

2.2 Degree-one summands and pairings

We record the coefficient descent needed to pass from complex eigenspaces to rational algebraic maps. If T is a finite-dimensional rational vector space, $P \subset T_{\mathbb{C}}$ is complex-linear, and $t \in T_k$ satisfies $\gamma t \in P$ for every $\gamma \in \operatorname{Gal}(k/\mathbb{Q})$, then

$$t \in (T \cap P) \otimes_{\mathbb{Q}} k. \tag{2.1}$$

Indeed, choose a finite Galois field K containing its coefficients. For $a \in K$, the trace $\operatorname{Tr}_{K/\mathbb{Q}}(at) = \sum_{\tau \in \operatorname{Gal}(K/\mathbb{Q})} \tau(a)\tau(t)$ is rational and belongs to P . If b_j is a basis of K and b_j^* is its trace-dual basis, these traces for $a = b_j^*$ are exactly the rational coefficients of t in the basis b_j . Here each τ is extended to k when applying the hypothesis. This proves (2.1). Conditions that a tensor have a specified Hodge type, or that a map preserve Hodge types, are complex-linear conditions, so the observation applies to them.

Let X be an abelian variety or a smooth connected projective curve. A subspace $W \subset H^1(X, k)$ is *admissible* if it is the image of a k -linear idempotent e such that every γe preserves the Hodge decomposition after extension to $\mathbb{C}$. A k -linear map between such spaces, or between finite external direct sums of them, is *admissible* if it preserves the Hodge decomposition after every scalar conjugation. We write $\bar{W}$ for the scalar complex conjugate of W .

A *pair tensor* between W and W' is a tensor $t \in W \otimes W'$ whose associated map $W^* \rightarrow W'$ is an isomorphism and for which every γt has Hodge type $(1, 1)$. For a tensor on W itself, symmetry

or alternation refers to the ordinary interchange of tensor factors, before the Koszul sign in product cohomology.

Lemma 2.2 (Algebraic maps and contractions). *Every admissible map is induced on the indicated summands by a k -linear combination of rational algebraic correspondences. Every pair tensor is algebraic over k , and contraction by its inverse pairing is an algebraic cohomological operation. If W is admissible, so is $\overline{W}$, and there is a pair tensor between W and $\overline{W}$.*

Proof. For an admissible map, compose with the source projector and the target inclusion to obtain a map on the ambient degree-one spaces. Equation (2.1) expresses it as a k -linear combination of rational Hodge maps. A rational Hodge map between the first cohomologies of abelian varieties is induced by a rational homomorphism in the contravariant direction: dualize to homology, multiply to preserve the lattices, and use the resulting complex-linear map of complex tori. A homomorphism of abelian varieties is algebraic. For a curve target use its Abel map to its Jacobian. For a curve source, the cup pairing identifies a degree-one Hodge map with a rational $(1, 1)$ class on the product, which is a divisor class by the Lefschetz $(1, 1)$ theorem. These arguments also apply componentwise to external direct sums. In particular an admissible projector commutes with the scalar extension of the rational Hodge group.

A pair tensor lies in the $H^1 \otimes H^1$ summand of degree two on the product. Scalar descent and the Lefschetz $(1, 1)$ theorem make it algebraic over k . Let $b : W \otimes W' \rightarrow k$ be its inverse pairing, with its bidegree $(-1, -1)$ understood. Extend b by the admissible projectors. For a rational polarization pairing ψ_X on the source $H^1(X, \mathbb{Q})$ there is a unique map $u : H^1(X', k) \rightarrow H^1(X, k)$ satisfying $b(x, y) = \psi_X(x, u(y))$. In every scalar conjugate, b pairs only complementary Hodge degrees; nondegeneracy of ψ_X therefore makes u an admissible map. The first part algebraizes u . The remaining contraction by ψ_X is cup product, an ample power if needed, and integration. It is algebraic.

The conjugate projector shows that $\overline{W}$ is admissible. The restriction of ψ_X to $W \times \overline{W}$ is perfect. After extension to $\mathbb{C}$, the positive Hermitian form associated with the polarization detects every nonzero vector of the Hodge-stable space $W_{\mathbb{C}}$ by pairing it with the conjugate of a vector of $W_{\mathbb{C}}$. Its inverse gives a tensor in $W \otimes \overline{W}$. For an arbitrary γ , conjugate the nonzero determinant of this restricted pairing. The resulting pairing on $\gamma W \times \gamma \overline{W}$ is still perfect and pairs complementary Hodge degrees, since both spaces are Hodge-stable and ψ_X is rational. Its inverse has type $(1, 1)$. This argument does not assume that γ commutes with complex conjugation. $\square$

For an admissible space W of dimension r , write

$$D(W) = \bigwedge^r W \subset W^{\otimes r} \subset H^r(X^r, k)$$

for its top alternating line on separate tensor slots; put $D(0) = k$ on a point. Ordinary permutations differ from geometric permutations only by rational Koszul signs. Direct-sum factorization and cancellation of these determinant lines can therefore be performed algebraically. More precisely, if an admissible isomorphism identifies W with $\bigoplus_j W_j$, alternation and the admissible inclusions and projections induce nonzero maps in both directions between $D(W)$ and $\bigotimes_j D(W_j)$. If W, W' have a pair tensor, the determinant of the inverse pairing contracts $D(W) \otimes D(W')$ nontrivially to the unit line. Conversely, alternating the appropriate tensor power of the pair tensor supplies $D(W) \otimes D(W')$ algebraically. For an alternating self-pair on a $2r$ -space, the r th exterior power supplies $D(W)$ algebraically. These statements follow in bases from the nonzero determinant, or Pfaffian, of the pair tensor and from Lemma 2.2.

2.3 The tensor criterion

For a polarizable rational Hodge structure H of weight one whose only types are $(1, 0)$ and $(0, 1)$, let $J = \text{Hdg}(H)$ be the smallest rational algebraic subgroup of $\text{GL}(H)$ containing the Hodge circle, which acts on $H^{1,0}$ and $H^{0,1}$ by z and $\bar{z}$ for $|z| = 1$. It lies in the symplectic group of the polarization. A tensor in $H^{\otimes 2p}$ is called *balanced* if it has Hodge type (p, p) . The group J is connected: its identity component is rational and contains the connected circle, so minimality applies. It is reductive: a rational invariant subspace is a Hodge substructure, and polarizability supplies a rational Hodge complement, which is again invariant by minimality. The faithful representation on H is thus completely reducible.

Lemma 2.3 (Hodge tensors determine the Hodge group). *Inside the symplectic group of H , the group $\text{Hdg}(H)$ is the pointwise stabilizer of all rational tensors of type (a, a) in $H^{\otimes 2a}$, for all $a \geq 0$.*

Proof. By minimality, the rational J -invariant tensors are precisely the rational tensors fixed by the Hodge circle. To see that these determine J , use Chevalley's theorem to choose a rational representation T of $\text{GL}(H)$ and a line $L \subset T$ whose stabilizer is J [27, Theorem 4.27]. Reductivity gives a J -equivariant projector $e : T \rightarrow L$. Rational representations of $\text{GL}(H)$ are subquotients of sums of mixed tensor powers of H [27, Theorem 4.14]. Complete reducibility for $\text{GL}(H)$ [27, Theorem 22.42] gives a $\text{GL}(H)$ -equivariant embedding of $\text{End}(T)$ as a direct summand of a finite sum of mixed tensor powers. The element e in this representation is J -invariant. Use the polarization to replace dual tensor factors by ordinary factors, equivariantly for the ambient symplectic group. The components of e are then rational tensors fixed by the circle, hence balanced tensors in even degrees. An element fixing every such tensor fixes e , preserves its image L , and therefore belongs to J . The reverse inclusion follows from minimality. $\square$

We also use the standard effective weight-one equivalence between toric Hodge groups and CM abelian varieties, including rational summands. The following form ensures an actual CM source.

Lemma 2.4 (Toric degree-one summands). *Let A be a complex abelian variety and let $H \subset H^1(A, \mathbb{Q})$ be a rational Hodge summand. If $\text{Hdg}(H)$ is a torus, then H is isomorphic as a rational Hodge structure to $H^1(B, \mathbb{Q})$ for a CM isogeny factor B of A .*

Proof. Choose a rational Hodge projector onto H . By the degree-one realization in Lemma 2.2, it is induced by a rational endomorphism of A . After clearing denominators, its image is an abelian isogeny factor B with $H^1(B, \mathbb{Q}) \simeq H$. Its Hodge group is a torus, so B is CM under our convention.

We also describe the embedding-line decomposition used below. Put $J(H) = \text{Hdg}(H)$. Its envelope in $\text{End}_{\mathbb{Q}}(H)$ is a commutative semisimple algebra E_0 : after splitting the torus it is an algebra of diagonal scalars. The polarization adjoint preserves E_0 and is positive, since these scalars commute with the Weil operator. A positive involution on a real finite étale algebra fixes each primitive real summand and is complex conjugation on each complex summand. Exchanging idempotents would give $e^*e = 0$, and the identity involution on a complex summand would give $i^*i = -1$, contradicting positivity. A real summand is impossible here: the connected compact circle can act on it only trivially, whereas weight one has only the characters $z, \bar{z}$. Thus the rational field factors of E_0 are CM fields. Splitting the associated E_0 -modules into field lines gives rank-one CM Hodge structures. The projectors onto these field lines are rational Hodge maps. The same degree-one realization gives abelian isogeny factors of B with toric Hodge groups. For a rank-one module over a CM field E , the idempotents of $E \otimes_{\mathbb{Q}} k = \prod_{\sigma: E \hookrightarrow k} k$ give one-dimensional embedding lines. Scalar conjugation permutes these lines, and each is pure of type $(1, 0)$ or $(0, 1)$. $\square$

For admissible W , define its imbalance at γ by

$$\epsilon_\gamma(W) = \dim_{\mathbb{C}}(\gamma W_{\mathbb{C}} \cap H^{1,0}) - \dim_{\mathbb{C}}(\gamma W_{\mathbb{C}} \cap H^{0,1}).$$

The determinant line $\gamma D(W)$ is pure of bidegree given by these two dimensions. The perfect complementary pairing in Lemma 2.2 gives

$$\epsilon_\gamma(\overline{W}) = -\epsilon_\gamma(W) \quad \text{for every } \gamma. \quad (2.2)$$

A self-pair forces equal Hodge dimensions, hence even rank and imbalance zero. Abstract self-duality of an SL_2 module alone does not supply an all-conjugate $(1,1)$ self-pair; such a block may still need to be treated as linear. These identities remain valid without commuting γ with complex conjugation.

Theorem 2.5 (A determinant criterion for all powers). *Let A be a complex abelian variety, $H = H^1(A, \mathbb{Q})$, $J = \mathrm{Hdg}(H)$, and $R = J^{\mathrm{der}}$. Suppose the following data exist.*

- (a) *A rational Hodge summand $H_0 \subset H$ has toric Hodge group. Choose a decomposition of $H_{0,k}$ into CM embedding lines, with repeated occurrences distinguished, and call this finite list $\mathcal{E}_0$.*
- (b) *The complement of $H_{0,k}$ is a direct sum of blocks of nonzero admissible spaces. In a symplectic or an orthogonal block, each space is equipped with an admissible isomorphism to a representative V_i chosen as an actual summand of that block, with, respectively, an alternating or a symmetric pair tensor on itself. A linear block has $\dim V_i \geq 2$, with V_i again an actual summand, and each space is equipped with an admissible isomorphism to either V_i or $\overline{V}_i$, paired with each other as in Lemma 2.2. The pair identifies $\overline{V}_i$ with $V_i^*(-1)$ as a Hodge-graded space and with the dual as a group representation.*
- (c) *Under these identifications the image of R_k contains*

$$P = \prod_{i \in I_{\mathrm{Sp}}} \mathrm{Sp}(V_i) \times \prod_{i \in I_{\mathrm{SO}}} \mathrm{SO}(V_i) \times \prod_{i \in I_{\mathrm{SL}}} \mathrm{SL}(V_i).$$

Each factor acts by its standard representation on its own copies, by the dual representation on the opposite copies of a linear block, and trivially on every other block and on $H_{0,k}$. The symplectic and orthogonal forms are the inverse pairings.

- (d) *Let $W_1, \dots, W_s$ be any list, with arbitrary repetitions, from*

$$\{V_i : i \in I_{\mathrm{SO}}\} \cup \{V_i, \overline{V}_i : i \in I_{\mathrm{SL}}\}.$$

Whenever there are $\ell_1, \dots, \ell_r \in \mathcal{E}_0$, again with arbitrary repetitions, such that

$$\sum_{j=1}^s \epsilon_\gamma(W_j) + \sum_{a=1}^r \epsilon_\gamma(\ell_a) = 0 \quad \text{for every } \gamma \in \mathrm{Gal}(k/\mathbb{Q}), \quad (2.3)$$

the line $\bigotimes_{j=1}^s D(W_j)$ is CM-supplied on its separate tensor slots. Empty lists are allowed.

Then for every $p \geq 0$ every rational balanced tensor in $H^{\otimes 2p}$ is the class of a rational algebraic cycle on its separate A -slots. The rational Hodge conjecture holds in every codimension of A^m for every $m \geq 1$.

Call the data saturated if every singleton $D(V_i)$ for an orthogonal or linear block is CM-supplied, without any balance condition. For saturated data and any CM abelian variety M , the induced data on $A \times M$ are saturated. In particular the same tensor and Hodge-conjecture conclusions hold for every power of $A \times M$.

The list of CM lines in (2.3) is chosen once and then tested at every scalar conjugation. This is weaker than requiring individual determinant supply: special periods in Section 3 will require only certain products of column determinants. Saturation is useful when arbitrary CM companions are to be added.

Proof. We use the multilinear first fundamental theorems in their all-degree form. For a nondegenerate bilinear form b on V , let $t_b \in V \otimes V$ be its inverse-form, or coevaluation, tensor. The $\mathrm{Sp}(V)$ invariants in $V^{\otimes d}$ are spanned by permutations of products of t_b for the alternating form and vanish for odd d [23, Theorem 4.1 and Corollary 4.2]. For $\mathrm{SO}(V)$, products of t_b for the symmetric form and a top alternating tensor span all invariants [24, Theorem 2.1 and Corollary 4.2]. For $\mathrm{SL}(V)$ in mixed powers of V, V^* , the generators are the identity coevaluation tensor $\sum_j e_j \otimes e_j^* \in V \otimes V^*$, independent of the chosen basis, and the top alternating tensors of V and V^* [35, Section 5]. These are spanning statements in every degree; stable-range bounds concern independence, not spanning. They hold over k by scalar extension, since invariants are kernels of linear maps defined over k . For the product P , decompose each tensor slot into its block summands and apply these statements separately to each factor.

We need one elementary simultaneous-balance observation. Suppose a subspace $E \subset T_k$ of a rational pure weight- $2p$ Hodge structure is spanned by finitely many lines L_a , each pure in Hodge bidegree after every scalar conjugation. A rational vector $\alpha \in E$ of type (p, p) is spanned by those L_a whose every conjugate has type (p, p) . To prove this, for fixed γ group the original lines according to the bidegree of γL_a . Their group sums form a direct sum, as is seen by applying γ and using the Hodge decomposition. Let q_γ be the projection retaining the group of type (p, p) . Since $\gamma\alpha = \alpha$, this projection fixes α . Every L_a is an eigenline of every q_γ with eigenvalue zero or one. The projections therefore commute even if the spanning lines are dependent. Only finitely many distinct projections occur, because each is determined by the subset of the finite list it retains. Their product fixes α and has image spanned by exactly the lines retained by every conjugation. This proves the observation.

Fix a tensor degree. A rational balanced Hodge tensor is fixed by J , hence by P . The invariant theorems show that the P -invariant subspace is spanned by finitely many lines formed, up to slot permutations and the prescribed admissible identifications, from three types of factors:

$$\text{lines in } \mathcal{E}_0, \quad \text{pair tensors,} \quad D(V_i) \text{ for } i \in I_{\mathrm{SO}}, \quad D(V_i), D(\bar{V}_i) \text{ for } i \in I_{\mathrm{SL}}.$$

Every such line is pure after every scalar conjugation. This is part of the definition for pairs, follows from the top exterior power of a Hodge-graded space for determinants, and holds for CM embedding lines. The simultaneous-balance observation reduces a rational balanced tensor to generator lines balanced in all conjugates. Discarding the pair factors from the bidegree count leaves exactly (2.3), with the actual CM lines in the chosen generator. Assumption (d) supplies its determinant product. Lemma 2.4 supplies the toric summand by a CM isogeny factor, and the pair tensors and identifications are algebraic by Lemma 2.2. Lemma 2.1 therefore places the entire generator line in the scalar extension of the rational CM-supplied subspace. Rational membership descends, and the same lemma makes the original tensor algebraic.

For A^m , first cohomology is $H^{\oplus m}$. The same argument works with slots labelled by the copies: it merely repeats the old blocks, and assumption (d) already permits arbitrary repetitions. Finally $H^{2p}(A^m, \mathbb{Q}) = \bigwedge^{2p} H^1(A^m, \mathbb{Q})$. Normalized antisymmetrization gives a rational Hodge lift of a Hodge class into the ordinary tensor power. The tensor result makes that lift algebraic on separate slots. Pullback along the diagonal, equivalently cup product of the slots, returns the exterior class.

Suppose now that the data are saturated. Scalar conjugation supplies the opposite determinants, and products supply every determinant list, so (d) holds. Let T be the toric Hodge group of $H^1(M, \mathbb{Q})$ and let J' be the Hodge group of $H \oplus H^1(M, \mathbb{Q})$. It is a connected reductive subgroup

of $J \times T$ whose projection onto J is surjective by minimality. Its derived subgroup consequently projects onto R and trivially onto T . Thus, inside this representation,

$$(J')^{\text{der}} = R \times \{1\}.$$

The product P still acts on the old blocks and trivially on the enlarged toric summand $H_0 \oplus H^1(M, \mathbb{Q})$. The old singleton determinants remain supplied after pullback along $A \times M \rightarrow A$ on their slots. These are saturated data on $A \times M$, and the proved assertions apply. If all of H is toric, there are no classical blocks and the only determinant list is empty, so this case is included. $\square$

2.4 Passing from supplied cohomology to powers

The next elementary transfer passes from complete cohomological supply to Hodge classes on arbitrary products. Its hypothesis concerns products of the source varieties, not merely each source separately.

Lemma 2.6 (Correspondence transfer). *Let $\mathcal{S}$ be a collection of smooth projective varieties such that the rational Hodge conjecture holds on every finite product of members of $\mathcal{S}$. Let $Y_1, \dots, Y_v$ be smooth projective varieties. Suppose that, for each i and each degree q , $H^q(Y_i, \mathbb{Q})$ is spanned by images of maps*

$$\Gamma_* : H^{q-2j}(S, \mathbb{Q}) \longrightarrow H^q(Y_i, \mathbb{Q}), \quad \Gamma \in \text{CH}^{\dim S+j}(S \times Y_i)_{\mathbb{Q}}, \quad S \in \mathcal{S}.$$

Then the rational Hodge conjecture holds on every finite product of the Y_i , with arbitrary repetitions and with any additional members of $\mathcal{S}$.

The same conclusion holds if the spanning is given over k by restrictions of maps induced by k -linear combinations of rational algebraic correspondences, provided their images span the full target degree after scalar extension.

Proof. Fix a product Y allowed by the conclusion. In each degree, finite dimensionality makes a finite sum of the stipulated maps surjective. Tensor these maps and the identity maps of any additional source factors. The Künneth decomposition shows that, in each degree of the target product, a finite sum of maps from finite source products is surjective. One uses the indicated Künneth summands only to see that the images span; no algebraic Künneth projector on a source is required. External products and composition make all these maps algebraic correspondences. For a target degree $2p$, write j_ν for the sum of the individual shift parameters of one such external-product map, so its cohomological shift is $2j_\nu$, and write S_ν for its source product. The surjection is a morphism

$$\bigoplus_{\nu} H^{2p-2j_\nu}(S_\nu, \mathbb{Q})(-j_\nu) \longrightarrow H^{2p}(Y, \mathbb{Q})$$

of polarizable pure Hodge structures of weight $2p$. The splitting argument in Lemma 2.1 lifts a rational target Hodge class. Its ν th component before twisting has type $(p - j_\nu, p - j_\nu)$ on S_ν . It is algebraic by the source hypothesis, so its image is algebraic.

For the scalar variant, expand each scalar correspondence into finitely many rational correspondence cycles. Its image on any specified source subspace is contained in the scalar extension of the sum of the full images of those rational maps. If the scalar images span the full target degree, the quotient by that rational image sum becomes zero after scalar extension and was already zero over $\mathbb{Q}$. The first part therefore applies. $\square$

For later use, if A has saturated data, then the collection of varieties $A^a \times M$, where $a \geq 0$ and M is any CM abelian variety, satisfies the source hypothesis of Lemma 2.6. A finite product is $A^{\sum a} \times \prod M$, an abelian factor of a sufficiently high power of $A \times \prod M$, and its Hodge classes transfer by the algebraic factor inclusion and projection. For a curve C with $A = \text{Jac}(C)$, the Abel map supplies $H^1(C)$ from A , while a point and the fundamental class supply the even cohomology. If a smooth projective variety F has $\mathcal{C}^q(F) = H^q(F, \mathbb{Q})$ in every degree, finite dimensionality expresses each degree by finitely many rational correspondence images from CM abelian varieties. External products then supply the full cohomology of every product of copies of C and such varieties F from varieties $A^a \times M$ in the preceding source collection. Lemma 2.6 gives the corresponding Hodge-conjecture conclusion.

3 Quadratic actions at every period

We prove Theorems 1.1 and 1.2 by applying the determinant criterion of Theorem 2.5 to the Hodge group at the given period. The cycle input is algebraicity of the Weil plane on every hyperbolic Weil sixfold. We state it below and defer its geometric proof to Sections 4–8.

3.1 The Weil-plane input

An action of an imaginary quadratic field on an abelian variety will always mean a unital embedding

$$E \hookrightarrow \text{End}^0(A).$$

The action need not be the full endomorphism algebra, and it need not be central on an isotypic power. We first specify the polarization convention in the six-dimensional result.

Write $E = \mathbb{Q}(u)$, where $u^2 = -d$ and $d \in \mathbb{Z}_{>0}$, and put $V = H_1(A, \mathbb{Q})$. A rational Riemann form e for a polarization is *compatible* with E if

$$e(ax, y) = e(x, \bar{a}y) \quad (a \in E).$$

In particular $e(ax, ay) = \text{Nm}_{E/\mathbb{Q}}(a)e(x, y)$, the norm-pullback condition for a polarized variety of Weil type. It then defines the Hermitian form, conjugate-linear in the first variable,

$$h_E(x, y) = e(x, uy) + ue(x, y) \in E. \tag{3.1}$$

This is the usual homological convention [40, Lemma 5.2]. The adjoint identity and alternation show directly that h_E is Hermitian and that

$$e(x, y) = \text{Tr}_{E/\mathbb{Q}}((2u)^{-1}h_E(x, y));$$

in particular it is nondegenerate. For an embedding $\sigma : E \hookrightarrow k$, write $H^1(A, k)_\sigma$ for the simultaneous σ -eigenspace of the pullback E -action. It is admissible: its spectral projector is

$$\frac{u^* - \bar{\sigma}(u) \text{id}}{\sigma(u) - \bar{\sigma}(u)},$$

and every scalar conjugate is a Hodge endomorphism. If $\sigma(u) = i\sqrt{d}$, the signature of h_E is (p, q) , where

$$p = \dim_{\mathbb{C}}((H^1(A, k)_\sigma)_{\mathbb{C}} \cap H^{1,0}(A)), \quad q = \dim_{\mathbb{C}}((H^1(A, k)_{\bar{\sigma}})_{\mathbb{C}} \cap H^{1,0}(A)).$$

Here $p + q = \dim A$. On $V_{\mathbb{R}}$ viewed via σ as an $E \otimes_{\mathbb{Q}} \mathbb{R} \simeq \mathbb{C}$ -space, the complex structure I has eigenspaces with eigenvalues $i, -i$. On them $u = \sqrt{d}I$ and $u = -\sqrt{d}I$, respectively, so $h_E(x, x) = e(x, ux)$ has the indicated positive and negative signs by the Riemann positivity $e(x, Ix) > 0$. The eigenspaces are Hermitian-orthogonal because h_E is I -invariant. The corresponding holomorphic σ - and $\bar{\sigma}$ -forms have dimensions p and q . A nondegenerate rank- $2b$ Hermitian form is called *hyperbolic* if it has a totally isotropic E -subspace of dimension b .

Every quadratic action admits a compatible polarization. Starting with a rational Riemann form e_0 , set

$$e(x, y) = e_0(x, y) + d^{-1}e_0(ux, uy). \quad (3.2)$$

Both summands satisfy Riemann positivity because u is an invertible complex-linear map. Direct substitution gives $e(ux, y) = -e(x, uy)$. A positive integral multiple clears denominators and is the Riemann form of an ample polarization.

For a sixfold, $H^1(A, \mathbb{Q})$ has dimension twelve and is an E -space of rank six. Thus its two spectral eigenspaces each have dimension six. The direct sum

$$\bigwedge^6 H^1(A, k)_{\sigma} \oplus \bigwedge^6 H^1(A, k)_{\bar{\sigma}} \subset H^6(A, k)$$

is stable under every scalar conjugation: scalar conjugation merely permutes the two spectral eigenspaces and their exterior powers. Finite-dimensional Galois descent therefore gives a unique rational two-dimensional subspace $\mathcal{W}_E(A) \subset H^6(A, \mathbb{Q})$ whose scalar extension is this sum. When h_E has signature $(3, 3)$, each determinant line has type $(3, 3)$.

Theorem 3.1 (Algebraicity of the hyperbolic Weil plane). *Let E be an imaginary quadratic field. Let A be a complex abelian sixfold with a unital E -action and a compatible polarization whose actual homological Hermitian form h_E is hyperbolic of signature $(3, 3)$. Then*

$$\mathcal{W}_E(A) \subset \text{im}(\text{CH}^3(A)_{\mathbb{Q}} \longrightarrow H^6(A, \mathbb{Q})).$$

The assertion holds at every compatible period, including nonsimple varieties and periods with additional endomorphisms.

Markman states this theorem in the discriminant- -1 formulation [25, Theorem 1.5.1]; his published account also states it for all split Weil sixfolds [26, Theorem 1.2]. The later sections give the secant-sheaf and spin proof with the fixed conventions used here.

The theorem supplies the total E -determinant of a split sixfold. At a special period that determinant can split into several column determinants for the larger endomorphism algebra. Our task is to show that the products required by Theorem 2.5 still have CM supply. We first obtain determinant supply by adding CM elliptic factors, then determine the actual small-dimensional derived groups, and finally check the determinant products.

3.2 Determinants supplied by a split sixfold

For a Hermitian form h , write $[\det h]$ for its determinant class in $\mathbb{Q}^{\times} / \text{Nm}_{E/\mathbb{Q}}(E^{\times})$. Changing an E -basis multiplies the determinant by a norm.

Lemma 3.2 (A rational hyperbolicity test). *Let E be imaginary quadratic. A nondegenerate E -Hermitian form of rank $2b$ and signature (b, b) , $b \geq 1$, is hyperbolic if and only if*

$$[\det h] = [(-1)^b].$$

Every hyperbolic form of this rank is isometric to the sum of b planes with matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Proof. For necessity, let L be a totally isotropic b -space and choose $0 \neq v \in L$. Choose w with $h(v, w) = 1$ and replace w by $w - \frac{1}{2}h(w, w)v$. The two vectors now span a hyperbolic plane. The projection of L to the orthogonal complement of this plane has isotropic dimension $b - 1$: for $x \in L$, replace x by $x - h(w, x)v$. Induction gives the displayed sum of planes, each with determinant -1 . This proves necessity and the isometry assertion.

Conversely, $x \mapsto h(x, x)$ is a nondegenerate rational quadratic form of dimension $4b$ and signature $(2b, 2b)$. Nondegeneracy follows from the trace pairing and trace duality. When $b \geq 2$, Meyer's theorem gives a nonzero isotropic vector [31, Corollary 3.3.4]. Split off its hyperbolic plane. The complement has signature $(b - 1, b - 1)$ and determinant class $[(-1)^{b-1}]$, so one may repeat until rank two. Write the last form diagonally as $\langle a, b' \rangle$, with $a, b' \in \mathbb{Q}^\times$. Its determinant condition says $-ab' = \text{Nm}(t)$ for some $t \in E^\times$. The vector $(t/a, 1)$ is isotropic because

$$a \text{Nm}(t/a) + b' = 0.$$

It splits the last hyperbolic plane. □

The following construction adjoins auxiliary factors and then removes their determinant lines by algebraic pairings. Schoen [34, Proposition 10] uses a related padding and extraction argument for Weil fourfolds; here the auxiliary factors are CM elliptic curves and the output is determinant supply on separate slots.

Lemma 3.3 (Quadratic determinant supply). *Let A be a complex abelian variety with an action of an imaginary quadratic field E , and let a compatible rational Riemann form have associated Hermitian form of signature (p, q) . Put $n = \dim A = p + q$. Both lines*

$$D(H^1(A, k)_\sigma), \quad D(H^1(A, k)_{\bar{\sigma}})$$

are CM-supplied in either of the following cases:

1. $n = 6$, $(p, q) = (3, 3)$, and the form is hyperbolic;
2. $n < 6$ and $p, q \leq 3$.

The second assertion permits every compatible rational Riemann form. Both assertions hold at every period.

Proof. Theorem 3.1 proves that the rational Weil plane is algebraic on every polarized imaginary-quadratic sixfold whose actual homological Hermitian form is hyperbolic of signature $(3, 3)$. Its convention is exactly (3.1), so it applies directly to the first case, including special periods. It supplies the rational Weil two-plane, not all Hodge classes at such a period.

For a sixfold X in the first case, the definition of $\mathcal{W}_E(X)$ shows that algebraicity of this rational plane makes each of its two embedding determinant lines algebraic over k . To pass to the determinant on separate slots, let $\mu : X^6 \rightarrow X$ be addition. For an integer $t \geq 2$, the interpolation polynomial

$$\pi_1^X = \prod_{\substack{0 \leq r \leq 2 \dim X \\ r \neq 1}} \frac{[t]^* - t^r \text{id}}{t - t^r}$$

is an algebraic cohomological projector to $H^1(X)$, since $[t]^*$ acts by t^r on $H^r(X)$. For a basis $v_1, \dots, v_6$ of either embedding space, expanding $\mu^*v = \sum_i \text{pr}_i^*v$ shows that

$$(\pi_1^X)^{\boxtimes 6} \mu^*(v_1 \wedge \dots \wedge v_6) = \sum_{\tau \in S_6} \text{sgn}(\tau) v_{\tau(1)} \otimes \dots \otimes v_{\tau(6)}.$$

The right side is a nonzero vector in the top alternating line. Thus both determinant lines in the first assertion are algebraic over k , and hence are CM-supplied from a point.

Now start with an arbitrary A in the second assertion. Add $3 - p$ rank-one E -Hodge structures of signature $(1, 0)$ and $3 - q$ of signature $(0, 1)$. They are realized by CM elliptic curves $C_1, \dots, C_{6-n}$ with the chosen or the conjugate E -action. In an E -basis their Hermitian coefficients may have any positive rational magnitude and the prescribed signs: positive rational rescaling of an elliptic polarization leaves the Hodge structure unchanged.

The orthogonal sum on $X = A \times C_1 \times \dots \times C_{6-n}$ has signature $(3, 3)$. Its determinant d_0 in chosen bases is negative. At least one line has been added, so multiply one added coefficient by $-1/d_0 > 0$. The new determinant is -1 , and Lemma 3.2 makes the form hyperbolic. Clearing denominators scales the form by a positive rational number; in rank six this changes the determinant by a norm and preserves the required discriminant. The first part therefore supplies the determinant of this actual product.

Put $L_j = H^1(C_j, k)_\sigma$. The σ -space of X is

$$H^1(A, k)_\sigma \oplus L_1 \oplus \dots \oplus L_{6-n}.$$

Project its determinant to successive blocks of slots belonging to these summands. The alternation of a concatenated basis maps to a nonzero multiple of

$$D(H^1(A, k)_\sigma) \otimes L_1 \otimes \dots \otimes L_{6-n}.$$

These block projections are induced by the product inclusions and projections, so are algebraic. Each opposite line $\bar{L}_j$ is supplied by the identity on the CM curve C_j . Tensoring with these lines and contracting the perfect elliptic polarization pairings removes all L_j . The contractions are nonzero and algebraic by Lemma 2.2. Lemma 2.1 now supplies the required determinant. Scalar conjugation supplies its opposite. The construction began with the fixed A , so no period-generality condition was used. $\square$

3.3 Columns and their pairings at the given period

We first decompose first cohomology into admissible irreducible summands and determine their pairings. The endomorphism algebra in this calculation is the actual full algebra at the given period. We will then determine the derived image on each summand and prove that the resulting images act independently, as required by Theorem 2.5.

Let B be a simple non-CM abelian variety, $g = \dim B$, and set

$$H_B = H^1(B, \mathbb{Q}), \quad J_B = \text{Hdg}(H_B), \quad R_B = J_B^{\text{der}}, \quad D = \text{End}_{\text{Hdg}}(H_B).$$

The degree-one realization in Lemma 2.2 identifies D with the opposite of $\text{End}^0(B)$, a division algebra. It is also the commutant of J_B : the rational centralizer of an endomorphism commuting with the circle contains J_B by minimality. The group J_B is connected reductive as in Section 2. If $Z = Z(D)$ and $[D : Z] = s^2$, base change of the commutant and complete reducibility give

$$D_k = \prod_{\tau: Z \hookrightarrow k} M_s(k), \quad H_{B,k} = \bigoplus_{\tau: Z \hookrightarrow k} W_\tau \otimes U_\tau, \quad \dim U_\tau = s. \quad (3.3)$$

Here J_B acts irreducibly on W_τ and trivially on U_τ ; distinct τ give nonisomorphic full- J_B modules. Choose one primitive matrix idempotent in each central block and identify W_τ with its image in $H_{B,k}$. We call this actual summand a *column*. The other primitive idempotents select its matrix

copies, and matrix units identify them with the chosen column. These idempotents and maps belong to D_k , so the spaces and identifications are admissible. The pairings below will be transported to these chosen copies; the restriction of the original polarization to an arbitrary copy need not be nondegenerate.

A column remains irreducible on R_B : the connected center acts by scalars, and the product of that center with R_B surjects onto J_B . A column is *mixed* if it has both circle weights 1, -1 , and *pure* if it has only one. These terms refer to the chosen embedding: scalar conjugation can take a pure column to a mixed one. A self-pair means a pair tensor on the same column.

Lemma 3.4 (Albert columns and their pairings). *In Types I–III let $F = Z(D)$ be totally real of degree f . In Type IV let the CM center $K = Z(D)$ have degree $2f$, and write $[D : K] = b^2$. The columns in (3.3) have the following data:*

Albert type	necessary divisibility	$\dim_k W$	pair tensor
I	$f \mid g$	$2g/f$	alternating on W
II	$2f \mid g$	g/f	alternating on W
III	$2f \mid g$	g/f	symmetric on W
IV	$fb^2 \mid g$	$g/(fb)$	between opposite columns.

The listed tensors are pair tensors in the sense of Section 2, and all copy identifications are admissible. In Type IV the opposite column is the dual R_B -module.

Proof. We use the structural Albert classification with its positive Rosati involution [10, Sections 10.13–10.14]. In Types II and III, D is quaternionic over F , so its rational dimension is $4f$. Since H_B is a vector space over D , $2f \mid g$. In Type IV the rational dimension of D is $2fb^2$, giving $fb^2 \mid g$. Dividing a central block by its split matrix multiplicity gives the displayed dimensions. In Type I the initial column dimension $2g/f$ must be even by the alternating form proved next, giving $f \mid g$.

Choose a rational polarization pairing ψ on H_B . On a first-kind block $W \otimes U$, the Rosati involution on $\text{End}(U)$ is adjoint to a nondegenerate form b_U . The real-center idempotents are selfadjoint, so these central blocks are orthogonal and the restriction of ψ to each is perfect. Fixing two W -arguments in ψ gives a form c on U satisfying

$$c(Tu, v) = c(u, T^\dagger v) \quad (T \in \text{End}(U)).$$

Writing $c(u, v) = b_U(Su, v)$ shows that S commutes with every matrix, hence is scalar. Consequently

$$\psi|_{W \otimes U} = \beta_W \otimes b_U$$

for a nondegenerate J_B -invariant form β_W . Since ψ is alternating, the symmetry signs of the two factors are opposite.

In Type I, U is one-dimensional and b_U is symmetric. In Type II the positive first-kind involution on the split real quaternion algebra is orthogonal, so b_U is symmetric after splitting. One can see the sign from positivity: for a symplectic-adjoint involution on $M_2(\mathbb{R})$, a rank-one idempotent e would satisfy $e^\dagger = 1 - e$, giving $\text{Tr}(ee^\dagger) = 0$. In Type III the positive involution on the Hamilton quaternions is canonical conjugation, which becomes symplectic-adjoint on $M_2(k)$. The alternative first-kind involution fixes a nonzero pure quaternion v and would give $\text{Trd}(vv^\dagger) = \text{Trd}(v^2) < 0$. Thus b_U is alternating in Type III and β_W is symmetric.

These factorizations yield the required pair tensors on the chosen columns. For Types I and II first choose a multiplicity line nonisotropic for b_U . Restriction of ψ to its column copy is then

nondegenerate. Transport its inverse tensor to the chosen copy of W by an admissible matrix-unit isomorphism. For Type III choose two multiplicity lines with nonzero b_U -pairing, restrict ψ between their column copies, and identify those copies by a matrix unit. This gives a nonzero scalar multiple of the symmetric β_W ; restricting twice to one multiplicity line would instead give zero. Again transport the inverse tensor to the chosen column by a matrix-unit isomorphism. The projectors and matrix units are admissible, and the rational pairing ψ pairs complementary Hodge bidegrees. Its factored inverse tensors therefore have type $(1, 1)$ in every scalar conjugate. The chosen nonzero multiplicity pairing remains nonzero under those conjugations.

In Type IV positivity makes the involution on K complex conjugation. Thus ψ pairs only opposite central embeddings. Factoring the split matrix pairing gives a perfect pairing of their columns. For the chosen copy W , use the perfect W -to- $\bar{W}$ pair of Lemma 2.2, and identify $\bar{W}$ with the chosen copy at the opposite embedding by an admissible matrix-unit isomorphism. This supplies the pair on the chosen columns and identifies the opposite column with the derived dual. $\square$

3.4 The possible derived projections

The two Hodge-circle weights impose the standard minuscule restriction on the derived representations; compare [38, Section 3.2, Lemmas 4–5]. We give the argument in the circle normalization with weights $1, -1$, retaining the scalar-conjugate columns needed at special periods.

Lemma 3.5 (Two circle weights). *In the irreducible tensor decomposition of a column W_τ for the simple ideals of $\text{Lie}(R_B)_k$, every nontrivial simple factor representation is minuscule.*

Proof. The sum of a Galois orbit of simple ideals descends to a rational ideal. In each such orbit the complexified circle differential has a nonzero projection to at least one ideal. Otherwise the kernel of the rational adjoint action on that ideal would be a proper rational algebraic subgroup containing the circle, contrary to minimality of J_B .

Given an acting ideal, pass to a simultaneous scalar conjugate of it and W_τ for which this projection x is nonzero. The conjugated column is still admissible, so it has only circle weights $1, -1$. Fix eigenvectors in all other tensor factors. The semisimple traceless operator x then has at most two eigenvalues on its factor. It has at least two because a nonzero representation of a simple Lie algebra is faithful and a scalar traceless semisimple operator is zero.

Choose a Cartan subalgebra containing x . For any root α , a Weyl conjugate x' has $\alpha(x') \neq 0$: the Weyl orbit of a root spans the dual Cartan space. If a highest weight λ satisfied $\langle \lambda, \alpha^\vee \rangle \geq 2$ for a positive root, the root string $\lambda, \lambda - \alpha, \lambda - 2\alpha$ would yield three different eigenvalues on x' . Weyl conjugation does not change the spectrum, so this is impossible. Hence $\langle \lambda, \alpha^\vee \rangle \leq 1$ for every positive root, the minuscule condition. Scalar conjugation transports it back to the original factor. $\square$

The part of the minuscule classification needed here is particularly small [14, Sections 27.2, 30.4, and 32.1]:

dimension	simple nontrivial minuscule modules
2	A_1 standard, alternating;
3	A_2 standard and dual;
4	A_3 standard and dual; $C_2 = B_2$ standard/spin, alternating;
5	A_4 standard and dual;
6	A_5 standard and dual; C_3 standard, alternating; A_3 on $\wedge^2 k^4$, symmetric.

The unmarked standard and dual modules are nonselfdual. The low-rank identifications $D_3 = A_3$ and $B_2 = C_2$ are included. Every selfdual module in this table has even dimension.

Lemma 3.6 (Small derived projections). *An alternating self-paired column of dimension 2, 4, or 6 has the full symplectic derived image. A symmetric self-paired column cannot have dimension two; in dimensions four and six its derived image is, respectively, the full SO_4 and SO_6 .*

If $D = K$ is an imaginary quadratic field, B is non-CM, and $\dim_K H_B \leq 6$, then the derived image on either embedding column is the full special linear group, with dual action on the opposite column.

Proof. A self-paired column is a selfdual derived module. Every factor in its irreducible tensor decomposition is then selfdual, by restricting an isomorphism with the dual to each simple ideal. In dimensions at most six the only nontrivial tensor decompositions are $2 \cdot 2$ and $2 \cdot 3$. The latter is not selfdual. The former is a tensor product of two alternating A_1 standards and has a symmetric form. The displayed table and Lemma 3.5 now give exactly the asserted alternating images, the symmetric $2 \otimes 2$ image in dimension four, and the symmetric $\bigwedge^2 k^4$ image in dimension six. These last two are SO_4 and SO_6 . A connected subgroup of SO_2 is a torus and cannot act irreducibly on a two-dimensional space over k , excluding that dimension. Connectedness promotes the Lie-algebra images to the stated groups.

For the quadratic assertion write $W, \bar{W}$ for the two columns. Both are mixed: if one were pure, the circle on $W \oplus \bar{W}$ would lie in the rational K -scalar torus, making B CM. They are derived duals by Lemma 3.4. If W were derived-selfdual, $\mathrm{Hom}_{R_B}(W, \bar{W})$ would be one-dimensional and stable under J_B , since R_B is normal. Choose a nonzero element of this line; it is an isomorphism by Schur's lemma. This intertwiner shifts the entire circle weight spectrum by a scalar character. Both spectra are $\{-1, 1\}$, so the shift is zero. This remains true for its scalar conjugates, since there are only the two mixed embedding columns. Extending by the source projector and applying (2.1) gives a full-Hodge-group endomorphism joining the two distinct central embeddings, contrary to the commutant K_k . Thus W is not derived-selfdual.

Every simple ideal of $\mathrm{Lie}(R_B)_k$ acts on W : the action on $W \oplus \bar{W} = H_{B,k}$ is faithful, and the two actions are dual. Each Galois orbit has an ideal with nonzero circle differential, as in Lemma 3.5. There cannot be two such active ideals, since two nonconstant factor spectra have a sumset of at least three elements, whereas W has only two weights. Hence all acting ideals form one Galois orbit and have the same Lie type. The only nonselfdual tensor candidate in dimensions at most six is $2 \otimes 3$, with types A_1 and A_2 , so it cannot occur. The remaining nonselfdual modules in the table are exactly the standard and dual standard special linear modules. This proves the assertion. $\square$

Lemmas 3.4–3.6 give the permitted representations once the column dimensions are known. The quadratic action and the total dimension now restrict which columns can actually occur.

Lemma 3.7 (The small column list). *Let A have an imaginary-quadratic E -action and $\dim A \leq 6$. Write $A \sim \prod_B B^{r_B}$ with pairwise nonisogenous simple factors, and discard the CM factors. Put $g = \dim B$. After identifying matrix copies, the remaining columns have the following possibilities.*

1. *In Types I and II they have full symplectic derived images and dimensions 2, 4, 6.*
2. *In Type III, $g = 4$ or 6, the center is $\mathbb{Q}$, and $r_B = 1$. The unique column has full SO_g image. Each eigenspace of the given E -action is one copy of this column and has equal Hodge multiplicities $g/2, g/2$.*
3. *In Type IV, $D = K$ is a CM field. If E admits no $\mathbb{Q}$ -embedding into K , then K is imaginary quadratic, $g = 3$, and $r_B = 2$; the K -columns have the full SL_3 and dual images.*

If E admits a $\mathbb{Q}$ -embedding into K , put $[K : \mathbb{Q}] = 2f$ and $a = g/f$. The possibilities are

$$f = 1, \quad 3 \leq a = g \leq 6; \quad (f, a, g) = (2, 2, 4), (2, 3, 6), (3, 2, 6).$$

Every column has the full SL_a image, paired with the dual at the opposite embedding. In the cases with $a = 2$, some embedding pairs are mixed and some are pure.

All these are projections of the actual derived Hodge group of A .

Proof. The action restricts to each isotypic factor: the central isotypic projectors commute with it, and a unital homomorphism from a field to each matrix factor is injective.

In Type I it extends to an action of the quadratic field $F \otimes_{\mathbb{Q}} E$ on F^{r_B} , since F is totally real. Thus r_B is even and $g \leq 3$. The table in Lemma 3.4 gives alternating column dimensions 2, 4, 6. In Type II the same dimensions follow from $2f \mid g \leq 6$. Lemma 3.6 gives the full symplectic images.

In Type III the initial symmetric dimensions are also 2, 4, 6. Lemma 3.6 excludes two and gives the orthogonal images for four and six. Since the column dimension is g/f and $gr_B \leq 6$, either remaining dimension forces $f = r_B = 1$. The injection $E \hookrightarrow D$ becomes $k \times k \hookrightarrow M_2(k)$; its two nonzero complementary idempotents each have rank one. Its eigenspaces are therefore the two separate matrix copies of the column. The self-pair matches the two Hodge weights perfectly, proving the asserted equal multiplicities.

For Type IV first eliminate noncommutative D . If $b > 1$, the divisibility $fb^2 \mid g \leq 6$ leaves only

$$(g, f, b, \dim W) = (4, 1, 2, 2).$$

More generally, a two-dimensional column cannot have every scalar conjugate mixed. If it did, any nonzero tensor in $D(W)$ would be a nondegenerate alternating pair tensor of type $(1, 1)$ in every conjugate. If its isomorphism is $t_W : W^* \rightarrow W$ and the polarization isomorphism is $t_{\mathrm{opp}} : W^* \rightarrow \bar{W}$, then $t_{\mathrm{opp}} t_W^{-1} : W \rightarrow \bar{W}$ is an admissible Hodge isomorphism, since the two Hodge shifts cancel. By (2.1) it belongs, after extending by the source projector, to D_k . This is impossible because D_k preserves the distinct center embeddings.

If every Type IV column is pure, the circle acts by a scalar on each $W_{\tau} \otimes U_{\tau}$, with opposite scalars at opposite embeddings. It therefore lies in the rational torus of center scalars. Minimality makes the Hodge group toric, so B is CM by Section 2. A non-CM rank-two factor must consequently have both pure and mixed embedding pairs. For $f = 1$ there is only one pair, excluding the tuple above and also the field case $f = 1, a = 2$. A one-dimensional column is likewise pure at every embedding and gives CM.

We now have $D = K$. The K -algebra $K \otimes_{\mathbb{Q}} E$ is a quadratic field L/K exactly when E does not admit a $\mathbb{Q}$ -embedding into K ; otherwise it is $K \times K$. The given action on $\mathrm{Mat}_{r_B}(K)$ extends to a unital map from this K -algebra. In the field case K^{r_B} is a vector space over L , so r_B is even. Hence $g \leq 3$; the exclusions just proved leave only $f = 1, g = 3, r_B = 2$, with quadratic K distinct from E . In the split case its two idempotents have ranks $r_+, r_- \geq 0$ summing to r_B . Thus the action may be noncentral, as in $a \mapsto \mathrm{diag}(a, \bar{a})$ on K^2 when $K = E$. Diagonalizing the idempotents by a K -matrix only changes how the matrix copies in each opposite pair of K -columns are assigned to the two E -eigenspaces; these matrix identifications are admissible.

In the split case $fa = g \leq 6, a \geq 2$, and the exclusion of $f = 1, a = 2$ give exactly the displayed list. When $f = 1$ the column is mixed and Lemma 3.6 applies. When $f > 1$ its dimension is two or three. Its restriction to R_B is irreducible by the observation following (3.3); since its dimension is greater than one, that action is nontrivial. Lemma 3.5 and the displayed small table force SL_2 or SL_3 , since two nontrivial factors already have dimension at least four. This holds also at a pure column: the active circle differential may be at another scalar-conjugate column. The rank-two pure/mixed assertion was proved above.

Finally, let ρ_B be the restriction of $\mathrm{Hdg}(H^1(A))$ to H_B . Its rational algebraic image contains the circle of B , hence contains J_B by minimality. Conversely, $\rho_B^{-1}(J_B)$ is a rational algebraic subgroup

of $\text{Hdg}(H^1(A))$ containing the circle of A . Minimality of $\text{Hdg}(H^1(A))$ forces this inverse image to be the whole group. Thus the image is J_B , and the derived projection is also surjective. The images just computed are therefore the projections of the actual group of A . $\square$

3.5 Independence of the derived images

Lemma 3.7 gives the individual projections, whereas assumption (c) of Theorem 2.5 requires their simultaneous product. A proper subdirect image would identify two projected simple ideals, so that the same derived factor acts on different column blocks. We exclude these graph links using the actual endomorphism algebra and the circle weights. This method is closely related to the product and intertwiner arguments of Moonen and Zarhin [28, (3.1) and Lemmas 3.3–3.4]; the proof below treats the column representations and partially pure cases in the present dimension range.

Lemma 3.8 (Independence of the small columns). *In Lemma 3.7, the listed symplectic, orthogonal, and special linear derived images are independent, after identifying matrix copies and pairing opposite CM embeddings. In other words, they satisfy assumption (c) of Theorem 2.5.*

Proof. Split each $\mathfrak{so}_4$ target into its two simple ideals. The Lie image of the derived group of A is a semisimple subdirect algebra of the resulting product of simple target algebras. A projection of a simple source ideal onto a simple target is either zero or an isomorphism; exactly one source ideal maps nontrivially to each target. Indeed its image is an ideal, and the images of different source ideals commute. Thus failure of independence is exactly a graph link identifying two isomorphic target ideals. The link relation is invariant under scalar Galois transport.

First consider columns mixed after every scalar conjugation, apart from the two factors of an SO_4 target and the $\mathfrak{so}_6 \simeq \mathfrak{sl}_4$ case treated next. An isomorphism of the target Lie algebras in our list identifies their standard modules up to duality in the special linear case. Replace one column by its opposite in the dual case. The intertwiner equations are defined over k ; choose a nonzero derived intertwiner t over k . Its space is a line by irreducibility, and normality of the derived group makes this line stable under the full Hodge group. Writing $h(z)$ for the Hodge-circle action, we have

$$h(z)th(z)^{-1} = z^m t$$

for an integer m . Because t is an isomorphism, the target weight spectrum is the source spectrum translated by m . Both sets are $\{-1, 1\}$, so $m = 0$. The same argument applies to every scalar conjugate of t . Extending it by its source projector and using (2.1) puts it in $\text{End}_{\text{Hdg}}(H^1(A, \mathbb{Q})) \otimes k$. This commutant may mix only matrix copies within one central embedding and one simple isogeny type. The putative link contradicts the choice of different columns. Self-pairs force mixedness in every conjugate, and a non-CM quadratic-center column is mixed in both conjugates. Thus this argument covers all ordinary links outside the partially pure cases.

The only unequal standard representations of isomorphic simple targets in this range, outside SO_4 , arise from $\mathfrak{so}_6 \simeq \mathfrak{sl}_4$. A Type III SO_6 factor has dimension six and exhausts A , so no other non-CM factor can provide such a link.

Consider next a Type IV factor with both pure and mixed columns. We first exclude links among its own columns. Here $r_B = 1$, and the columns over one embedding of E are the $f = 2$ or 3 embeddings of K/E . The action of $\text{Gal}(k/E)$ on them is transitive. An invariant equivalence relation on a prime-sized transitive set is either discrete or universal, since its equivalence classes have equal size. On a column the derived part of the circle differential is its traceless part: the connected center acts by scalars. It vanishes on a pure column and is nonzero on a mixed column. A graph link identifies the two projections of the same circle differential, so it preserves whether

that differential is zero. The universal relation would link a pure column to a mixed one and is impossible. Thus all ideals within this factor are independent.

We must also exclude links to another isogeny factor. Only the quartic-center fourfold can have an external non-CM partner; the other partially pure cases have dimension six. Its two A_1 ideals are interchanged by Galois. The remainder has dimension at most two and inherits E . By Lemma 3.7, a non-CM A_1 target there can only come from a non-CM elliptic curve repeated twice or from a Type II surface. In either case there is only one absolute A_1 target and it is Galois-stable. Linking it to one member of the interchanged pair would link it to the other as well, contradicting their independence.

It remains to treat SO_4 . Its column is $k^2 \otimes k^2$ for its two A_1 ideals. A nonzero semisimple traceless differential on k^2 has spectrum $\{a, -a\}$, $a \neq 0$. If both factor differentials were nonzero, the tensor spectrum $\{\pm a \pm b\}$ would have at least three values. Hence at most one factor differential is nonzero. The self-pair makes the column mixed, so at least one is nonzero. These are the two absolute ideals of this rational factor, since its action on its sole column, repeated twice, is faithful. If both ideals were separately Galois-stable, the one with zero differential would descend to a rational adjoint quotient on which the circle acts trivially, contrary to minimality. Hence Galois interchanges them. The same remainder argument as in the preceding paragraph excludes an external link to either one.

All graph links have been excluded. The Lie image is therefore the full product. A connected closed subgroup of the product of the connected classical groups with this full Lie algebra is the product itself. This proves the required independent derived image; no independence of central characters has been asserted. $\square$

3.6 The balanced determinant products

We now verify assumption (d) of Theorem 2.5. Choose an isogeny $f : A' = \prod_B B^{r_B} \rightarrow A$ as in Lemma 3.7. Transport E to $\text{End}^0(A')$ by conjugation with the rational inverse of f . Pulling a compatible Riemann form back by $f_* : H_1(A', \mathbb{Q}) \rightarrow H_1(A, \mathbb{Q})$ identifies the rational E -Hermitian spaces isometrically, so their signatures and hyperbolicity agree. Isogenies and factor projections are algebraic correspondences, and pullback and pushforward along an isogeny transfer the Hodge conjecture in both directions. We may therefore work with this product.

Take H_0 to be the sum of its CM isotypic factors. Once a decomposition into CM embedding lines is chosen, this choice and Lemmas 3.4, 3.7, and 3.8 give assumptions (a)–(c). In each application below the decomposition is chosen from the geometry of A , before any determinant or CM list is tested.

The determinant requirements in the two theorems below have three forms. Symplectic blocks contribute only algebraic pair tensors, and an orthogonal column is admissibly isomorphic to an E -eigenspace whose determinant is supplied by Lemma 3.3. For a linear block with quadratic center, the same lemma supplies the individual determinants when both signature entries are at most three; in the remaining odd-rank cases the factor exhausts A , and balance forces cancellation of opposite determinants. For higher centers, balance will force equal net exponents on the columns above one E -embedding. Their product is the full E -determinant, to which the supply lemma applies.

For any such fixed decomposition, consider any determinant list and one CM list satisfying (2.3). An orthogonal column has imbalance zero because of its self-pair. For each special linear representative W_j , let $n_j \in \mathbb{Z}$ be the number of its determinant entries minus the number of opposite entries. Equation (2.2) rewrites balance as

$$\sum_j n_j \epsilon_\gamma(W_j) + c_\gamma = 0 \quad \text{for every } \gamma \in \text{Gal}(k/\mathbb{Q}), \quad (3.4)$$

where c_γ is the sum of the imbalances of the actual CM lines in the fixed list. Opposite determinant pairs are algebraically supplied by the determinant of their pair tensor, as explained after Lemma 2.2. Thus only the net exponents require further supply. We write $d_j = \epsilon_1(W_j) = 2 \dim W_j^{1,0} - \dim W_j$, where 1 denotes the identity scalar conjugation and $W_j^{1,0} = (W_j)_\mathbb{C} \cap H^{1,0}$. When considering columns above one E -embedding, we identify its image with $E \subset k$; then $\text{Gal}(k/E)$ preserves that set of columns. More precisely, $\gamma \in \text{Gal}(k/E)$ sends the central embedding indexing W_j to another one, which we denote by γj . The image of its primitive idempotent may select a different matrix copy, but the admissible copy identifications preserve Hodge dimensions. Consequently

$$\epsilon_\gamma(W_j) = d_{\gamma j}.$$

The balance equations below therefore use permutations of this fixed imbalance vector, together with the conjugates of the same fixed CM list.

Proof of Theorem 1.2. In dimension at most five, Types I and II require only their algebraic pair tensors. A Type III factor has dimension four; its orthogonal column is admissibly isomorphic to an E -eigenspace of signature $(2, 2)$. Lemma 3.3 supplies the determinant of that eigenspace for every compatible rational form. The matrix-unit identification transports this supply to the chosen column's volume.

In Type IV the case where E does not embed in K is absent, since it requires a threefold with multiplicity two. In the split $f = 1$ case identify K with E . The mixed signatures in dimensions three and four have both entries at most three, so their determinants are supplied individually by Lemma 3.3. A five-dimensional factor exhausts A . Its single determinant imbalance is $2p - 5 \neq 0$. There is no CM remainder, so (3.4) gives zero net exponent. No determinant of signature $(1, 4)$ or $(4, 1)$ is required.

The only remaining possibility is $(f, a, g) = (2, 2, 4)$. Of the two columns W_1, W_2 above a chosen E -embedding, one is mixed and one pure. Their imbalance vector is therefore $(0, 2)$ or $(0, -2)$, up to order. A possible remainder has dimension one and inherits a unital E -action. Its first cohomology is rank one over E , so the circle acts through the E -scalar torus; it is an E -CM elliptic curve. Choose its two E -embedding lines as the fixed decomposition of this part of H_0 , before testing any list. Every scalar automorphism fixing E fixes these lines and their contribution c_γ .

Such an automorphism exchanges the two embeddings of K/E . Subtracting its equation (3.4) from the identity equation gives

$$(n_1 - n_2)(d_1 - d_2) = 0.$$

Since $d_1 \neq d_2$, $n_1 = n_2$. After supplying opposite pairs, the remaining line is a power of $D(W_1) \otimes D(W_2)$, or of its opposite. The direct-sum determinant maps of Section 2 identify the basic product $D(W_1) \otimes D(W_2)$, by nonzero algebraic operations, with the determinant of the full E -embedding space of B . Its signature is $(1, 3)$ or $(3, 1)$, so Lemma 3.3 supplies it.

Every determinant list satisfying balance is now CM-supplied. Theorem 2.5 gives the conclusion. $\square$

Proof of Theorem 1.1. The central isotypic projectors are selfadjoint for the compatible polarization. Indeed a polarization identifies each simple isogeny type with its own dual type, and there are no homomorphisms between distinct types. The isotypic summands are therefore orthogonal for e , and also for h_E because their projectors commute with E . Their signatures add to $(3, 3)$. We do not assert that a proper summand of the hyperbolic form is itself hyperbolic.

Types I and II again require only pairs. In Type III, a four-dimensional column is admissibly isomorphic to an E -eigenspace of signature $(2, 2)$, whose determinant is supplied for its arbitrary

restricted form. A six-dimensional Type III factor exhausts A , so the determinant of either E -eigenspace is supplied for the given hyperbolic form. Both statements follow from Lemma 3.3; the admissible matrix-unit identifications transport the supply to the chosen column's volume.

Suppose first that a Type IV factor has E not embedding into K . It is a quadratic-center threefold with multiplicity two and exhausts A . Its K -column is mixed of rank three, so its imbalance is nonzero. There is no CM remainder, and (3.4) forces its net determinant exponent to vanish.

Now consider the split $f = 1$ case and identify K with E . For multiplicity one in dimensions three, four, and five, orthogonal addition of signatures to $(3, 3)$ puts each signature entry at most three. The only repeated non-CM case is $g = 3, r_B = 2$. For it use the central $K = E$ action on B , of mixed signature $(1, 2)$ or $(2, 1)$, and any compatible polarization on B . The possibly noncentral given action on B^2 changes only the multiplicities of its two column types, as proved in Lemma 3.7. Thus Lemma 3.3 supplies the required determinants in all these ranks. If $g = 6$, B exhausts A , and the given hyperbolic determinant supplies both orientations.

For the quartic-center fourfold $(f, a, g) = (2, 2, 4)$, replace u by $-u$ if necessary and retain the convention $\sigma(u) = i\sqrt{d}$, so that its restricted Hermitian form has signature $(1, 3)$. This replacement negates h_E , preserving hyperbolicity and the total signature $(3, 3)$ while interchanging the two possible restricted signatures. The complementary two-dimensional isotypic sum T has signature $(2, 0)$. In $H_1(T, \mathbb{Q})$, of rank two over E , the complex structure is the same E -scalar on the whole space. Every E -line in an E -basis is therefore a rational Hodge substructure of the same rank-one CM type. The degree-one realization makes T isogenous to a square of an E -CM elliptic curve. Choose the resulting E -embedding lines as the fixed decomposition of its contribution to H_0 , before testing any list. Their imbalance contribution is fixed by automorphisms fixing E .

The two column imbalances are $(0, -2)$, up to order. Exchanging the two embeddings of K/E and subtracting (3.4) gives $n_1 = n_2$. The remaining product is a power of $D(W_1) \otimes D(W_2)$, or its opposite. Algebraic direct-sum determinant maps identify the basic product $D(W_1) \otimes D(W_2)$ with the full E -determinant of B , of signature $(1, 3)$, which is supplied by Lemma 3.3.

It remains to consider the six-dimensional higher centers. Here $B = A$, so $c_\gamma = 0$. For $(f, a) = (2, 3)$, the two holomorphic multiplicities above one E -embedding sum to three. The alternatives $(0, 3)$ and $(3, 0)$ make every column pure and the factor CM. Otherwise the vector is $(1, 2)$, up to order, with imbalance $(-1, 1)$. Equation (3.4) gives $n_1 = n_2$.

For $(f, a) = (3, 2)$, the three multiplicities belong to $\{0, 1, 2\}$ and sum to three. The vector $(1, 1, 1)$ is excluded by the rank-two argument in Lemma 3.7. The remaining vector is a permutation of $(0, 1, 2)$, with imbalance a permutation of $(-2, 0, 2)$. The Galois action over E on the three embeddings of K/E is transitive and therefore contains a three-cycle. Order the columns so that the imbalance is $(-2, 0, 2)$. Applying (3.4) to the three cyclic conjugates gives, up to reversing their order,

$$-n_1 + n_3 = 0, \quad -n_2 + n_1 = 0, \quad -n_3 + n_2 = 0.$$

Thus $n_1 = n_2 = n_3$.

In either case, after opposite pairs have been supplied, only a power of $\bigotimes_{j=1}^f D(W_j)$, or its opposite, remains. The center projectors are admissible algebraic degree-one maps. Alternation and those projectors identify the basic product $\bigotimes_{j=1}^f D(W_j)$ nontrivially with the determinant of the full E -embedding space of A . The given hyperbolic form supplies this determinant by Lemma 3.3.

These cases exhaust Lemma 3.7. The equalities above concern arbitrary integer net exponents, so they verify determinant supply in every tensor degree. Products, opposite pairs, and the direct-sum maps preserve CM supply. Assumption (d) of Theorem 2.5 is therefore satisfied, and that theorem proves the result on every self-power. $\square$

4 The geometric seed for the Weil-plane theorem

We begin the proof of Theorem 3.1, the algebraicity input used in Lemma 3.3. We follow Markman’s secant-sheaf and spin construction [25, 26]. The argument will retain the actual polarized Hermitian space and the full derived obstruction group throughout descent and deformation.

We first construct a coherent perfect sheaf on the product of a genus-three Jacobian and its dual, with a coherent action of a finite group of translations combined with line tensors. An exact Hochschild–Atiyah trace identity and a dimension calculation then show that the semiregularity map is injective on the *entire* invariant derived Ext^2 group. A separate finite Clifford calculation identifies the normalized rational Betti character of the same sheaf: it has a nonzero Weil component, and all its components remain Hodge on the full compatible marked domain.

We next descend the sheaf to a twisted sheaf on the finite quotient. A projective-bundle construction turns it into an ordinary coherent sheaf while retaining the full obstruction group. We prove the restricted smooth-projective-germ semiregularity theorem needed to deform that ordinary sheaf, and recover the normalized character by algebraic projective-bundle operations. Finally, proper Hilbert parameter spaces and the Baire theorem extend the local algebraicity to every marked period; rational isogenies pass from the seed lattice to every hyperbolic quadratic lattice. The proof of Theorem 3.1 is completed in Subsection 8.4.

The geometric input is a pair of finite orbits of Abel curves on a genus-three Jacobian. We choose the orbits so that their ideals have no sections after any degree-zero twist of twice the principal polarization. This vanishing will control the fiber cohomology of a fixed Fourier–Mukai transform and allow its shifted derived dual to be an ordinary coherent sheaf. We construct that sheaf, its coherent line-twisted translation action, and the finite quotient used in the deformation argument. The source ideal and transform are those of Markman [25, (6.2.1), (6.2.3), and (9.2.1)]; the compact-type degeneration below supplies the required orbits.

All varieties in this section are over $\mathbb{C}$. If V is a group variety, $t_v(z) = z + v$, and $t_v D$ denotes the image translate of a divisor D . Thus $t_v^* \mathcal{O}(D) = \mathcal{O}(t_{-v} D)$.

4.1 The Jacobian and the required Abel orbits

Let C be a smooth nonhyperelliptic curve of genus three and choose $p \in C$. Give

$$X = \mathrm{Pic}^2(C)$$

the group law with origin $\mathcal{O}_C(2p)$; equivalently, tensoring with $\mathcal{O}_C(2p)$ identifies $\mathrm{Pic}^0(C)$ with X . Put

$$\Theta = W_2(C), \quad H = \mathcal{O}_X(\Theta), \quad \widehat{X} = \mathrm{Pic}^0(X).$$

For $v \in X$ define

$$\lambda(v) = t_v^* H \otimes H^{-1}, \quad a(v) = \mathcal{O}_X(t_v \Theta - \Theta) = t_{-v}^* H \otimes H^{-1} = \lambda(-v). \quad (4.1)$$

The theorem of the square makes these homomorphisms, and the canonical principal polarization makes both isomorphisms. We use $a(v)$ both for a point of $\widehat{X}$ and for its represented line bundle. More generally, for a line bundle D on an abelian variety or scheme, $\phi_D(v) = t_v^* D \otimes D^{-1}$.

In cohomology, Θ will also denote the normalized integral Betti class $c_{1,B}(H)$, so that $\int_X \Theta^3/3! = 1$. It does not denote a raw first-jet or Hodge class. Here is the sign convention that will be used for this polarization. Set

$$L = H^1(X, \mathbb{Q}), \quad U = L^\vee = H_1(X, \mathbb{Q}) = H^1(\widehat{X}, \mathbb{Q}), \quad \vartheta : U \longrightarrow L, \quad y \longmapsto \iota_y \Theta. \quad (4.2)$$

The last identification is the one defined by the normalized Poincaré bundle $\mathcal{P}$ on $X \times \widehat{X}$, whose restriction to $X \times \{N\}$ is N . If e_i and f_i are dual bases of L and U , then $c_{1,B}(\mathcal{P}) = \sum_i e_i \wedge f_i$. The normalized biextension identity is

$$(1 \times \lambda)^* \mathcal{P} \simeq m_X^* H \otimes \text{pr}_1^* H^{-1} \otimes \text{pr}_2^* H^{-1} \otimes \underline{H}_0, \quad (4.3)$$

where m_X is addition and the constant line $\underline{H}_0$ rigidifies the two zero axes. Comparing the mixed Künneth terms in (4.3) gives

$$\lambda^* = \vartheta, \quad a^* = -\vartheta, \quad \lambda_* = -\vartheta, \quad a_* = \vartheta. \quad (4.4)$$

Indeed, if $\Theta = \frac{1}{2} \sum_{i,j} c_{ij} e_i \wedge e_j$, its mixed term after addition is $\sum_{i,j} c_{ij} e_i^{(1)} \wedge e_j^{(2)}$. Consequently $\lambda^* f_i = \sum_j c_{ij} e_j = \iota_{f_i} \Theta$. The two homology maps are the ordinary transposes of the cohomology maps, and $\vartheta^t = -\vartheta$, proving the last two signs.

Define the two Abel curves

$$C_p = \{\mathcal{O}_C(p+q) : q \in C\}, \quad \Sigma_p = \{\omega_C(-p-q) : q \in C\}. \quad (4.5)$$

They have the normalized Betti class and degree

$$[C_p] = [\Sigma_p] = \frac{\Theta^2}{2}, \quad \Theta \cdot C_p = \Theta \cdot \Sigma_p = 3. \quad (4.6)$$

For clarity, these normalizations follow directly from the Jacobian lattice. In the analytic description $J(C) = H^0(C, \omega_C)^* / H_1(C, \mathbb{Z})$, the Abel map induces the identity on the displayed lattice. Take a symplectic basis e_i, f_i of $H^1(C, \mathbb{Z})$, identified with $H^1(X, \mathbb{Z})$ by Abel pullback. The canonical principal class is $\sum_{i=1}^3 e_i \wedge f_i$. Pairing $\Theta^2/2$ with each basis element of $\Lambda^2 H^1(X, \mathbb{Z})$ gives the same value as integration of its Abel pullback on C . Poincaré duality proves the first formula for C_p . The negative Abel map has the same action on degree-two cohomology, giving the formula for Σ_p . The degree in (4.6) is then $\int_X \Theta^3/2 = 3$. This lattice description also shows that restriction

$$\text{Pic}^0(X) \longrightarrow \text{Pic}^0(C_p) \quad (4.7)$$

is an isomorphism: its integral lattice map and its map on $H^{0,1}$ are the Abel pullback isomorphisms. For a translate of Σ_p the analogous map is the negative of this isomorphism.

We will repeatedly use curve Riemann–Roch and duality in the following form. For a proper Gorenstein curve D and an invertible sheaf N ,

$$\chi(D, N) = \deg N + \chi(D, \mathcal{O}_D), \quad H^1(D, N)^\vee = H^0(D, \omega_D \otimes N^{-1}). \quad (4.8)$$

The numerical Riemann–Roch formula is the rank-one case of [39, Lemma 53.5.2, Tag 0BS6]; the canonical duality isomorphism is stated in [39, Remark 53.4.3, Tag 0BS4]. We will apply them to smooth curves.

The temporary restriction $d \geq 3$ loses no imaginary quadratic field: the final proof of Theorem 3.1 replaces a purely imaginary generator u_0 by $u = 2u_0$, scaling the associated Hermitian form without changing its isotropic subspaces.

Proposition 4.1 (Simultaneous equivariant seeds). *For every integer $d \geq 3$, put $n = d + 1$. There exist a smooth nonhyperelliptic genus-three curve C , a point $p \in C$, and cyclic subgroups $G_1, G_2 \subset X = \text{Pic}^2(C)$ of order n , with $G_1 \cap G_2 = 0$, as follows. There are cosets $\{s_1, \dots, s_n\} = s_0 + G_2$ and $\{t_1, \dots, t_n\} = t_0 + G_1$ such that*

$$C_i = t_{s_i} C_p, \quad \Sigma_j = t_{t_j} \Sigma_p, \quad A = \prod_{i=1}^n C_i, \quad \Sigma = \prod_{j=1}^n \Sigma_j \quad (4.9)$$

are disjoint within each of the two indicated unions, and

$$H^0(X, I_A \otimes H^{\otimes 2} \otimes N) = H^0(X, I_\Sigma \otimes H^{\otimes 2} \otimes N) = 0 \quad \text{for every } N \in \hat{X}. \quad (4.10)$$

The uniform vanishing in the proposition has a specific role in the sheaf construction. For every $x \in X$, the product ideal $I_A I_{t-x\Sigma}$ is contained in I_A , so every degree-zero twist of this ideal by $H^{\otimes 2}$ has zero H^0 . Its quotient in $\mathcal{O}_X$ has one-dimensional support; the ideal sequence and ample-line vanishing will then give zero H^3 . For general x , the two orbit unions are disjoint. The restriction of $H^{\otimes 2}$ to each component has degree six, greater than $2g(C) - 2 = 4$, which also gives zero H^2 for these parameters.

We will prove that these cohomology groups are the derived fibers of the fixed transform. The first two vanishings give a local model $K^1 \rightarrow K^2$, and the generic H^2 -vanishing makes its dual differential generically injective. Integrality of the parameter space then makes that dual differential injective everywhere. Its cokernel is the coherent sheaf we seek. We first construct the orbits required by the proposition.

4.2 Constructing the seed by a compact-type degeneration

On a product of three elliptic curves, the central support of the limiting Abel curve will be the union of the three coordinate axes. The next two lemmas show that four suitably translated copies of this star impose more vanishing conditions than a nonzero section of a product of degree-two lines can satisfy. We will carry this vanishing to a smooth Jacobian through a polarized family and the proper relative Picard scheme.

4.2.1 Vanishing on the elliptic star

Lemma 4.2 (Singular marked points of a trilinear form). *Let V_1, V_2, V_3 be two-dimensional vector spaces and let $0 \neq T \in V_1 \otimes V_2 \otimes V_3$. Suppose a finite indexed collection*

$$(x_h, y_h, z_h) \in \mathbb{P}(V_1^*) \times \mathbb{P}(V_2^*) \times \mathbb{P}(V_3^*)$$

has at most two indices in every fiber of each coordinate projection. At most three indices satisfy all three equations

$$T(-, y_h, z_h) = T(x_h, -, z_h) = T(x_h, y_h, -) = 0. \quad (4.11)$$

Indices, rather than distinct triples, are counted.

Proof. Choose bases and write $T(x, y, z) = y^t M(x) z$, where $M(x) = x_0 A + x_1 B$ is a pencil of two-by-two matrices. Put $D(x) = \det M(x)$. Equations (4.11) imply

$$M(x)z = 0, \quad y^t M(x) = 0, \quad y^t A z = y^t B z = 0. \quad (4.12)$$

Rank two for $M(x)$ is impossible. If its rank is one, its adjugate is a nonzero multiple of zy^t , so the determinant derivative formula and (4.12) give $\partial_{x_0} D(x) = \partial_{x_1} D(x) = 0$. If $M(x) = 0$, its adjugate is zero and the same conclusion holds. For a nonzero binary quadratic D , the gradient is a nonzero linear map in characteristic zero and has at most one projective zero. All qualifying indices then have the same x , and there are at most two.

It remains to consider $D \equiv 0$. Every nonzero matrix in the pencil has rank one. A one-dimensional pencil makes T a product of three linear forms. For a two-dimensional pencil write $A = uv^t$ and $B = u'v'^t$. The equality

$$0 = \det(A + B) = \det[u, u'] \det[v, v']$$

shows that the pencil has a common left or right factor. After permuting the variables, both cases have the form $T = l(x)q(y, z)$. If the bilinear form q has rank two, the last two partial contractions force $l(x) = 0$, again allowing at most two indices. If q has rank one, write $T = l(x)m(y)k(z)$. Equations (4.11) then hold precisely when at least two of $l(x), m(y), k(z)$ vanish. Each of the three distinguished projective coordinate values occurs at most twice. If N indices qualify, counting these incidences gives $2N \leq 6$, hence $N \leq 3$. $\square$

Lemma 4.3 (The elliptic star). *Let $J_0 = E_1 \times E_2 \times E_3$ be a product of elliptic curves with origins, and let Z_0 be the reduced union of the three coordinate elliptic axes through zero. Let $K \subset J_0$ be a finite subgroup of order at least four whose projection to every E_i is injective. The translates $Z(g) = g + Z_0$, $g \in K$, are pairwise disjoint. For line bundles D_i of degree two on E_i ,*

$$H^0\left(J_0, I_{\bigcup_{g \in K} Z(g)} \otimes (D_1 \boxtimes D_2 \boxtimes D_3)\right) = 0. \quad (4.13)$$

The same vanishing holds for any closed subscheme with that support.

Proof. If axis i at g met axis j at $h \neq g$, the coordinates of g and h outside $\{i, j\}$ would agree. There is at least one such coordinate, contradicting injectivity there. This proves disjointness.

Put $V_i = H^0(E_i, D_i)$. Formula (4.8) gives $\dim V_i = 2$ and $h^0(D_i(-q)) = 1$ for every $q \in E_i$. Evaluation is therefore basepoint-free and defines a degree-two map $\varphi_i : E_i \rightarrow \mathbb{P}(V_i^*)$, each of whose fibers has at most two distinct points. The marks

$$(\varphi_1(g_1), \varphi_2(g_2), \varphi_3(g_3)), \quad g = (g_1, g_2, g_3) \in K,$$

have at most two indices in each coordinate fiber, since $K \rightarrow E_i$ is injective. The product formula for sections identifies a section of $D_1 \boxtimes D_2 \boxtimes D_3$ with a tensor in $V_1 \otimes V_2 \otimes V_3$. Its restriction to axis i at g is contraction by evaluation at the other two coordinates. Vanishing on $Z(g)$ is exactly (4.11). Lemma 4.2 forbids this for four or more indices for a nonzero section. Finally, the ideal of a closed subscheme with this support is contained in the radical ideal of the reduced union, which proves the last assertion. $\square$

4.2.2 Finite cohomology models and base change

We need to pass from vanishing on the limiting product to vanishing on nearby Jacobians, uniformly over their Picard twists. Later we will identify every derived fiber of the fixed Fourier–Mukai kernel. The following finite-model statement supplies both semicontinuity for the first step and arbitrary base change for the second. It also defines the determinant line that identifies the limiting polarization.

Lemma 4.4 (Finite models and arbitrary base change). *Let B be a regular Noetherian scheme of finite dimension, let $f : V \rightarrow B$ be projective, and let $\mathcal{F}$ be a coherent sheaf flat over B . Then $Rf_*\mathcal{F}$ is locally represented by a bounded complex $K^\bullet$ of finite-rank locally free sheaves. For every morphism $h : T \rightarrow B$ there is a derived base-change isomorphism*

$$Lh^*Rf_*\mathcal{F} \simeq Rf_{T,*}(\mathcal{F}|_{V \times_B T}), \quad (4.14)$$

where the restriction on the right is ordinary, and the left side is represented locally by $K^\bullet \otimes_{\mathcal{O}_T}$. The functions $b \mapsto \dim_{\kappa(b)} H^i(V_b, \mathcal{F}_b)$ are upper semicontinuous. If all derived fibers have cohomology in degrees $[c, e]$, the local model can be chosen in precisely those degrees.

Proof. Over an affine $U = \operatorname{Spec} R \subset B$, choose a finite affine cover V_i of V_U . All finite intersections are affine because V_U is separated. Its Čech complex has terms

$$C_R^q = \prod_{i_0 < \dots < i_q} \Gamma(V_{i_0} \cap \dots \cap V_{i_q}, \mathcal{F}). \quad (4.15)$$

It is bounded. Each term is R -flat: for a module on an affine scheme, flatness over R is checked at the total-space stalks, where it is the assumed relative flatness. For any R -algebra R' , $C_R^\bullet \otimes_R R'$ is exactly the affine Čech complex of the ordinary pullback of $\mathcal{F}$ to $V_U \times_R R'$. A bounded complex of flat modules is K -flat, so this tensor product also computes the derived tensor product. Affine Čech cohomology therefore proves (4.14) for affine base changes, and those isomorphisms glue for an arbitrary T .

The cohomology modules of $C_R^\bullet$ are finite over R , by proper coherence for a projective scheme over a Noetherian affine base [39, Lemma 30.16.1(3), Tag 0B5T]. A finite module over the regular Noetherian finite-dimensional ring R has a finite resolution by finite projective modules [39, Section 10.110, Tag 065U]. The truncation triangles, applied to the finitely many cohomology modules of $C_R^\bullet$, consequently make its derived object perfect. After localizing, it has the stated finite free model.

For such a model the dimension of the i -th fiber cohomology is

$$\operatorname{rank} K^i - \operatorname{rank}(d^i \otimes \kappa(b)) - \operatorname{rank}(d^{i-1} \otimes \kappa(b)). \quad (4.16)$$

Ranks of matrices are lower semicontinuous by their minors; (4.16) proves upper semicontinuity. Finally, at a term below c , the first remaining differential is injective on the fiber. Invert a maximal minor, split off this term, and replace the next term by its locally free cokernel. At a term above e , do the dual operation using a fiberwise surjective last differential. Repeating at a chosen point and shrinking its neighborhood gives a model in $[c, e]$. $\square$

We also use the perfect determinant in a concrete form. For a finite locally free complex put $\det K^\bullet = \bigotimes_i (\det K^i)^{(-1)^i}$. The determinant of an acyclic complex is canonically trivialized by the exact sequences of its cycles. Applying these trivializations to cones makes the determinants independent of the finite model, so they glue and commute with base change. In particular, a model $K^0 \rightarrow K^1$ of equal ranks has a regular determinant section of $(\det K^\bullet)^{-1}$; its zero set is where the fiber complex is not acyclic.

4.2.3 From the polarized star to a smooth seed

Proof of Proposition 4.1. The polarized elliptic limit. We first realize the elliptic product with its product principal polarization as the central Jacobian of a smoothing. This identifies the central twice-theta line with the degree-two product lines tested in Lemma 4.3.

Start with the stable compact-type curve

$$C_0 = R \cup E_1 \cup E_2 \cup E_3, \quad R \simeq \mathbb{P}^1, \quad (4.17)$$

where the elliptic tails E_i meet R at three distinct nodes e_i and have no other intersections. We choose a projective one-parameter smoothing $\mathcal{C} \rightarrow S$ over a smooth pointed complex algebraic curve $(S, 0)$ with regular total surface and with smooth nonhyperelliptic noncentral fibers.

Here is the existence check, for the unmarked stable curve. The local presentation of Ω_{C_0} at a node has length one, its local $\mathcal{E}xt^1$ is supported at the nodes, and the local-to-global sequence gives $\operatorname{Ext}^2(\Omega_{C_0}, \mathcal{O}_{C_0}) = 0$. The global tangent contribution is the sum of the H^1 of the tangent lines on the normalized components, with their node points removed. It is zero for $T_R(-e_1 - e_2 - e_3) = \mathcal{O}_{\mathbb{P}^1}(-1)$,

and has dimension one for each $T_{E_i}(-e_i) = \mathcal{O}_{E_i}(-e_i)$. The three node-smoothing parameters add three more dimensions. Thus the unmarked versal germ is smooth of dimension six. The algebraic input is the smooth Deligne–Mumford stable-curve stack and density of its smooth-curve locus [39, Theorem 109.25.3, Tag 0E99]; it gives an algebraic versal chart here. Smooth hyperelliptic genus-three curves have a five-dimensional locus, as seen from eight branch points on $\mathbb{P}^1$ modulo PGL_2 . Its closure is proper in that chart. A general algebraic curve through the chosen point is not contained in this closure and has nonzero first derivative in each node parameter. The local total-space equation at each node is then $uv = t$ times a unit, so the total surface is regular. Shrinking S gives the asserted noncentral fibers. Choose a smooth point on R only now. Smoothness near it lifts it, after an étale localization of S , to a section σ . This point was not included in the six-dimensional unmarked germ.

Let $\mathcal{J}$ be the componentwise degree-zero relative Picard space of $\mathcal{C}/S$. The exact Picard input used here is [13, Section 2, p. 136]: for a stable family this open group space is separated, smooth, and of finite type, and its fibers are semiabelian. The normalization sequence for a connected nodal curve with dual graph Δ gives

$$1 \longrightarrow \mathbb{G}_m^{b_1(\Delta)} \longrightarrow \mathrm{Pic}^0(C_0) \longrightarrow \prod_v \mathrm{Pic}^0(C_v) \longrightarrow 1. \quad (4.18)$$

Indeed, its kernel is the set of gluing scalars at the nodes modulo rescaling on each component; one scalar remains per graph cycle. Our graph is a tree and $\mathrm{Pic}^0(R) = 0$, so $\mathcal{J}_0 = \prod_i \mathrm{Pic}^0(E_i)$. All noncentral fibers are smooth Jacobians. Thus every geometric fiber is proper and connected. Deligne’s [13, Section 1.1, pp. 131–132] states both that a separated flat finite-presentation space with proper connected geometric fibers is proper, and that a generic polarization on an abelian space over a normal integral base extends uniquely and makes that space projective. Consequently $\mathcal{J}$ is an abelian scheme and the generic canonical polarization extends to $\lambda_{\mathcal{J}} : \mathcal{J} \rightarrow \hat{\mathcal{J}}$. Its degree is locally constant and is one generically, so it is principal. We use only the componentwise degree-zero Picard space; no separatedness of the unrestricted relative Picard functor is asserted.

The decomposition of the central Jacobian as an abelian variety does not yet identify its polarization. To obtain the product polarization needed for the vanishing lemma, we compute a theta line by a determinant of cohomology. After an étale localization, the relative Picard space has a universal line bundle; this is the precise universal-line input from [11, Section 2.1, last paragraph]. Restrict it to $\mathcal{C} \times_S \mathcal{J}$ and rigidify it along σ ; denote it by $\mathcal{U}$. With p_C the projection to $\mathcal{J}$, put

$$\mathcal{T} = (\det R p_{C,*}(\mathcal{U} \otimes \mathcal{O}_{\mathcal{C}}(2\sigma)))^{-1}. \quad (4.19)$$

The base $\mathcal{J}$ is regular, and this is a flat projective curve family with a line bundle. Lemma 4.4 and the determinant construction following it define (4.19) and its base changes. Every fiber has Euler characteristic zero and cohomology in degrees zero and one; locally it has a two-term equal-rank model, with determinant section. On a smooth nonhyperelliptic fiber the section vanishes exactly on the effective degree-two locus W_2 . Its generic multiplicity there is one: at an effective degree-two line the groups H^0 and H^1 have dimension one, and the first-order differential is cup product

$$H^1(\mathcal{O}_C) \longrightarrow \mathrm{Hom}(H^0(N), H^1(N)).$$

Its dual is multiplication $H^0(N) \otimes H^0(\omega_C N^{-1}) \rightarrow H^0(\omega_C)$, which is nonzero on nonzero sections. Splitting the invertible blocks of a local two-term model shows that this nonzero first-order differential is the linear term of its remaining one-by-one determinant. Thus the determinant section has a simple zero there. It follows that the homomorphism $\phi_{\mathcal{T}}$ associated to $\mathcal{T}$ is the generic canonical

polarization. It equals $\lambda_{\mathcal{J}}$ over S , since the two morphisms agree on the schematically dense generic fiber and the target is separated.

On C_0 , the rigidification makes $\mathcal{U}$ trivial on $R \times \mathcal{J}_0$; its restrictions to the elliptic components are the Poincaré lines $\mathcal{P}_i$, normalized at e_i . The normalization exact sequence for $\mathcal{U} \otimes \mathcal{O}(2\sigma)$ factors (4.19), up to constant lines from R and the nodes, as

$$\mathcal{T}_0 \simeq \bigotimes_{i=1}^3 \left(\det R \operatorname{pr}_{\widehat{E}_i, *}(\mathcal{P}_i) \right)^{-1}. \quad (4.20)$$

For an elliptic curve the i -th factor is $\mathcal{O}_{\widehat{E}_i}(0)$. Its fiber cohomology vanishes off the origin. At the origin the one-dimensional H^0 and H^1 have the nonzero first-order cup-product differential $H^0(\mathcal{O}_{E_i}) \otimes H^1(\mathcal{O}_{E_i}) \rightarrow H^1(\mathcal{O}_{E_i})$; the same determinant argument gives a simple zero. Equation (4.20) therefore proves

$$(\mathcal{J}_0, \lambda_{\mathcal{J}, 0}) = \prod_{i=1}^3 (\operatorname{Pic}^0(E_i), \lambda_i) \quad (4.21)$$

as principally polarized abelian varieties. In particular, the polarization on this product is the product polarization.

Let $\mathcal{P}_{\mathcal{J}}$ be the normalized relative Poincaré line and set

$$\mathcal{D} = (1, \lambda_{\mathcal{J}})^* \mathcal{P}_{\mathcal{J}} \quad \text{on } \mathcal{J}. \quad (4.22)$$

The biextension identity gives $\phi_{\mathcal{D}} = \lambda_{\mathcal{J}} + \lambda_{\mathcal{J}}^{\vee} = 2\lambda_{\mathcal{J}}$. Concretely, translating the graph in (4.22) by x produces the two variable line factors $\mathcal{P}_{\mathcal{J}}(-, \lambda_{\mathcal{J}}x)$ and $\mathcal{P}_{\mathcal{J}}(x, \lambda_{\mathcal{J}}(-))$; the remaining factor is a constant fiber. These represent $\lambda_{\mathcal{J}}x$ and $\lambda_{\mathcal{J}}^{\vee}x$. On a smooth fiber it follows that $\mathcal{D}_s \otimes \mathcal{O}(-2\Theta_s) \in \operatorname{Pic}^0$: their associated homomorphisms, equivalently their integral Riemann forms, agree. Centrally, (4.21) and the product Poincaré bundle give $\mathcal{D}_0 = D_1 \boxtimes D_2 \boxtimes D_3$, up to a constant line, with $\deg D_i = 2$. Every line in $\operatorname{Pic}^0(\mathcal{J}_0)$ is the external product of three degree-zero lines.

The flat Abel limit and the torsion orbits. We next construct a flat family of Abel images whose central support is the elliptic star. Support is enough for the vanishing lemma, since the ideal of any thickening of that star is contained in its reduced ideal.

The pointed Abel map on the smooth fibers, $q \mapsto \mathcal{O}(q - \sigma)$, extends to a morphism $\alpha : \mathcal{C} \rightarrow \mathcal{J}$. We give the extension argument to record the support of its limit. The map is a rational S -map, defined on the smooth-fiber open. Elimination of indeterminacy for a regular surface and a proper target [39, Lemma 54.4.3, Tag 0C5H] gives a sequence of blowups at closed points of the central fiber on which it becomes a morphism. Each reduced fiber of this sequence over an indeterminacy point is a connected tree of rational curves. Every morphism $\mathbb{P}^1 \rightarrow \mathcal{J}_0$ is constant: pullback of the translation-invariant one-forms is zero, and those forms span every cotangent space, so the differential is zero in characteristic zero. The resolved map is therefore constant on each such tree. Its schematic graph image in $\mathcal{C} \times_S \mathcal{J}$ is integral, proper and birational over $\mathcal{C}$, with one underlying point in each fiber. A proper morphism with finite fibers is finite [39, Lemma 37.44.1, Tag 02LS]. A finite integral birational morphism to the normal surface $\mathcal{C}$ is an isomorphism, by integral closedness in its function field. This descends the resolved map to α . The relative first Abel morphism and its tail twists are also given by [11, Section 2.2, (7), (9), and Proposition 2.4] for a regular smoothing of a stable compact-type curve; the principal component in the present star is R .

We compute the specialization directly. For a smooth $q_0 \in E_i$, lift it étale-locally to a section q . The divisor E_i is Cartier on the regular total surface and

$$\mathcal{O}_{\mathcal{C}}(E_i)|_{E_i} = \mathcal{O}_{E_i}(-e_i), \quad \mathcal{O}_{\mathcal{C}}(E_i)|_R = \mathcal{O}_R(e_i), \quad \mathcal{O}_{\mathcal{C}}(E_i)|_{E_j} = \mathcal{O}_{E_j} \quad (j \neq i). \quad (4.23)$$

Thus $\mathcal{O}_{\mathcal{C}}(q - \sigma + E_i)$ has componentwise degree zero and agrees generically with the Abel point. Separatedness of $\mathcal{J}$ identifies its special value with $\alpha(q)_0$. The restriction on E_i is $\mathcal{O}_{E_i}(q_0 - e_i)$, all other elliptic restrictions are trivial, and $\mathcal{O}_R(e_i - \sigma_0)$ is trivial. There is no gluing torus in (4.18). A section through R maps to zero, and continuity includes the nodes. Identifying $\text{Pic}^0(E_i)$ with E_i using e_i , we get

$$\alpha_0(R) = 0, \quad \alpha_0(q \in E_i) = (0, \dots, \mathcal{O}_{E_i}(q - e_i), \dots, 0), \quad \alpha_0(|C_0|) = |Z_0|. \quad (4.24)$$

Here Z_0 is exactly the three-axis star of Lemma 4.3.

Let $\mathcal{A} \subset \mathcal{J}$ be the scheme-theoretic closure of the generic Abel image. Its ideal is the contraction of the generic ideal. At a stalk over any base DVR its quotient embeds in its localization obtained by inverting a uniformizer, and is therefore torsion-free over that DVR, hence flat. The ideal is coherent by Noetherianity. The morphism α is proper, and $\mathcal{C}_\eta$ is dense in $\mathcal{C}$. Its closed topological image is therefore the closure of its generic image, so (4.24) gives

$$|\mathcal{A}_0| = |Z_0|. \quad (4.25)$$

No reducedness of $\mathcal{A}_0$ is claimed. On the punctured base the relative Abel map is a closed immersion. Indeed, it is proper with finite fibers, since every smooth-fiber Abel map is an embedding, so it is finite by the cited finite-fiber criterion. Finite pushforward commutes with base change by its affine description, and the fiber maps $\mathcal{O}_{\mathcal{J}_s} \rightarrow \alpha_{s,*}\mathcal{O}_{\mathcal{C}_s}$ are surjective. Nakayama's lemma makes the total map surjective. Its image is flat there and thus has a saturated ideal, so it agrees with the restriction of the generic closure. Hence the noncentral $\mathcal{A}_s$ are the actual Abel curves schemewise.

Put $H_n = (\mathbb{Z}/n)^2$. Choose isomorphisms $H_n \simeq E_i[n]$ and embed H_n diagonally in $\mathcal{J}_0[n] = \prod_i E_i[n]$. Denote its two cyclic direct factors by G_1, G_2 . Multiplication by n on the abelian scheme is proper, has invertible differential in characteristic zero and finite geometric fibers, so its kernel is finite étale. After an étale localization, the chosen torsion points extend to sections with their group relations. They remain distinct, since equality of two sections of a finite étale scheme is open and closed. Thus the two cyclic factors retain order n and trivial intersection. The entire central H_n projects injectively into each E_i .

For one cyclic factor, the translates $t_g\mathcal{A}$ are flat over S , and their central supports are the disjoint stars of Lemma 4.3. A pairwise intersection is closed in the proper scheme $\mathcal{J}/S$, so its image in S is closed and misses 0. Removing finitely many such images makes all these translates disjoint. Their disjoint union $\mathcal{Z}$ is then a closed flat subscheme of $\mathcal{J}$, with actual Abel orbits on noncentral fibers and

$$|Z_0| = \left| \bigcup_g Z(g) \right|. \quad (4.26)$$

Apply the same argument to the negative closure $-\mathcal{A}$ for the other factor; its central support is still Z_0 . This reasoning uses the closure and its support, and does not assume that scheme-theoretic image or union formation commutes with the closed-fiber base change.

Vanishing for every Picard twist. We now have flat orbit closures and a line $\mathcal{D}$ representing twice the principal polarization on every fiber. It remains to make the central vanishing uniform on a neighborhood of the smoothing parameter. The ideal sequence for $\mathcal{Z}$ is exact after every base change to a fiber: both $\mathcal{O}_{\mathcal{J}}$ and $\mathcal{O}_{\mathcal{Z}}$ are S -flat. It also makes $I_{\mathcal{Z}}$ S -flat. On $\mathcal{J} \times_S \hat{\mathcal{J}}$ the coherent sheaf

$$\text{pr}_{\mathcal{J}}^*(I_{\mathcal{Z}} \otimes \mathcal{D}) \otimes \mathcal{P}_{\mathcal{J}} \quad (4.27)$$

is flat over $\hat{\mathcal{J}}$: locally it is the base change of an S -flat module, followed by tensoring with a line. The projection to the regular scheme $\hat{\mathcal{J}}$ is projective. Lemma 4.4 makes the locus

$$\mathcal{V} = \{N \in \hat{\mathcal{J}}_s \text{ for some } s : H^0(\mathcal{J}_s, I_{\mathcal{Z}_s} \otimes \mathcal{D}_s \otimes N) \neq 0\} \quad (4.28)$$

closed. On the central fiber the exact ideal sequence embeds I_{Z_0} in $\mathcal{O}_{\mathcal{J}_0}$, and (4.26) gives $I_{Z_0} \subset I_{\bigcup_g Z(g)}$. Lemma 4.3, applied to every external degree-two triple, therefore makes $\mathcal{V} \cap \widehat{\mathcal{J}}_0$ empty. Properness of $\widehat{\mathcal{J}}/S$ makes the image of $\mathcal{V}$ closed. Removing it leaves a neighborhood of 0 on which the vanishing holds for every Picard twist on every fiber. Repeat for the negative orbit.

Choose a noncentral complex point of this neighborhood and put $C = \mathcal{C}_s$, $p = \sigma_s$. Under the group identification $\text{Pic}^0(C) \simeq X$, the positive Abel curve is C_p . The negative Abel point $\mathcal{O}_C(3p - q)$ differs from $\omega_C(-p - q)$ by the fixed degree-zero line $\mathcal{O}_C(4p) \otimes \omega_C^{-1}$; hence the negative curve is a translate of Σ_p . The difference between $\mathcal{D}_s$ and $H^{\otimes 2}$ is absorbed by the arbitrary Picard twist. The two factors thus give exactly (4.9)–(4.10). $\square$

Only the two within-part disjointness assertions are required; curves from the two different parts may meet. They never coincide. In fact, if $t_s C_p = t_t \Sigma_p$, then the fixed degree-two line $\omega_C(-2p - s + t)$ has an effective divisor containing every varying $q \in C$. It must have at least two independent sections, contrary to nonhyperellipticity. Hence every C_i and every $t_{-x} \Sigma_j$ have zero-dimensional intersection or are disjoint, for every $x \in X$.

4.3 The relative Fourier–Mukai kernel and its coherent dual

Fix a seed supplied by Proposition 4.1. We now realize the cohomological argument following that proposition: first identify the fixed kernel as a flat family of product ideals, then use its fiber amplitude to construct the coherent dual.

4.3.1 The fixed transform and its kernel

Put $M = X \times \widehat{X}$. A point (x, N) of this product will parameterize a translation and a Picard twist in one fixed kernel on the chosen Jacobian X . The flatness proved below concerns this family over M ; deformation of the sixfold itself is treated later. We use the derived category of bounded coherent complexes and the following fixed equivalence:

$$\begin{aligned} \Psi(F) &= R\text{pr}_{\widehat{X},*}(\text{pr}_X^* F \otimes \mathcal{P}^{-1})[3], \\ \mu : X \times X &\longrightarrow X \times X, \quad \mu(x, y) = (x + y, y), \\ \Phi &= (1 \times \Psi) \circ \mu^*, \quad \widetilde{\Phi} = \Phi \circ ((H \boxtimes H) \otimes -). \end{aligned} \tag{4.29}$$

Here $1 \times \Psi$ is the relative transform on the second factor. Mukai’s inversion theorem [29, Theorem 2.2] gives the equivalence for the normalized Poincaré kernel. Pulling that kernel by inversion on one factor and shifting by [3] are autoequivalences, so it also gives the stated equivalence for Ψ . The shear and tensor factors in (4.29) are equivalences as well. Thus (4.29) specifies the transform completely; no postcomposition shear is implicit.

Use the source and its transform

$$I = I_\Sigma \boxtimes I_A, \quad G = \widetilde{\Phi}(I)[-3]. \tag{4.30}$$

On $X \times M$ use coordinates (y, x, N) , and let

$$\pi(y, x, N) = (x, N), \quad \alpha_X(y, x, N) = y + x.$$

Define

$$\begin{aligned} \mathcal{I} &= \text{pr}_y^* I_A, \quad \mathcal{J} = \alpha_X^* I_\Sigma, \\ \mathcal{H} &= \text{pr}_y^* H \otimes \alpha_X^* H \otimes \text{pr}_{y,N}^* \mathcal{P}^{-1}, \quad \mathcal{F} = \mathcal{H} \otimes (\mathcal{I} \otimes^L \mathcal{J}). \end{aligned} \tag{4.31}$$

At first the last tensor product is derived. Expanding the shear in (4.29) changes $I_\Sigma H \boxtimes I_A H$ into $\alpha_X^*(I_\Sigma H) \otimes^L \text{pr}_y^*(I_A H)$; the second Fourier–Mukai kernel then contributes $\mathcal{P}^{-1}[3]$. The shift in (4.30) cancels this [3]. Consequently, as globally defined derived objects,

$$G = R\pi_* \mathcal{F}. \quad (4.32)$$

This is the fixed-kernel form of [25, (6.2.3) and (9.2.1)].

Lemma 4.5 (Local Tor and product ideals). *Let V be a smooth threefold and let $D_1, D_2 \subset V$ be smooth curves with finite, possibly empty, intersection. Then*

$$\text{Tor}_k^{\mathcal{O}_V}(I_{D_1}, I_{D_2}) = 0 \quad (k > 0), \quad I_{D_1} \otimes I_{D_2} \hookrightarrow \mathcal{O}_V \quad (4.33)$$

by multiplication, with image $I_{D_1} I_{D_2}$. The quotient from $\mathcal{O}_V / I_{D_1} I_{D_2}$ to the structure sheaf of the reduced union has zero-dimensional kernel. The same statements hold when each D_i is a disjoint union of smooth curves.

Proof. At an intersection point choose a regular sequence f_1, f_2 for D_1 . Its Koszul resolution restricted to the regular one-dimensional local ring of D_2 has injective left arrow: at least one $\bar{f}_i$ is nonzero because the intersection is zero-dimensional, and that local ring is a domain. Thus $\text{Tor}_k(\mathcal{O}_{D_1}, \mathcal{O}_{D_2}) = 0$ for $k \geq 2$. The two ideal sequences give

$$\text{Tor}_k(I_{D_1}, I_{D_2}) = \text{Tor}_{k+2}(\mathcal{O}_{D_1}, \mathcal{O}_{D_2}) = 0 \quad (k > 0)$$

and $\text{Tor}_1(\mathcal{O}_{D_1}, I_{D_2}) = 0$. Tensoring the first ideal sequence with I_{D_2} proves the injection and product image in (4.33). Away from the intersection the product ideal equals the reduced-union ideal, proving the assertion about its kernel. For disjoint unions, at any point at most one component from each part is present, so the same local calculation applies. This is the local fact behind [25, Appendix 11, pp. 91–93], in the form needed here. $\square$

Proposition 4.6 (The relative kernel and all its base changes). *For the seed of Proposition 4.1, the derived tensor in (4.31) is the ordinary coherent sheaf $\mathcal{I} \subset \mathcal{O}_{X \times M}$. Both this ideal and its quotient in $\mathcal{O}_{X \times M}$ are flat over M . Ideal multiplication and both ideal sequences commute with every base change $T \rightarrow M$. In particular, $\mathcal{F}$ is M -flat, and at $m = (x, N)$ it has the ordinary fiber*

$$\begin{aligned} \mathcal{F}_m &= I_{Z_x} \otimes \mathcal{M}_{x,N}, \\ I_{Z_x} &= I_A I_{\Sigma_x}, \quad \Sigma_x = \coprod_j t_{-x} \Sigma_j, \quad \mathcal{M}_{x,N} = H^{\otimes 2} \otimes a(-x) \otimes N^{-1}. \end{aligned} \quad (4.34)$$

Moreover $G = R\pi_* \mathcal{F}$ has a local finite locally free model, and for every $h : T \rightarrow M$,

$$Lh^* G \simeq R\pi_{T,*}(\mathcal{F}|_{X \times T}). \quad (4.35)$$

Proof. The subscheme $A \times M$ is M -flat, and the subscheme defined by $\mathcal{I}$ is isomorphic to $\Sigma \times M$ under $(y, x, N) \mapsto (y+x, x, N)$. Their ideal sequences make $\mathcal{I}$ and $\mathcal{J}$ M -flat. Locally along $A \times M$, its regular codimension-two embedding gives the ideal resolution

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}^{\oplus 2} \longrightarrow \mathcal{I} \longrightarrow 0, \quad (4.36)$$

whose restriction to each fiber is the corresponding curve-ideal resolution. Tensor (4.36) with $\mathcal{J}$. On every fiber the left arrow is injective by Lemma 4.5.

We use the following precise local flatness criterion. For a local map $R \rightarrow S'$ with S' Noetherian, a finite S' -module Q , an R -flat module P , and an R -linear map $u : Q \rightarrow P$, injectivity modulo the maximal ideal of R implies injectivity of u and R -flatness of its cokernel [39, Lemma 10.99.1, Tag 00MD]. Its mechanism is useful here: flatness compares the maximal-ideal filtrations and gives injectivity modulo every power; Krull intersection gives injectivity. Repeating modulo every ideal of R and using the Tor criterion gives flatness of the cokernel. Apply it at a total-space stalk to $\mathcal{J} \rightarrow \mathcal{J}^{\oplus 2}$, with $R = \mathcal{O}_{M,m}$. It proves that the tensor of (4.36) is exact on the left and has M -flat cokernel. Where $\mathcal{J} = \mathcal{O}$ the conclusion is immediate. Thus the derived tensor is the ordinary M -flat tensor.

Its multiplication map to $\mathcal{O}_{X \times M}$ is injective on every fiber by Lemma 4.5. The same criterion, now with the M -flat target $\mathcal{O}_{X \times M}$, gives an injection with M -flat quotient. Tensoring either exact sequence with any $\mathcal{O}_T$ stays exact because its quotient is M -flat. This proves multiplication and arbitrary base change before pushforward. The fiber formula follows from $t_x^* H = H \otimes a(-x)$ and $\mathcal{P}|_{X \times \{N\}} = N$. Finally M is smooth and regular and π is projective. Lemma 4.4, applied to the actual flat $\mathcal{F}$, gives the local finite model and (4.35). $\square$

The scheme Z_x may have embedded points at intersections. Lemma 4.5 says that its reduction is $A \cup \Sigma_x$ and that the difference has zero-dimensional support. In particular Z_x always has one-dimensional support.

4.3.2 The shifted dual is an ordinary sheaf

Proposition 4.7 (The actual coherent dual sheaf). *Let $D_M(-) = R\mathcal{H}om_M(-, \mathcal{O}_M)$. Every derived fiber of G has cohomology only in degrees one and two, and its degree-two cohomology vanishes on a nonempty open set. Consequently*

$$E = D_M(G)[-1] \quad (4.37)$$

is an actual coherent perfect sheaf on M , locally with a length-one locally free resolution, of rank

$$r = 8d > 0. \quad (4.38)$$

No reflexivity or codimension assertion is required for this claim.

Proof. For $m = (x, N)$, Proposition 4.6 gives

$$0 \longrightarrow \mathcal{F}_m \longrightarrow \mathcal{M}_{x,N} \longrightarrow \mathcal{O}_{Z_x} \otimes \mathcal{M}_{x,N} \longrightarrow 0. \quad (4.39)$$

The line $\mathcal{M}_{x,N}$ is ample: it is $H^{\otimes 2}$ tensored by a topologically trivial line. Kodaira vanishing applies on the smooth projective abelian threefold, whose canonical bundle is trivial, and gives $H^k(X, \mathcal{M}_{x,N}) = 0$ for $k > 0$ [22, Theorem 3, p. 1272]. Hirzebruch–Riemann–Roch for a line bundle on a smooth projective variety [20, Main Theorem (M), p. 111], with the trivial tangent bundle of X , gives

$$h^0(X, \mathcal{M}_{x,N}) = \chi(X, \mathcal{M}_{x,N}) = \int_X \frac{(2\Theta)^3}{3!} = 8. \quad (4.40)$$

Since $I_{Z_x} \subset I_A$, (4.10) gives $H^0(\mathcal{F}_m) = 0$. Since Z_x has one-dimensional support, (4.39) gives

$$H^3(\mathcal{F}_m) = 0, \quad H^2(\mathcal{F}_m) = H^1(Z_x, \mathcal{M}_{x,N}|_{Z_x}). \quad (4.41)$$

Each difference image $\Sigma_j - C_i$ is the closed image of the proper surface $\Sigma_j \times C_i$, and has dimension at most two. Hence

$$M^\circ = \left(X \setminus \bigcup_{i,j} (\Sigma_j - C_i) \right) \times \hat{X} \quad (4.42)$$

is nonempty and open. For $m \in M^\circ$, Z_x is the disjoint union of $2n$ smooth genus-three curves. By (4.6), $\mathcal{M}_{x,N}$ has degree six on each. Formula (4.8) gives zero H^1 on each component since $6 > 2 \cdot 3 - 2$. Thus (4.41) vanishes on M° , for every N . Equation (4.35) identifies all these cohomology groups with the cohomology of the derived fibers of G .

The last assertion of Lemma 4.4 now gives a local model

$$K^1 \xrightarrow{u} K^2 \quad (4.43)$$

for G . On M° its fiber map is surjective. Thus $u^* : (K^2)^* \rightarrow (K^1)^*$ is generically injective. The variety M is integral, and the source of u^* is locally free; its generically zero kernel must therefore be zero. Dualizing (4.43) and shifting as in (4.37) leaves its sole cohomology in degree zero, with resolution

$$0 \longrightarrow (K^2)^* \xrightarrow{u^*} (K^1)^* \longrightarrow E \longrightarrow 0. \quad (4.44)$$

These local sheaves are the sole cohomology sheaf of the globally defined object $D_M(G)[-1]$, so they glue. This constructs the actual sheaf, rather than assuming $E = (\mathcal{H}^1 G)^*$.

Finally, on M° , each curve restriction in (4.39) has Euler characteristic $6 + 1 - 3 = 4$. Equations (4.39)–(4.40) give

$$\chi(\mathcal{F}_m) = 8 - 2n \cdot 4 = -8d. \quad (4.45)$$

The Euler characteristic of the finite model is $-\text{rank } K^1 + \text{rank } K^2$. The rank of (4.44) is therefore $8d$. $\square$

4.4 The coherent translation action and the quotient

For the group action, enumerate the source cosets in (4.9) by G_2 and G_1 , respectively. Translation on a closed invariant subscheme gives its ideal the canonical translation linearization, so $I = I_\Sigma \boxtimes I_A$ has a genuine $G_1 \times G_2$ -linearization for the source translations. The word “coherent” in the next statement includes the composition isomorphisms, not only one isomorphism for each group element.

Proposition 4.8 (Coherent conjugation and the graph quotient). *For $g = (g_1, g_2) \in G_1 \times G_2$, put*

$$\begin{aligned} b_g &= (g_1 - g_2, a(-g_1 - g_2)) \in M, \\ N_g &= \text{pr}_X^* a(-g_1) \otimes \text{pr}_{\widehat{X}}^* \mathcal{P}_{-g_2}, \quad \mathcal{P}_v = \mathcal{P}|_{\{v\} \times \widehat{X}}. \end{aligned} \quad (4.46)$$

The conjugation of the source translations by $\tilde{\Phi}$ has explicit functors

$$T_g(-) = N_g \otimes t_{b_g, *}(-). \quad (4.47)$$

There are unital associative line isomorphisms

$$N_g \otimes t_{b_g, *} N_h \simeq N_{g+h} \quad (4.48)$$

and compatible isomorphisms $T_g G \simeq G$. Perfect duality gives actual sheaf isomorphisms

$$N_g^{-1} \otimes t_{b_g, *} E \simeq E \quad (4.49)$$

with the corresponding unital associative line compositors. The homomorphism $g \mapsto b_g$ is injective. Its image $\bar{G}$ has order n^2 , acts freely by translations, and

$$q : M \longrightarrow Y_0 = M/\bar{G} \quad (4.50)$$

is a finite étale isogeny of abelian sixfolds.

Proof. We verify the conjugation at kernels, which also accounts for its compositors. For $\ell \in \widehat{X}$ write $L_\ell = \mathcal{P}|_{X \times \{\ell\}}$. The normalized biextension has the kernel identities

$$\begin{aligned} (t_v \times 1)^* \mathcal{P}^{-1} &\simeq \mathcal{P}^{-1} \otimes \mathrm{pr}_{\widehat{X}}^* \mathcal{P}_{-v}, \\ (1 \times t_{-\ell})^* \mathcal{P}^{-1} &\simeq \mathcal{P}^{-1} \otimes \mathrm{pr}_X^* L_\ell. \end{aligned} \quad (4.51)$$

Projection formula and flat base change for translation, applied to these kernels, give

$$\Psi \circ t_{v,*} \simeq (\mathcal{P}_{-v} \otimes -) \circ \Psi, \quad \Psi \circ (L_\ell \otimes -) \simeq t_{\ell,*} \circ \Psi. \quad (4.52)$$

These are the inverse-Poincaré versions of the translation–tensor exchange in [29, Section 3, (3.1)]. Tensor conjugation on the source is

$$(H \otimes -) \circ t_{v,*} \circ (H^{-1} \otimes -) \simeq (a(-v) \otimes -) \circ t_{v,*}. \quad (4.53)$$

For the shear,

$$\mu^*(a(u) \boxtimes a(v)) \simeq a(u) \boxtimes a(u+v), \quad \mu^{-1} t_{(g_1, g_2)} \mu = t_{(g_1 - g_2, g_2)}. \quad (4.54)$$

Thus tensor conjugation and shear give the line $a(-g_1) \boxtimes a(-g_1 - g_2)$ and the translation $(g_1 - g_2, g_2)$. Applying (4.52) on the second factor gives (4.46)–(4.47). Commuting the target translation past $\mathcal{P}_{-g_2}$ introduces only a constant one-dimensional biextension fiber; choosing its trivialization changes the graph kernel identification by a scalar.

All the displayed identities are identities of integral kernels: they are obtained by pullback, projection formula, and the normalized biextension multiplication. That multiplication is associative. Convolution with the fixed kernel of $\widetilde{\Phi}$ and its inverse therefore transports the genuine source-translation compositors and their associativity diagrams. The kernel of each functor in (4.47) is an invertible sheaf on the graph of t_{b_g} . The product kernel and the kernel for $g + h$ have the same graph, since $b_{g+h} = b_g + b_h$. A degree-zero derived isomorphism between these line sheaves is an actual line isomorphism. Any two choices differ by a scalar, since $H^0(M, \mathcal{O}_M)^\times = \mathbb{C}^\times$. Transporting the compositors through the chosen graph identifications consequently gives (4.48) with its unital associativity, and transports the source linearization to $T_g G \simeq G$. This does not infer a scalar composition from simplicity of G or E ; the scalars come from the line kernels and their compositors.

Duality commutes with translation by an isomorphism and changes a line factor to its inverse. Dualize the preceding compatible isomorphisms and invert them. With the shift in (4.37) this gives (4.49) and the inherited associative line compositors. Because (4.37) has only its degree-zero cohomology sheaf, these are actual sheaf isomorphisms.

Finally, $b_g = 0$ implies $g_1 = g_2$ and $2g_1 = 0$, since a is injective. The intersection $G_1 \cap G_2 = 0$ therefore makes the kernel trivial. Nonzero translations have no fixed points. The quotient by this finite flat closed subgroup exists [13, Section 2, p. 136]; its quotient map is a $\bar{G}$ -torsor, hence finite étale because that holds after the trivializing base change. The quotient is a proper smooth connected group space. Over $\mathbb{C}$ it is a quasi-projective group scheme [13, Section 1.1(a), p. 131], hence is a projective abelian variety. This proves (4.50). $\square$

We have constructed the positive-rank coherent perfect sheaf E , its coherent line-twisted action (4.49), and the finite étale quotient (4.50). The action is defined on the entire sheaf by the kernel construction; neither simplicity nor reflexivity was needed. In the subsequent obstruction and character calculations we write

$$\mathcal{I} = I, \quad \mathcal{G} = G, \quad \mathcal{E} = E.$$

The obstruction calculation applies to $\mathcal{G}$, while the character and deformation arguments use its coherent shifted dual $\mathcal{E}$. The flat family constructed here belongs to the fixed seed. Subsequent sections use semiregularity to deform an associated ordinary sheaf locally, and then propagate the resulting algebraic classes through the period domain.

5 Hochschild evaluation and the invariant obstruction space

The geometric construction has produced a coherent perfect sheaf with a finite line-twisted action. To use its quotient in a deformation argument, we must control the entire invariant derived obstruction space. We first prove the exact Atiyah trace identity and its transport through the fixed Fourier–Mukai equivalence, and then compute the invariant space for the Abel-orbit seed.

5.1 Hochschild evaluation and the Atiyah trace

We will relate the evaluation of a Hochschild class on a perfect complex to contraction against its Chern character, and then compare these two maps under a fixed Fourier–Mukai equivalence. This is the compatibility used in the obstruction argument of [25, Diagram (8.3.4) and Lemma 8.3.10]. The proof below calculates the normalized Hochschild–Kostant–Rosenberg map on the formal diagonal of an abelian variety. It then obtains the trace identity from the diagonal adjunction. In particular, the calculation fixes one convention for the bivector sign on both sides.

5.1.1 Grading, contraction, and the first-jet class

Let A be a complex abelian variety of dimension d , let $\Delta : A \hookrightarrow A \times A$ be the diagonal, and let $Q \in \text{Perf}(A)$. Write $\mathcal{O}_\Delta = \Delta_* \mathcal{O}_A$. We use cohomological shifts, so a vector bundle shifted by $[1]$ lies in degree -1 . Put

$$V = H^0(A, T_A), \quad \mathcal{S}_A = \text{Sym}_{\mathcal{O}_A}(\Omega_A^1[1]) = \bigoplus_{k=0}^d \Omega_A^k[k].$$

The complex on the right has zero differential. Translation gives $T_A = \mathcal{O}_A \otimes_{\mathbb{C}} V$. Our gradings are

$$\begin{aligned} HT^r(A) &= \bigoplus_{a+b=r} H^a(A, \bigwedge^b T_A), \\ H\Omega_m(A) &= \bigoplus_{k-h=m} H^h(A, \Omega_A^k), \\ \mathbb{H}^r(A, \mathcal{S}_A) &= H\Omega_{-r}(A). \end{aligned} \tag{5.1}$$

All exterior degrees in these sums are between 0 and d .

We specify contraction including its order. If $\iota_\xi : \bigwedge^k V^* \rightarrow \bigwedge^{k-1} V^*$ is ordinary interior multiplication, define

$$\iota_{\xi_1 \wedge \dots \wedge \xi_b}^L = \iota_{\xi_1} \circ \dots \circ \iota_{\xi_b}, \quad s_b = (-1)^{b(b-1)/2}. \tag{5.2}$$

Composition is read right to left. Thus $\iota_{u \wedge v}^L = \iota_u^L \iota_v^L$, and

$$\iota_{\xi_1 \wedge \dots \wedge \xi_b}^L(\vartheta_1 \wedge \dots \wedge \vartheta_b) = s_b \det(\vartheta_j(\xi_i)).$$

The factor s_b is the graded dual pairing of $\text{Sym}^b(V[-1])$ with $\text{Sym}^b(V^*[1])$: interleaving their odd factors requires $b(b-1)/2$ crossings. The pairing of one vector and one covector is positive.

Use invariant Dolbeault representatives, with the antiholomorphic cochain before the holomorphic form. For $\eta \in H^a(A, \mathcal{O}_A)$, $u \in \bigwedge^b V$, $\omega \in H^h(A, \mathcal{O}_A)$, and $\vartheta \in \bigwedge^k V^*$, define

$$(\eta \otimes u) \lrcorner (\omega \otimes \vartheta) = (-1)^{bh} (\eta \wedge \omega) \otimes \iota_u^L \vartheta. \quad (5.3)$$

The products used with this action are

$$\begin{aligned} (\eta \otimes u)(\eta' \otimes v) &= (-1)^{ba'} (\eta \wedge \eta') \otimes (u \wedge v), & \eta' \in H^{a'}(A, \mathcal{O}_A), \\ (\omega \otimes \vartheta)(\omega' \otimes \vartheta') &= (-1)^{kh'} (\omega \wedge \omega') \otimes (\vartheta \wedge \vartheta'), & \omega' \in H^{h'}(A, \mathcal{O}_A). \end{aligned} \quad (5.4)$$

These are the total-complex Koszul rules; in particular (5.3) is a left action for the first product. They also determine the meaning of powers of a $(1, 1)$ -class. For example, if $j(\xi) = \xi \lrcorner \vartheta$ for such a class, then $\xi \lrcorner \vartheta^2 = 2j(\xi)\vartheta$.

Write $\mathcal{I}_\Delta$ for the diagonal ideal. Fix the conormal identification

$$df = f \otimes 1 - 1 \otimes f \pmod{\mathcal{I}_\Delta^2}.$$

Let a_Δ be the class of the first infinitesimal diagonal:

$$0 \longrightarrow \Delta_* \Omega_A^1 \longrightarrow \mathcal{O}_{A \times A} / \mathcal{I}_\Delta^2 \longrightarrow \mathcal{O}_\Delta \longrightarrow 0. \quad (5.5)$$

For a kernel M from X to Y , we use $\Phi_M(-) = R p_{Y*}(L p_X^*(-) \otimes^{\mathbf{L}} M)$. Evaluation of a_Δ by this operation defines the raw first-jet class

$$a_Q : Q \longrightarrow Q \otimes \Omega_A^1[1].$$

For a bundle it is the class of its first jet sequence. For a perfect complex it is the same derived natural transformation. The power $a_Q^b : Q \rightarrow Q \otimes \Omega_A^b[b]$ means successive applications of this class, with each new one-form appended on the right, followed by the exterior projection. This fixes both the Yoneda and exterior orders; a_Q^0 is the identity.

For a coefficient complex C , the perfect partial trace is

$$\mathrm{Tr}_{Q,C} : \mathrm{Hom}(Q, Q \otimes C) \longrightarrow \mathrm{Hom}(\mathcal{O}_A, C).$$

Precisely, it is the composite

$$\mathcal{O}_A \xrightarrow{\mathrm{coev}} Q \otimes Q^\vee \xrightarrow{\mathrm{braid}} Q^\vee \otimes Q \xrightarrow{1 \otimes f} Q^\vee \otimes Q \otimes C \xrightarrow{\mathrm{ev} \otimes 1} C$$

for $f : Q \rightarrow Q \otimes C$. These are the duality and symmetry maps of perfect complexes, with their usual Koszul signs. On a bounded locally free representative this is supertrace. The definition applies without change to shifted C , and it does not require a cohomology sheaf of Q to be locally free.

5.1.2 The normalized HKR map and the trace identity

Set

$$H_A = L\Delta^* \mathcal{O}_\Delta, \quad HH^r(A) = \mathrm{Ext}_{A \times A}^r(\mathcal{O}_\Delta, \mathcal{O}_\Delta).$$

We use the derived adjunction $L\Delta^* \dashv \Delta_*$, with unit

$$\eta_\Delta : \mathcal{O}_\Delta \longrightarrow \Delta_* H_A.$$

The usual sheafified local bar map has degree- n formula

$$[f_1] \cdots [f_n] \mapsto \frac{df_1 \wedge \cdots \wedge df_n}{n!},$$

with the two outer coefficients multiplied. This formula specifies the normalized bar morphism $\mathbf{I}_A : H_A \rightarrow \mathcal{S}_A$; see [7, Section 4 and Corollary 4.2] for the local bar construction. An element of $HT^r(A)$ is a morphism $\alpha : \mathcal{S}_A \rightarrow \mathcal{O}_A[r]$ by the graded dual pairing in (5.2). Define its adjoint Hochschild class by

$$\mathrm{HKR}_A(\alpha) = \Delta_* \alpha \circ \Delta_* \mathbf{I}_A \circ \eta_\Delta : \mathcal{O}_\Delta \longrightarrow \mathcal{O}_\Delta[r]. \quad (5.6)$$

Let $\chi_Q : HH^r(A) \rightarrow \mathrm{Ext}_A^r(Q, Q)$ denote evaluation of a diagonal morphism on Q . For $\alpha^{a,b} \in H^a(A, \bigwedge^b T_A)$, contraction with an Atiyah power means $\alpha^{a,b} \lrcorner a_Q^b = (1_Q \otimes \alpha^{a,b}) \circ a_Q^b$, using the graded-dual morphism $\alpha^{a,b} : \Omega_A^b[b] \rightarrow \mathcal{O}_A[a+b]$. Define the raw Hodge Chern components by

$$\mathrm{ch}_{H,j}(Q) = \mathrm{Tr}_{Q, \Omega_A^j[j]} \left(\frac{a_Q^j}{j!} \right) \in H^j(A, \Omega_A^j), \quad \mathrm{ch}_H(Q) = \sum_{j=0}^d \mathrm{ch}_{H,j}(Q). \quad (5.7)$$

Proposition 5.1 (Hochschild–Atiyah trace identity). *Let A be a complex abelian variety of dimension d , and let $Q \in \mathrm{Perf}(A)$. Use the raw first-jet class a_Q from (5.5), and the left-interior and HKR conventions (5.3) and (5.6). Define*

$$\mathrm{ob}_Q = \chi_Q \circ \mathrm{HKR}_A : HT^2(A) \longrightarrow \mathrm{Ext}_A^2(Q, Q).$$

For $\alpha = \sum_{a+b=2} \alpha^{a,b}$, this evaluation is

$$\mathrm{ob}_Q(\alpha) = \sum_{a+b=2} \alpha^{a,b} \lrcorner \frac{a_Q^b}{b!}.$$

On the full group $\mathrm{Ext}_A^2(Q, Q)$, define

$$\begin{aligned} \sigma_Q &= (\sigma_{Q,j})_{0 \leq j \leq d-2} : \mathrm{Ext}_A^2(Q, Q) \longrightarrow H\Omega_{-2}(A) = \bigoplus_{j=0}^{d-2} H^{j+2}(A, \Omega_A^j), \\ \sigma_{Q,j}(v) &= \mathrm{Tr}_{Q, \Omega_A^j[j+2]} \left(\left(\frac{a_Q^j}{j!} \right) [2] \circ v \right). \end{aligned} \quad (5.8)$$

The direct sum is zero if $d < 2$. Set

$$c_Q(\alpha) = \alpha \lrcorner \mathrm{ch}_H(Q) \in H\Omega_{-2}(A).$$

Then

$$\sigma_Q \circ \mathrm{ob}_Q = c_Q. \quad (5.9)$$

More precisely, with $\mathrm{ch}_{H,k}(Q) = 0$ for $k \notin [0, d]$, each component is

$$\sigma_{Q,j}(\mathrm{ob}_Q(\alpha)) = \sum_{a+b=2} \alpha^{a,b} \lrcorner \mathrm{ch}_{H,j+b}(Q) \quad \text{in } H^{j+2}(A, \Omega_A^j). \quad (5.10)$$

The proof uses two identities on the diagonal. We first show that the normalized bar map carries the Hochschild action to the fixed left contraction. We then show that the adjunction unit becomes the exponential of the first-jet class. Naturality of the unit and of perfect trace will give the stated triangle.

5.1.3 The normalized HKR action on the formal diagonal

We next verify globally the normalization and action of these maps on an abelian variety.

The difference isomorphism

$$A \times A \longrightarrow A \times A, \quad (x, y) \longmapsto (y, x - y)$$

takes Δ to $A \times \{0\}$, and hence identifies the formal completion along Δ with $A \times \widehat{A}_0$. In characteristic zero a commutative formal group has logarithm coordinates. In this case they can be obtained by integrating the closed invariant formal one-forms term by term, with constant term zero. The differential of the difference between the logarithm of the formal sum and the sum of the two logarithms is zero, and its constant term is zero; thus the logarithm carries the formal group law to addition. Its linear term is invertible. Choose coordinates $z_1, \dots, z_d$ whose linear terms are a basis of invariant one-forms. The completed structure sheaf is then

$$\mathcal{R} = \mathcal{O}_A \widehat{\otimes}_{\mathbb{C}} \mathbb{C}[[z_1, \dots, z_d]].$$

These are formal normal coordinates. No global nonconstant regular or holomorphic functions on $A \times A$ cutting out Δ are asserted.

We explain why this completion computes the derived operations that will be used. Noetherian completion is flat. If R is a Noetherian local ring along the diagonal, J is its diagonal ideal, and $F \rightarrow R/J$ is a finite free resolution, then $\widehat{R} \otimes_R F$ remains a resolution and

$$\mathrm{Hom}_{\widehat{R}}(\widehat{R} \otimes_R F, R/J) = \mathrm{Hom}_R(F, R/J). \quad (5.11)$$

The same equality holds with a locally free R/J -module as target. Such finite resolutions exist locally because the diagonal of the smooth variety A is a regular immersion. The identifications in (5.11) commute with composition and with tensoring by R/J . They therefore identify the derived sheaf Hom complexes on the diagonal, their Yoneda pairings, and the action on $L\Delta^* \mathcal{O}_\Delta$. After sheaf cohomology they identify the corresponding global operations. In particular, the diagonal-supported morphisms with locally free diagonal coefficients below can be compared after completion and then descended by these identifications.

Let e_i have cohomological degree -1 and correspond to dz_i , and let ξ_i be the dual invariant vector. A global finite free resolution on the formal completion is

$$K = \left(\mathcal{R} \otimes \bigwedge^\bullet V^*, d_K = \sum_i z_i C_i \right), \quad C_i = \iota_{\xi_i}, \quad \varepsilon : K \longrightarrow \mathcal{O}_A. \quad (5.12)$$

Here $K^{-n} = \mathcal{R} \otimes \bigwedge^n V^*$. The formal variables form a regular sequence, so K resolves the diagonal. Its pullback and derived Hom are represented by

$$\begin{aligned} \mathcal{O}_A \otimes_{\mathcal{R}} K &= \mathcal{O}_A \otimes \bigwedge^\bullet V^* = \mathcal{S}_A, \\ \mathrm{Hom}_{\mathcal{R}}(K, \mathcal{O}_A) &= \mathcal{O}_A \otimes \bigwedge^\bullet V, \end{aligned} \quad (5.13)$$

with zero internal differentials and with the graded dual identification just specified.

It remains to identify this splitting with the normalized bar map, rather than merely with some splitting of cohomology. Complete the sheafified local bar resolution on the small diagonal in A^{n+2} . Taking the last point as base, use the formal logarithm to write $z^{(0)}, \dots, z^{(n)}$ for the logarithms of the other points minus the last point. The outer $\mathcal{R}$ -action uses $z^{(0)}$. Each completed bar term is a formal power series extension of the Noetherian sheaf $\mathcal{R}$, hence is flat over $\mathcal{R}$. The ordinary bar

contracting homotopy after forgetting one outer action inserts a unit. It is continuous for these formal-adic topologies, so its identities extend to the completion. Thus the augmented completed bar complex is a flat resolution of the diagonal.

There is a global map from K to this resolution whose term in degree $-n$ is

$$F_n(e_{i_1} \wedge \cdots \wedge e_{i_n}) = \sum_{\tau \in S_n} \text{sgn}(\tau) z_{i_{\tau(1)}}^{(1)} \cdots z_{i_{\tau(n)}}^{(n)}. \quad (5.14)$$

An interior bar face identifies two successive middle points, so the alternating determinant on the right becomes zero. The last face sets $z^{(n)} = 0$, again giving zero. The first face sets $z^{(1)} = z^{(0)}$. Expansion along that column gives, for $I = (i_1, \dots, i_n)$,

$$bF_n(e_I) = \sum_{\ell=1}^n (-1)^{\ell-1} z_{i_\ell}^{(0)} F_{n-1}(e_{I \setminus i_\ell}) = F_{n-1}(d_K e_I).$$

In degree zero the map is the identity on the augmentation. It is therefore a comparison of resolutions. It is global on the formal neighborhood: the logarithm makes every face substitution just used compatible on that neighborhood.

The normalized local bar formula defining $\mathbf{I}_A$ restricts to this completed resolution. Indeed, it uses only the first jet in each middle variable along the small diagonal, so is continuous for the completion; naturality of differentials identifies the formulas on overlaps. After pullback to the diagonal, each term of (5.14) maps to $dz_{i_1} \wedge \cdots \wedge dz_{i_n}/n!$: the permutation sign and the exterior sign cancel. The $n!$ terms sum to dz_I . Consequently $\mathbf{I}_A$ is the identity $e_I \mapsto dz_I$ on the first complex in (5.13). In particular it is a quasi-isomorphism. Because the comparison (5.14) is global, this conclusion has no additional off-diagonal or higher-cohomology component. This is the restriction of the usual sheaffied bar HKR morphism, not a newly chosen splitting.

We now compute its cap action. The C_i have degree 1, satisfy

$$C_i C_j = -C_j C_i, \quad d_K C_i + C_i d_K = 0,$$

and, for $u = \xi_1 \wedge \cdots \wedge \xi_b$, give a closed lift $C_u^L = C_{\xi_1} \cdots C_{\xi_b}$. Its augmentation on K^{-b} is s_b times determinant evaluation. More explicitly, the map

$$\mathcal{O}_A \otimes \bigwedge^\bullet V \longrightarrow \mathcal{E}nd_{\mathcal{R}}(K), \quad u \longmapsto C_u^L$$

is a morphism of differential graded algebras and is a quasi-isomorphism. To see the last assertion, postcompose with ε . The resulting map to $\mathcal{H}om_{\mathcal{R}}(K, \mathcal{O}_A)$ is the basis isomorphism given by the graded dual pairing. The map $\mathcal{E}nd_{\mathcal{R}}(K) \rightarrow \mathcal{H}om_{\mathcal{R}}(K, \mathcal{O}_A)$ is itself a quasi-isomorphism, because K is bounded free and ε is a quasi-isomorphism. This gives global closed lifts for the sheaf-level Yoneda and cap pairings.

For clarity about global cohomology, first perform the derived Hom and derived pullback with the finite free complex K . The two resulting complexes in (5.13) are finite complexes of trivial $\mathcal{O}_A$ -bundles. The closed lifts C_u^L compute their sheaf-level pairings. Resolve only these $\mathcal{O}_A$ -coefficients by the ordinary Dolbeault resolution $\mathcal{A}_A^{0,\bullet}$, and totalize the pairings with Dolbeault wedge. The Dolbeault Poincaré lemma computes their sheaf cohomology. Projective GAGA for coherent cohomology and Ext identifies these cohomology groups and pairings with their algebraic counterparts. Neither a Dolbeault resolution of the infinite formal sheaf nor GAGA for such a sheaf is used.

Every class in $H^a(A, \mathcal{O}_A)$ has an invariant Dolbeault representative; these representatives form $\bigwedge^a H^1(A, \mathcal{O}_A)$. This standard computation for a complex torus, together with the triviality of T_A ,

shows that decomposable classes $\eta \otimes u$ span $HT^r(A)$. For such a class the totalized operator on a normal Koszul term k and a Dolbeault cochain ω is

$$L_{\eta,u}(\omega \otimes k) = (-1)^{b|\omega|}(\eta \wedge \omega) \otimes C_u^L(k). \quad (5.15)$$

This formula denotes the induced pairing on the finite diagonal complexes above. The normal lift is closed by its anticommutation with d_K ; totalization is closed because η is closed. Its augmentation is the cochain for the graded dual pairing. The sign moves the degree- b normal operator past the Dolbeault cochain.

Take the Yoneda convention $\beta \smile \gamma = \beta[|\gamma|] \circ \gamma$. Composing (5.15) gives the first product in (5.4), because $C_u^L C_v^L = C_{u \wedge v}^L$. Tensoring the same lift with the diagonal gives precisely (5.3). The roof for η_Δ on K has denominator $\varepsilon : K \rightarrow \mathcal{O}_A$ and numerator the projection $K \rightarrow \mathcal{O}_A \otimes_{\mathcal{R}} K$. The augmentation just calculated is therefore the adjoint class (5.6). Derived adjunction and the quasi-isomorphism $\mathbf{I}_A$ show that $\mathrm{HKR}_A : HT^r(A) \rightarrow HH^r(A)$ is an isomorphism. The calculation proves, in particular,

$$\mathbf{I}_A((L\Delta^* \mathrm{HKR}_A(\alpha))h) = \alpha \lrcorner \mathbf{I}_A(h), \quad \alpha \in HT^2(A), \quad h \in \mathbb{H}^0(A, H_A). \quad (5.16)$$

Here a derived endomorphism acts on hypercohomology by composition. The same calculation of the pairings accounts for every summand of the target $H\Omega_{-2}(A)$, since invariant representatives span the relevant groups.

The cochain normalization can also be read directly from (5.14). In unshifted differential-operator notation it is

$$\mathrm{HKR}_A(\xi_1 \wedge \cdots \wedge \xi_b) = \frac{s_b}{b!} \sum_{\tau \in S_b} \mathrm{sgn}(\tau) \xi_{\tau(1)} \otimes \cdots \otimes \xi_{\tau(b)} \quad \text{in } HH^b(A). \quad (5.17)$$

Indeed, evaluation on F_b gives s_b times determinant evaluation, the augmentation of C_u^L . This is the cochain convention in [6, Equation (3.20)]. In particular $s_2 = -1$; replacing left interior multiplication by determinant-order contraction would precompose both evaluation and contraction by $\mathrm{diag}(1, 1, -1)$ on $H^2(\mathcal{O}_A) \oplus H^1(T_A) \oplus H^0(\wedge^2 T_A)$. We make no such replacement.

The formal coordinates, their comparison, and (5.15) are fixed by diagonal translation. Translations also fix invariant Dolbeault representatives. Thus translations act trivially on $HT^r(A)$, and HKR_A identifies this with their action on $HH^r(A)$. Finally, T_A is holomorphically trivial, so its positive Chern classes and its Atiyah class vanish and $\mathrm{td}(A) = \sqrt{\mathrm{td}(A)} = 1$. The explicit identity (5.16) is the abelian specialization of the compatibility in [6, Theorem 1.4]; it was proved here without using the general formality theorem or a separate convention for dual chains.

5.1.4 The exponential first-jet class and evaluation

The preceding comparison fixes the homological map. We next compute its adjoint on a perfect complex; this supplies the factorials in the obstruction and trace maps.

On K^{-1} , the extension (5.5) is represented by $e_i \mapsto dz_i$. The sign can be checked by mapping $K^0 = \mathcal{R}$ to $\mathcal{R}/(z)^2$ and K^{-1} to $(z)/(z)^2$ by $e_i \mapsto z_i$. The square with the Koszul differential commutes, and the conormal identification is exactly the one in (5.5). A closed lift is

$$\mathcal{A}(k) = \sum_i C_i(k) \otimes dz_i : K \longrightarrow (K \otimes V^*)[1].$$

Successive application, with the forms appended in the order fixed above, followed by exterior projection, is the lift

$$\mathcal{A}^{(b)}(k) = \sum_{i_1, \dots, i_b} C_{i_b} \cdots C_{i_1}(k) \otimes dz_{i_1} \wedge \cdots \wedge dz_{i_b}. \quad (5.18)$$

It represents the spliced b -fold first-jet class: composing the next closed lift splices the next extension, while the displayed form order is the appended order. On $e_{j_1} \wedge \cdots \wedge e_{j_b}$, its nonzero augmented terms are indexed by permutations. The contraction and exterior factors have the same permutation sign, which cancels. The result is $b! dz_{j_1} \wedge \cdots \wedge dz_{j_b}$. The augmentation is zero on other Koszul degrees.

On the other hand, the degree- b component of $\Delta_* \mathbf{I}_A \circ \eta_\Delta$ is the projection from K^{-b} followed by $e_I \mapsto dz_I$. The preceding calculation and (5.11) therefore prove, on the original diagonal, the identity of morphisms

$$\Delta_* \mathbf{I}_A \circ \eta_\Delta = \sum_{b=0}^d \frac{a_\Delta^b}{b!} : \mathcal{O}_\Delta \longrightarrow \bigoplus_{b=0}^d \Delta_* \Omega_A^b[b]. \quad (5.19)$$

This is a direct abelian proof of the universal exponential identity of [7, Proposition 4.4], with its sign and divided powers specified by the same Koszul comparison.

Let

$$AH_Q = \Phi_{\eta_\Delta}(Q) : Q \longrightarrow Q \otimes H_A.$$

The projection formula for a diagonal kernel identifies its target as written. Applying the derived integral-kernel operation to (5.19) gives

$$(1_Q \otimes \mathbf{I}_A) \circ AH_Q = \sum_{b=0}^d \frac{a_Q^b}{b!} : Q \longrightarrow Q \otimes \mathcal{S}_A. \quad (5.20)$$

For a vector bundle, the first infinitesimal diagonal is finite over the second factor, so its evaluation is the ordinary first jet sequence. For a perfect complex the same equality is the derived kernel operation, computed locally on a bounded complex of vector bundles. It commutes with splicing and exterior projection. Consequently the powers in (5.20) are the raw powers a_Q^b already defined, also when Q is shifted.

Equations (5.6) and (5.20) give

$$\chi_Q(\mathrm{HKR}_A(\alpha)) = (1_Q \otimes \alpha) \circ \sum_{b=0}^d \frac{a_Q^b}{b!}. \quad (5.21)$$

If $\alpha^{a,b} \in H^a(A, \wedge^b T_A)$, its morphism from $\mathcal{S}_A$ is zero on all summands except $\Omega_A^b[b]$. Its value in (5.21) is exactly $\alpha^{a,b} \lrcorner (a_Q^b/b!)$, of Ext degree $a+b$, with the graded dual and total-complex signs fixed above. This proves the evaluation identity in the present perfect abelian setting; compare [21, Theorem A].

The components in (5.7) are the unscaled $d \log$ Hodge components. To identify them with the ordinary character, first consider a line bundle. Its first jet extension is the $d \log$ transition cocycle; thus (5.7) is the exponential of that first Hodge Chern class. Perfect trace is additive for a bounded locally free exact triangle, as is seen from supertrace on its cone. Pullback naturality of the first-jet class and the projective bundle formula allow a vector bundle to be pulled to its flag bundle, where its successive line quotients give the line-bundle calculation. Pullback on the Hodge cohomology in question is injective by that formula. Finally, the resolution property of a smooth projective variety represents every perfect class by an alternating sum of vector bundles; additivity treats Q . This gives the character identification of [7, Theorem 4.5] in the present convention.

Let $\mathrm{ch}_{B,j}(Q) \in H^{2j}(A, \mathbb{Q})$ denote the rational Betti Chern component. Under the Dolbeault–de Rham identification $H^j(A, \Omega_A^j) = H^{j,j}(A) \subset H^{2j}(A, \mathbb{C})$, the raw comparison is

$$\mathrm{ch}_{H,j}(Q) \longmapsto (2\pi i)^j \mathrm{ch}_{B,j}(Q). \quad (5.22)$$

Indeed, the exponential sequence with kernel $2\pi i\mathbb{Z}$ sends the $d\log$ cocycle to $2\pi i c_1^B$, and the line splitting and additivity just used give the formula in every weight. Dividing by $(2\pi i)^j$ is the Tate-normalized comparison. The trace identity (5.9) uses ch_H throughout.

5.1.5 Proof of the trace identity

Proof of Proposition 5.1. For $\beta : \mathcal{O}_\Delta \rightarrow \mathcal{O}_\Delta[2]$, write $a_A(\beta) = L\Delta^*\beta : H_A \rightarrow H_A[2]$. Naturality of the adjunction unit, with its canonical shift isomorphisms, is the equality

$$\Delta_* a_A(\beta) \circ \eta_\Delta = \eta_\Delta[2] \circ \beta.$$

Evaluate this equality of diagonal morphisms on Q . The projection formula gives

$$(1_Q \otimes a_A(\beta)) \circ AH_Q = AH_Q[2] \circ \chi_Q(\beta). \quad (5.23)$$

This is an identity of derived morphisms before any trace is taken.

Define

$$h_Q = \text{Tr}_{Q, H_A}(AH_Q) \in \mathbb{H}^0(A, H_A), \quad \sigma_Q^H(v) = \text{Tr}_{Q, H_A[2]}(AH_Q[2] \circ v) \in \mathbb{H}^2(A, H_A).$$

The perfect partial trace commutes with a morphism of coefficients: for $f : C \rightarrow C'$ and $w : Q \rightarrow Q \otimes C$,

$$\text{Tr}_{Q, C'}((1_Q \otimes f) \circ w) = f \circ \text{Tr}_{Q, C}(w). \quad (5.24)$$

This follows directly from its coevaluation–braiding–evaluation definition: f can be placed after evaluation since it acts only on the coefficient factor. Apply it to (5.23). We obtain

$$\sigma_Q^H(\chi_Q(\beta)) = a_A(\beta) \circ h_Q.$$

There is no odd exchange sign, since the degree here is two.

By (5.20) and (5.24),

$$\mathbf{I}_A(h_Q) = \text{ch}_H(Q), \quad \mathbf{I}_A(\sigma_Q^H(v)) = \text{Tr}_Q \left(\left(\sum_{j=0}^d \frac{a_Q^j}{j!} \right) [2] \circ v \right).$$

In the second equality the component with coefficient $\Omega_A^j[j+2]$ is $\sigma_{Q,j}$; those with $j > d-2$ vanish because their sheaf cohomology degree exceeds d . Take $\beta = \text{HKR}_A(\alpha)$ and apply (5.16). This proves (5.9). For the component statement, contraction of $\text{ch}_{H,j+b}(Q) \in H^{j+b}(A, \Omega_A^{j+b})$ removes b holomorphic forms and adds cochain degree a . Its target is $H^{j+b+a}(A, \Omega_A^j) = H^{j+2}(A, \Omega_A^j)$. This gives (5.10). In particular the factors $1/b!$ in evaluation, $1/j!$ in $\sigma_{Q,j}$, and $1/(j+b)!$ in the Chern component have already been accounted for; there is no further scalar. $\square$

The factorization just proved is the perfect-complex specialization of the character and module identities in [5, Proposition 6.2.1, Corollary 6.3.2, and Proposition 6.4.4]. Their differential-operator convention has $a_Q = -\text{At}_Q^{BF}$. Thus their terms $(-1)^j(\text{At}_Q^{BF})^j/j!$ are exactly $a_Q^j/j!$. There is no residual sign in (5.10). The proof above establishes the identity from the diagonal unit and the fixed bar comparison, rather than importing a compatibility between two unspecified HKR normalizations.

5.2 Fourier–Mukai transport of evaluation and contraction

We now show that a specified Fourier–Mukai equivalence preserves the equality $\ker \text{ob}_Q = \ker c_Q$. Evaluation is transported by convolution of kernels. For character contraction we construct a homological map using perfect Serre pairings; this also records the sign of the shift in the fixed transform.

5.2.1 Enhanced convolution and evaluation

Let A and B be smooth projective complex abelian varieties of dimensions d_A, d_B . Let $\Phi = \Phi_K : \text{Perf}(A) \rightarrow \text{Perf}(B)$ be a Fourier–Mukai equivalence with perfect kernel K , equipped with a fixed enhanced inverse kernel L , inverse adjunction units and counits, and their triangle identities. Here *enhanced* means that convolution, morphisms, and the isomorphisms commuting with cohomological shifts are those of the derived enhancement, not just chosen identifications of objects in the triangulated category.

For kernels P from X to Y and R from Y to Z , write

$$R \star P = R p_{XZ*} (L p_{XY}^* P \otimes^{\mathbf{L}} L p_{YZ}^* R).$$

Thus $\Phi_{R \star P} = \Phi_R \circ \Phi_P$. We use the standard enhanced associativity and projection formula for this convolution. The products are smooth proper and the projections are proper perfect morphisms, so these operations send perfect kernels to perfect kernels. The fixed adjunctions give

$$K \star L \simeq \mathcal{O}_{\Delta_B}, \quad L \star K \simeq \mathcal{O}_{\Delta_A}.$$

Conjugation

$$\mathcal{C}_\Phi(M) = K \star M \star L : \text{Perf}(A \times A) \longrightarrow \text{Perf}(B \times B)$$

is an exact enhanced equivalence, with the opposite conjugation as inverse. Let $t : \mathcal{C}_\Phi(\mathcal{O}_{\Delta_A}) \xrightarrow{\sim} \mathcal{O}_{\Delta_B}$ be its counit identification. For $\beta \in HH^r(A)$, define

$$\Phi^{HH}(\beta) = t[r] \circ \mathcal{C}_\Phi(\beta) \circ t^{-1} \in HH^r(B). \quad (5.25)$$

It is a graded isomorphism and preserves our Yoneda convention, because every displayed shifted morphism uses the enhanced shift isomorphism.

For $Q \in \text{Perf}(A)$, let $\Phi_{\text{Ext}} : \text{Ext}_A^2(Q, Q) \xrightarrow{\sim} \text{Ext}_B^2(\Phi Q, \Phi Q)$ be the induced map with the same shift structure. Associativity of convolution, evaluated on Q , gives

$$\chi_{\Phi Q}(\Phi^{HH} \beta) = \Phi_{\text{Ext}}(\chi_Q(\beta)), \quad \beta \in HH^2(A). \quad (5.26)$$

Indeed, evaluation on ΦQ is the composite $K \star \beta \star L \star K$ evaluated on Q . The inverse unit and counit cancel the adjacent $L \star K$ by the adjunction triangle identities. The outer t then gives exactly the morphism obtained by applying Φ to the evaluation on Q . This proves the equality on morphisms, including their shifts, rather than only identifying the dimensions of Ext groups.

Use the fixed HKR maps of Proposition 5.1 to put

$$\Phi^{HT} = (\text{HKR}_B)^{-1} \Phi^{HH} \text{HKR}_A : HT^2(A) \xrightarrow{\sim} HT^2(B).$$

The evaluation square is already established. The contraction map has the corresponding compatibility, with the homological comparison map constructed below.

Proposition 5.2 (Transport of evaluation and contraction). *For the fixed enhanced Fourier–Mukai equivalence $\Phi : \text{Perf}(A) \rightarrow \text{Perf}(B)$, there exists an isomorphism*

$$\Phi_{HH}^H : H\Omega_{-2}(A) \xrightarrow{\sim} H\Omega_{-2}(B)$$

depending only on that enhanced equivalence, not on a choice of perfect complex, such that for every $Q \in \text{Perf}(A)$,

$$\begin{aligned} \Phi_{\text{Ext}} \circ \text{ob}_Q &= \text{ob}_{\Phi Q} \circ \Phi^{HT}, \\ \Phi_{HH}^H \circ c_Q &= c_{\Phi Q} \circ \Phi^{HT}. \end{aligned} \quad (5.27)$$

Here ob and c have the conventions of Proposition 5.1, and Φ_{Ext} is the induced enhanced map on degree-two Ext. Consequently

$$\begin{aligned}\Phi^{HT}(\ker \text{ob}_Q) &= \ker \text{ob}_{\Phi Q}, \\ \Phi^{HT}(\ker c_Q) &= \ker c_{\Phi Q}.\end{aligned}\tag{5.28}$$

In particular $\ker \text{ob}_Q = \ker c_Q$ implies $\ker \text{ob}_{\Phi Q} = \ker c_{\Phi Q}$.

We construct Φ_{HH}^H using perfect Serre pairings and then prove the two squares. The construction will also retain the Chern-character sign of a shift in the equivalence.

5.2.2 Serre pairings and the homological map

We state explicitly the duality used to define homological transport. On a smooth proper variety X , Serre duality is the perfect natural pairing

$$\text{Hom}(P, R)^* \simeq \text{Hom}(R, S_X P), \quad S_X P = P \otimes \omega_X[\dim X],$$

for perfect P, R ; write $\text{Tr}_{X,P} : \text{Hom}(P, S_X P) \rightarrow \mathbb{C}$ for its trace. We use its standard expression as the Serre trace on $\mathcal{O}_X$ after the perfect partial trace. In particular it is supertrace on a shifted complex.

Put

$$\Sigma_A = \Delta_*(\omega_A[d_A]), \quad \Sigma_A^{-1} = \Delta_!\mathcal{O}_A = \Delta_*(\omega_A^{-1}[-d_A]).$$

Here $\Delta_! = S_{A \times A}^{-1} \Delta_* S_A$ is left adjoint to $L\Delta^*$. This is the adjunction obtained by applying Serre duality to $L\Delta^* \dashv \Delta_*$. The displayed formula for $\Delta_!\mathcal{O}_A$ follows from $\Delta^*\omega_{A \times A} = \omega_A^{\otimes 2}$. Define Hochschild homology by

$$HH_m(A) = \text{Hom}_{A \times A}(\Sigma_A^{-1}[m], \mathcal{O}_{\Delta_A}) \simeq \mathbb{H}^{-m}(A, H_A),\tag{5.29}$$

where the isomorphism is this adjunction. For $\beta \in HH^2(A)$ and $h : \Sigma_A^{-1}[m] \rightarrow \mathcal{O}_{\Delta_A}$, set

$$\beta \cap h = (\beta \circ h)[-2] \in HH_{m-2}(A).$$

Naturality of the adjunction in (5.29) identifies this with the action of $L\Delta^*\beta$ used in (5.16).

For $m = 0, -2$, let

$$\mathcal{T}^m(A) = \text{Hom}_{A \times A}(\mathcal{O}_{\Delta_A}, \Sigma_A[m]).$$

The Serre functor of $A \times A$ takes $\Sigma_A^{-1}[m]$ to $\Sigma_A[m]$. Therefore Serre duality gives the perfect pairing

$$\langle h, \delta \rangle_{A,m} = \text{Tr}_{A \times A, \Sigma_A^{-1}[m]}(\delta \circ h) : HH_m(A) \otimes \mathcal{T}^m(A) \longrightarrow \mathbb{C}.\tag{5.30}$$

This kernel description and its categorical character are the ones studied in [8, Theorems 6 and 10, Section 6.2]. We give the required transport and its cap verification in these pairings.

The Grothendieck-duality adjoint formulas for a perfect kernel on smooth proper varieties are

$$K^R = K^\vee \otimes p_A^* \omega_A[d_A], \quad K^L = K^\vee \otimes p_B^* \omega_B[d_B],$$

viewed after exchange of the two factors as kernels from B to A . We use these formulas with their adjunction units and counits. They give

$$K^R \simeq \Sigma_A \star K^L \star \Sigma_B^{-1}.$$

For an equivalence the fixed inverse L carries both inverse adjunctions. Taking the mate of the last isomorphism with these adjunctions gives the kernel Serre isomorphism

$$\theta : K \star \Sigma_A \xrightarrow{\sim} \Sigma_B \star K.$$

Its evaluation is the trace-preserving Serre isomorphism: for $f : P \rightarrow S_A P$,

$$\mathrm{Tr}_{B, \Phi P}(\theta_P \circ \Phi f) = \mathrm{Tr}_{A, P}(f). \quad (5.31)$$

To see why the coherence matters, transport the natural Serre duality pairing through the equivalence. Its uniqueness determines the isomorphism $\Phi S_A \rightarrow S_B \Phi$ making those pairings commute; evaluation at the identity gives (5.31). The adjoint formulas with the same units and counits realize this isomorphism by θ . An isomorphism between K^R and K^L as unstructured objects would not specify it. Smoothness, properness, and perfection are exactly the hypotheses for these duality and adjunction statements.

Let $u : \mathcal{C}_\Phi(\Sigma_A) \xrightarrow{\sim} \Sigma_B$ be $\theta \star L$ followed by $\Sigma_B \star t$, and set

$$\Psi^m(\delta) = u[m] \circ \mathcal{C}_\Phi(\delta) \circ t^{-1} : \mathcal{T}^m(A) \xrightarrow{\sim} \mathcal{T}^m(B) \quad (m = 0, -2).$$

For $\delta \in \mathcal{T}^{-2}(A)$ and $\beta \in HH^2(A)$, associativity and the middle cancellation $t^{-1}t$ give

$$\Psi^0(\delta[2] \circ \beta) = (\Psi^{-2}\delta)[2] \circ \Phi^{HH}(\beta). \quad (5.32)$$

For $\delta \in \mathcal{T}^0(A)$, the same associativity and the inverse adjunction triangle identities give the evaluation identity

$$(\Psi^0\delta)_{\Phi Q} = \theta_Q \circ \Phi(\delta_Q). \quad (5.33)$$

Here $\delta_Q : Q \rightarrow S_A Q$ is evaluation of the Serre-valued diagonal morphism. In this second equality the triangles cancel the adjacent $L \star K$ in the evaluation on Q . Thus both identities use the same t , u , and inverse adjunctions.

Define the homological isomorphisms for $m = 0, -2$ by the perfect pairings:

$$\langle \Phi_*^{HH} h, \Psi^m \delta \rangle_{B, m} = \langle h, \delta \rangle_{A, m} \quad (h \in HH_m(A), \delta \in \mathcal{T}^m(A)). \quad (5.34)$$

Existence, uniqueness, and invertibility follow from the nondegeneracy of (5.30) and the invertibility of Ψ^m .

The homological comparison map asserted in Proposition 5.2 is now defined by

$$\Phi_{HH}^H = \mathbf{I}_B \Phi_*^{HH} \mathbf{I}_A^{-1} : H\Omega_{-2}(A) \xrightarrow{\sim} H\Omega_{-2}(B). \quad (5.35)$$

The adjunctions (5.29) are understood. This map depends only on the fixed enhanced equivalence and may mix the individual Hodge bidegrees within the displayed spaces.

Proof of Proposition 5.2. The first equality in (5.27) follows from (5.26), (5.21), and the definition of Φ^{HT} .

For the second square, we verify compatibility with the cap action and the categorical character. If $\beta \in HH^2(A)$, $h \in HH_0(A)$, and $\delta \in \mathcal{T}^{-2}(A)$, the definition of cap and the trace of a shifted perfect object give

$$\langle \beta \cap h, \delta \rangle_{A, -2} = \langle h, \delta[2] \circ \beta \rangle_{A, 0}.$$

The shift is even, so its supertrace sign is $+1$. Applying this on A and B , and using (5.32) and (5.34), gives

$$\begin{aligned}\langle \Phi_*^{HH}(\beta \cap h), \Psi^{-2}\delta \rangle_{B,-2} &= \langle \beta \cap h, \delta \rangle_{A,-2} \\ &= \langle h, \delta[2] \circ \beta \rangle_{A,0} \\ &= \langle \Phi_*^{HH}h, (\Psi^{-2}\delta)[2] \circ \Phi^{HH}\beta \rangle_{B,0} \\ &= \langle \Phi^{HH}\beta \cap \Phi_*^{HH}h, \Psi^{-2}\delta \rangle_{B,-2}.\end{aligned}$$

The tests $\Psi^{-2}\delta$ exhaust $\mathcal{T}^{-2}(B)$. Nondegeneracy proves the needed transport, and only in this degree:

$$\Phi_*^{HH}(\beta \cap h) = \Phi^{HH}(\beta) \cap \Phi_*^{HH}(h), \quad \beta \in HH^2(A), \quad h \in HH_0(A). \quad (5.36)$$

Let $\mathfrak{h}_Q \in HH_0(A)$ correspond, under (5.29), to $h_Q = \text{Tr}_{Q, H_A}(AH_Q)$. We verify its trace characterization using the same AH_Q as in Proposition 5.1. For $\delta : \mathcal{O}_\Delta \rightarrow \Sigma_A$, let $\delta' : H_A \rightarrow \omega_A[d_A]$ be its adjoint, so $\delta = \Delta_*\delta' \circ \eta_\Delta$. The adjunction $\Delta_! \dashv L\Delta^*$, defined by Serre duality above, is trace-compatible. Thus (5.24) and the standard partial-trace expression of the Serre trace give

$$\begin{aligned}\langle \mathfrak{h}_Q, \delta \rangle_{A,0} &= \text{Tr}_{A, \mathcal{O}_A}(\delta' \circ h_Q) \\ &= \text{Tr}_{A, \mathcal{O}_A}(\text{Tr}_Q((1_Q \otimes \delta') \circ AH_Q)) \\ &= \text{Tr}_{A, Q}(\delta_Q).\end{aligned} \quad (5.37)$$

The last equality uses evaluation of $\Delta_*\delta' \circ \eta_\Delta$. It includes the supertrace when Q is shifted. Because the pairing is perfect, this characterizes $\mathfrak{h}_Q$ uniquely; it is the categorical Chern character in $HH_0(A)$. Its image under $\mathbf{I}_A$ is $\text{ch}_H(Q)$ by (5.20).

For a test $\delta_B = \Psi^0\delta_A$, equations (5.34), (5.37), (5.33), and (5.31) yield

$$\begin{aligned}\langle \Phi_*^{HH}\mathfrak{h}_Q, \delta_B \rangle_{B,0} &= \langle \mathfrak{h}_Q, \delta_A \rangle_{A,0} \\ &= \text{Tr}_{A, Q}((\delta_A)_Q) \\ &= \text{Tr}_{B, \Phi Q}((\delta_B)_{\Phi Q}) \\ &= \langle \mathfrak{h}_{\Phi Q}, \delta_B \rangle_{B,0}.\end{aligned}$$

The tests exhaust $\mathcal{T}^0(B)$, so uniqueness gives

$$\Phi_*^{HH}(\mathfrak{h}_Q) = \mathfrak{h}_{\Phi Q}. \quad (5.38)$$

Now take $\alpha \in HT^2(A)$. By (5.16), the class $c_Q(\alpha)$ is the $\mathbf{I}_A$ -image of $\text{HKR}_A(\alpha) \cap \mathfrak{h}_Q$. Equations (5.36) and (5.38) take its homological transport to

$$\Phi^{HH}(\text{HKR}_A(\alpha)) \cap \mathfrak{h}_{\Phi Q}.$$

Apply $\mathbf{I}_B$ and (5.16) on B . The result is $c_{\Phi Q}(\Phi^{HT}\alpha)$, as required. The three comparison maps in (5.27) are isomorphisms. Taking kernels proves (5.28). $\square$

Source identification and shifts. The Serre-pairing definition also makes the source identification of homological transport explicit. Let $\theta_{\mathcal{C}}$ be the trace-preserving Serre isomorphism of the equivalence $\mathcal{C}_\Phi$ between the two kernel categories. There is a unique

$$v : \mathcal{C}_\Phi(\Sigma_A^{-1}) \xrightarrow{\sim} \Sigma_B^{-1}$$

such that

$$u = S_{B \times B}(v) \circ (\theta_C)_{\Sigma_A^{-1}} : \mathcal{C}_\Phi(\Sigma_A) \longrightarrow \Sigma_B.$$

Indeed, apply the inverse Serre functor to $u \circ (\theta_C)_{\Sigma_A^{-1}}^{-1}$. Preservation of the Serre trace by θ_C then shows that (5.34) is exactly

$$h \longmapsto t \circ \mathcal{C}_\Phi(h) \circ v[m]^{-1}.$$

The source identification is obtained from Serre and left–right adjunction coherence, whereas the target identification is t .

The distinction fixes the sign of a shift. Test it in $\text{Perf}(\mathbb{C})$ with $K = \mathbb{C}[s]$ and $L = \mathbb{C}[-s]$. The categorical trace of the identity of $\mathbb{C}[s]$ is $\text{ev} \circ \text{braid} \circ \text{coev} = (-1)^s$. Thus (5.31) forces $\theta = (-1)^s$ in this test. It follows that Ψ^0 is multiplication by $(-1)^s$, and (5.34) makes Φ_*^{HH} on HH_0 multiplication by $(-1)^s$ as well. In agreement with perfect trace,

$$\text{ch}_H(Q[s]) = (-1)^s \text{ch}_H(Q).$$

In particular a shift $[-3]$ reverses this character. Cancelling K and L on both ends of a homology morphism without the Serre source identification would lose this sign. Degree-two evaluation has an even shift, so no additional odd exchange sign occurs there; it still uses the enhanced shift isomorphisms in (5.26).

The transported maps have source HT^2 ; no naturality of σ_Q on arbitrary Ext classes is asserted. The map Φ_{HH}^H is defined by Hochschild homology and the fixed HKR maps, not identified with an independently normalized Betti integral transform.

5.3 The invariant obstruction space of the Abel-orbit seed

Retain the seed of Proposition 4.1. Thus C is a smooth nonhyperelliptic curve of genus three, $X = \text{Pic}^2(C)$ has the specified group origin and is identified with its Jacobian, and $H = \mathcal{O}_X(\Theta)$ is its principal theta line. Put

$$\theta_B = c_{1,B}(H) \in H^2(X, \mathbb{Q}), \quad \int_X \theta_B^3 = 6, \quad n = d + 1 \geq 4.$$

The two subschemes supplied there are disjoint unions

$$A = \coprod_{j=1}^n C_j, \quad \Sigma = \coprod_{j=1}^n \Sigma_j$$

of translates of an Abel curve and of its inverse, respectively. The cyclic groups $G_2, G_1 \subset X$, each of order n , act transitively on the components of A, Σ , respectively. Write

$$\mathcal{I} = I_\Sigma \boxtimes I_A \quad \text{on } X^2, \quad G_{\text{src}} = G_1 \times G_2.$$

Each ideal has its canonical translation linearization as the ideal of an invariant subscheme, and $\mathcal{I}$ has the product linearization. On $M = X \times \widehat{X}$, the fixed transform of Proposition 4.8 is

$$\widetilde{\Phi} = \Phi \circ ((H \boxtimes H) \otimes -), \quad \mathcal{G} = \widetilde{\Phi}(\mathcal{I})[-3].$$

Write $\bar{G} \simeq G_{\text{src}}$ for the underlying translation group of its coherent line-translation action.

We will prove that the semiregularity map of Proposition 5.1 is injective on the full invariant derived group $\text{Ext}_M^2(\mathcal{G}, \mathcal{G})^{\bar{G}}$. The proof has two computations: the invariant group for $\mathcal{I}$ has dimension 48, and contraction with its raw Hodge Chern character has rank 48. The exact Hochschild–Atiyah identity then identifies the entire invariant group with the evaluation image. This calculation concerns the Abel-orbit seed in Markman’s secant-sheaf construction [25, Sections 8.3–8.4]; we give the invariant calculation directly, using the independently movable orbit components.

5.3.1 The Abel class and its normal directions

An *Abel curve* in X will mean a translate of the image of an Abel embedding $C \hookrightarrow X$, or a translate of its image under $[-1]$, with the Jacobian and polarization conventions of Section 4.

Lemma 5.3. *Let $D \subset X$ be an Abel curve. Its normalized Betti class and Chern character are*

$$[D]_{\text{B}} = \frac{\theta_{\text{B}}^2}{2}, \quad \text{ch}_{\text{B}}(\mathcal{O}_D) = \frac{\theta_{\text{B}}^2}{2} - \frac{\theta_{\text{B}}^3}{3}.$$

Normal restriction of translation-invariant vector fields is an isomorphism

$$H^0(X, T_X) \xrightarrow{\sim} H^0(D, N_{D/X});$$

both spaces have dimension three. If $Z = \coprod_{j=1}^n D_j$ is a disjoint union of n Abel curves, then its ideal satisfies the full normalized Betti identity

$$\text{ch}_{\text{B}}(I_Z) = 1 - \frac{n}{2}\theta_{\text{B}}^2 + \frac{n}{3}\theta_{\text{B}}^3. \quad (5.39)$$

Proof. The fundamental-class formula is (4.6), proved there from the Jacobian lattice. Translation acts trivially on cohomology, and inversion acts by $+1$ on H^4 , so it holds for every Abel curve in the stated sense.

For the inclusion $i : D \hookrightarrow X$, use Grothendieck–Riemann–Roch in its smooth proper form

$$\text{ch}(i_*F) \text{td}(X) = i_*(\text{ch}(F) \text{td}(D))$$

for a closed immersion of smooth projective varieties; this is the form of the Riemann–Roch theorem recorded by Borel and Serre [3]. The tangent bundle of X is trivial, and $\text{td}(D) = 1 + \frac{1}{2}c_1(T_D)$. Since $\deg T_D = 2 - 2 \cdot 3 = -4$ and $[\text{pt}]_{\text{B}} = \theta_{\text{B}}^3/6$, it follows that

$$\text{ch}_{\text{B}}(\mathcal{O}_D) = i_* \text{td}(D) = [D]_{\text{B}} - 2[\text{pt}]_{\text{B}} = \frac{\theta_{\text{B}}^2}{2} - \frac{\theta_{\text{B}}^3}{3}.$$

For disjoint Z , one has $\mathcal{O}_Z = \bigoplus_j \mathcal{O}_{D_j}$. Additivity in $0 \rightarrow I_Z \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_Z \rightarrow 0$ gives (5.39).

It remains to calculate the normal sections. For the original Abel embedding the normal sequence is

$$0 \longrightarrow T_C \longrightarrow \mathcal{O}_C \otimes H^1(C, \mathcal{O}_C) \longrightarrow N_{C/X} \longrightarrow 0. \quad (5.40)$$

Here the middle bundle is $T_X|_C$. The dual of the first arrow is the evaluation map $H^0(C, K_C) \otimes \mathcal{O}_C \rightarrow K_C$: differentiating the Abel integral sends a tangent vector v to the functional $\omega \mapsto \omega(v)$. Consequently, under Serre duality on C , the dual of the map

$$H^1(C, T_C) \longrightarrow H^1(C, \mathcal{O}_C) \otimes H^1(C, \mathcal{O}_C)$$

in the long exact sequence of (5.40) is the multiplication map

$$H^0(C, K_C) \otimes H^0(C, K_C) \longrightarrow H^0(C, K_C^2).$$

This is the usual cohomological Serre duality [36], applied to the dual evaluation map.

For completeness, multiplication is onto here by an elementary genus-three argument. For an effective divisor D_2 of length two, curve Riemann–Roch, in the form $h^0(L) - h^0(K_C \otimes L^{-1}) = \deg L + 1 - g$, gives

$$h^0(K_C - D_2) = h^0(D_2) = 1.$$

The last equality uses nonhyperellipticity. Two independent sections of a degree-two effective line, after removing their common base divisor, would give a map to $\mathbb{P}^1$ of degree at most two. Degree one would make C rational, and degree two would make it hyperelliptic. Thus K_C separates every length-two subscheme. It has three sections and degree four, so it embeds C as an integral plane quartic. Such a curve is not contained in a conic. Restriction of quadrics therefore makes $\text{Sym}^2 H^0(K_C) \rightarrow H^0(K_C^2)$ injective. Both spaces have dimension six, the second by curve Riemann–Roch and $\deg K_C^2 = 8 > 2g - 2$. The multiplication map is consequently onto, and its dual is injective.

Since $H^0(C, T_C) = 0$, the long exact sequence of (5.40) now identifies $H^0(C, N_{C/X})$ with the middle bundle's constant sections $H^1(C, \mathcal{O}_C) = H^0(X, T_X)$. This is exactly normal restriction of translation-invariant vector fields. Translation and inversion carry the normal sequence and this isomorphism to every Abel curve. $\square$

5.3.2 The full invariant derived Ext group

We next compute the derived deformation space of one orbit ideal. The calculation keeps the local-to-global contribution of the normal bundle; it does not replace derived Ext by the cohomology of an endomorphism sheaf.

Lemma 5.4. *Let $\Gamma \subset X$ be a finite subgroup of order n , and let $Z = \coprod_{j=1}^n D_j$ be a disjoint union of Abel curves permuted transitively by Γ . Give I_Z its canonical linearization. There is an isomorphism of Γ -representations*

$$\text{Ext}_X^1(I_Z, I_Z) \simeq H^1(X, \mathcal{O}_X) \oplus \bigoplus_{j=1}^n H^0(X, T_X). \quad (5.41)$$

The first summand is trivial, and Γ acts on the second by its regular permutation action on the components and the identity on each translation space. In particular,

$$\dim_{\mathbb{C}} \text{Ext}_X^1(I_Z, I_Z)^{\Gamma} = 6, \quad \dim_{\mathbb{C}} \text{Ext}_X^2(I_Z, I_Z)^{\Gamma} = 6, \quad \text{Hom}_X(I_Z, I_Z)^{\Gamma} = \mathbb{C}.$$

Proof. At a point of Z , write its regular codimension-two ideal in the regular local ring $R = \mathcal{O}_{X,x}$ as $I = (f, g)$, where f, g are part of a regular system of parameters. It has the resolution

$$0 \longrightarrow R \xrightarrow{(-g, f)} R^2 \xrightarrow{(f, g)} I \longrightarrow 0.$$

If an endomorphism sends f, g to u, v , its relation is $gu = fv$. Regularity of f, g gives $u = fr$, $v = gr$ for a unique $r \in R$, so $\text{Hom}_R(I, I) = R$. Applying $\text{Hom}_R(-, I)$ to the resolution shows that the degree-one cokernel is I/I^2 . Under a change of generators, the dual of the leftmost generator acquires the inverse determinant. Consequently the global degree-one sheaf is

$$(I_Z/I_Z^2) \otimes \det(I_Z/I_Z^2)^{-1} \simeq (I_Z/I_Z^2)^{\vee} = N_{Z/X}.$$

The rank-two isomorphism sends $v \otimes \ell^{\vee}$ to the functional $w \mapsto \ell^{\vee}(v \wedge w)$, so it is compatible with changes of generators. We view $N_{Z/X}$ here as a sheaf on X supported on Z . Away from Z the ideal is $\mathcal{O}_X$, and the resolution has length one. Thus, globally,

$$\text{Hom}_X(I_Z, I_Z) = \mathcal{O}_X, \quad \mathcal{E}xt_X^1(I_Z, I_Z) = N_{Z/X}, \quad \mathcal{E}xt_X^q(I_Z, I_Z) = 0 \quad (q \geq 2). \quad (5.42)$$

The standard local-to-global spectral sequence

$$E_2^{p,q} = H^p(X, \mathcal{E}xt_X^q(I_Z, I_Z)) \implies \text{Ext}_X^{p+q}(I_Z, I_Z)$$

is the hypercohomology spectral sequence of $R\mathcal{H}om_X(I_Z, I_Z)$. Its five-term sequence and (5.42) give

$$0 \longrightarrow H^1(X, \mathcal{O}_X) \longrightarrow \mathrm{Ext}_X^1(I_Z, I_Z) \xrightarrow{e} H^0(Z, N_{Z/X}) \longrightarrow H^2(X, \mathcal{O}_X). \quad (5.43)$$

The first arrow is also the map induced by the perfect unit $\mathcal{O}_X \rightarrow R\mathcal{H}om(I_Z, I_Z)$: the perfect trace composed with this unit is multiplication by the rank, which is one.

Translate the n curves independently. The locus in X^n where these translated proper curves remain disjoint is an open neighborhood of the original tuple. Over it their union is a disjoint union of flat translated families. Its ideal is flat as well, by the exact sequence from the structure sheaf of the union. Restriction of this family to first-order tangent vectors gives a linear map

$$s : \bigoplus_{j=1}^n H^0(X, T_X) \longrightarrow \mathrm{Ext}_X^1(I_Z, I_Z).$$

Indeed a flat deformation over $\mathbb{C}[\epsilon]/(\epsilon^2)$ gives the extension $0 \rightarrow I_Z \rightarrow I_{Z,\epsilon} \rightarrow I_Z \rightarrow 0$. For a vector field ξ , the moving generators are $f - \epsilon\xi(f)$ and $g - \epsilon\xi(g)$. Lifting the displayed relation $(-g, f)$ gives

$$(-g)(f - \epsilon\xi(f)) + f(g - \epsilon\xi(g)) = \epsilon(g\xi(f) - f\xi(g)).$$

Thus the local Ext cocycle is represented by $\overline{g\xi(f) - f\xi(g)} \in I/I^2$, up to the single sign in the chosen Ext boundary convention. Choose $\ell = f \wedge \bar{g}$ and $\ell^\vee(\ell) = 1$. Under the written determinant identification $v \otimes \ell^\vee \mapsto (w \mapsto \ell^\vee(v \wedge w))$, this displayed representative sends $\bar{f}, \bar{g}$ to $-\xi(\bar{f}), -\xi(\bar{g})$, respectively. Reversing the boundary convention reverses both signs. Consequently the edge map e sends the geometric deformation to its normal restriction up to one fixed sign, independent of the vector field and the component. By Lemma 5.3, $e \circ s$ is therefore the direct sum of the normal-restriction isomorphisms multiplied by that fixed sign. It follows that (5.43) reduces to the split short exact sequence

$$0 \longrightarrow H^1(X, \mathcal{O}_X) \longrightarrow \mathrm{Ext}_X^1(I_Z, I_Z) \xrightarrow{e} H^0(Z, N_{Z/X}) \longrightarrow 0,$$

with splitting given by s after this signed normal-restriction identification.

This splitting is equivariant. Translation acts trivially on $H^1(X, \mathcal{O}_X)$, as is seen from translation-invariant Dolbeault representatives, and it acts as the identity on $H^0(X, T_X)$. It sends the deformation of D_j to the deformation of its translated component. Since a group of order n acts transitively on n components, that permutation action is regular. This proves (5.41); its invariant summands have dimensions three and three.

We also use Serre duality for perfect complexes on a smooth projective threefold:

$$\mathrm{Ext}^i(F, G)^\vee \simeq \mathrm{Ext}^{3-i}(G, F \otimes \omega_X).$$

This is the extension of the usual locally free Serre duality [36] by finite locally free resolutions. Choose a nonzero translation-invariant holomorphic three-form on X . For $F = G = I_Z$, the resulting perfect pairing is

$$\mathrm{Ext}^1(I_Z, I_Z) \otimes \mathrm{Ext}^2(I_Z, I_Z) \longrightarrow H^3(X, \mathcal{O}_X) \simeq \mathbb{C}.$$

It is Γ -invariant: translations preserve the chosen form, composition, and the perfect trace. Thus these two Ext groups are dual Γ -representations. Averaging over the finite group in characteristic zero identifies invariants of the dual with the dual of the invariants, so the invariant Ext^2 also has dimension six. Finally (5.42) gives $\mathrm{Hom}(I_Z, I_Z) = H^0(X, \mathcal{O}_X) = \mathbb{C}$, with trivial action. $\square$

Proposition 5.5. *For the source object $\mathcal{I} = I_\Sigma \boxtimes I_A$ and its canonical $G_{\text{src}} = G_1 \times G_2$ linearization,*

$$\dim_{\mathbb{C}} \text{Ext}_{X^2}^2(\mathcal{I}, \mathcal{I})^{G_{\text{src}}} = 6 + 36 + 6 = 48.$$

This is the invariant subspace of the full derived Ext group.

Proof. The ideal sheaves are perfect by their local length-one resolutions. The perfect Künneth isomorphism for their derived global Hom complexes is

$$\mathbf{R} \text{Hom}_{X^2}(\mathcal{I}, \mathcal{I}) \simeq \mathbf{R} \text{Hom}_X(I_\Sigma, I_\Sigma) \otimes_{\mathbb{C}}^{\mathbf{L}} \mathbf{R} \text{Hom}_X(I_A, I_A).$$

Taking degree two gives

$$\begin{aligned} \text{Ext}_{X^2}^2(\mathcal{I}, \mathcal{I}) &\simeq \text{Ext}_X^2(I_\Sigma, I_\Sigma) \otimes \text{Hom}_X(I_A, I_A) \\ &\quad \oplus \text{Ext}_X^1(I_\Sigma, I_\Sigma) \otimes \text{Ext}_X^1(I_A, I_A) \\ &\quad \oplus \text{Hom}_X(I_\Sigma, I_\Sigma) \otimes \text{Ext}_X^2(I_A, I_A). \end{aligned} \tag{5.44}$$

Here the isomorphism follows by taking cohomology of the tensor product over $\mathbb{C}$ of the two derived global Hom complexes; over the field $\mathbb{C}$ there are no additional Tor terms. It is equivariant for the independent actions of G_1, G_2 . Averaging in each factor gives $(V_1 \otimes V_2)^{G_1 \times G_2} = V_1^{G_1} \otimes V_2^{G_2}$. Applying Lemma 5.4 to the three summands of (5.44) gives 6, $6 \cdot 6$, and 6. Because (5.44) is a decomposition of derived Ext itself, the count retains every local-to-global contribution. $\square$

5.3.3 The rank of contraction with the raw Hodge character

We now compute the map that Proposition 5.1 factors through this 48-dimensional invariant group. Use the gradings (5.1); in particular, for an abelian variety B ,

$$H\Omega_{-k}(B) = \bigoplus_{q-p=k} H^q(B, \Omega_B^p) \quad (k \in \mathbb{Z}).$$

For a perfect object Q on B , extend character contraction to every degree $k \geq 0$ by

$$c_Q^k : HT^k(B) \longrightarrow H\Omega_{-k}(B), \quad c_Q^k(\alpha) = \alpha \lrcorner \text{ch}_H(Q).$$

Here ch_H is the raw first-jet character (5.7), and contraction uses the left-interior action (5.3), including its Dolbeault Koszul sign. Thus $c_Q^2 = c_Q$ in Proposition 5.1.

The normalization of the character matters for that identity. Put $\theta_H = a_H \in H^1(X, \Omega_X^1)$. Under the raw de Rham comparison,

$$\theta_H \longmapsto (2\pi i)\theta_B, \quad \text{ch}_{H,k}(Q) \longmapsto (2\pi i)^k \text{ch}_{B,k}(Q) \quad \text{in codimension } k. \tag{5.45}$$

This is the weightwise dictionary of Proposition 5.1; the raw first-jet class is the negative of the Buchweitz–Flenner Atiyah convention. In particular Lemma 5.3 gives for either orbit ideal

$$f_n := \text{ch}_H(I_Z) = 1 - \frac{n}{2}\theta_H^2 + \frac{n}{3}\theta_H^3. \tag{5.46}$$

There is no codimension-one component. Each term in (5.46) has the raw weight specified by (5.45).

Use the abelian Hodge-cohomology identifications

$$\mathbf{U} = H^1(X, \mathcal{O}_X), \quad \mathbf{W} = H^0(X, \Omega_X^1), \quad \mathbf{T} = H^0(X, T_X) = \mathbf{W}^\vee, \quad H^q(X, \Omega_X^p) = \wedge^q \mathbf{U} \otimes \wedge^p \mathbf{W}.$$

These follow from the invariant Dolbeault forms on the complex torus. The spaces $\mathbf{U}, \mathbf{W}, \mathbf{T}$ each have dimension three. The polarization class $\theta_{\mathbf{H}} \in \mathbf{U} \otimes \mathbf{W}$ is nondegenerate, so in the fixed contraction convention

$$j : \mathbf{T} \longrightarrow \mathbf{U}, \quad j(\xi) = \xi \lrcorner \theta_{\mathbf{H}}$$

is an isomorphism.

Lemma 5.6. *For either orbit ideal I_Z , the map $c_{I_Z}^1$ is injective of rank six, and $c_{I_Z}^2$ is surjective of rank six.*

Proof. An element of $HT^1(X) = \mathbf{U} \oplus \mathbf{T}$ is a pair (η, ξ) . Using (5.46), its projections under $c_{I_Z}^1$ to $H^1(\mathcal{O}_X)$ and $H^2(\Omega_X^1)$ are respectively

$$\eta, \quad -n j(\xi) \wedge \theta_{\mathbf{H}}. \quad (5.47)$$

The first uses the rank term of the character; the second uses $-n\theta_{\mathbf{H}}^2/2$. There is no contribution from ξ to the first projection or from η to the second, because the codimension-one character is zero. The signs in (5.47) are those of the specified left-interior convention.

The map $v \mapsto v \wedge \theta_{\mathbf{H}}$ on $\mathbf{U}$ is injective. If $v \neq 0$, choose bases with $v = u_1$ and $\theta_{\mathbf{H}} = \sum_{i=1}^3 u_i \otimes w_i$, absorbing the fixed contraction signs in the dual basis. The terms of $v \wedge \theta_{\mathbf{H}}$ with $i = 2, 3$ are linearly independent nonzero tensors. Therefore the two projections in (5.47) determine η and ξ , proving the first assertion.

The only summands of $H\Omega_{-2}(X)$ are

$$H^2(X, \mathcal{O}_X) = \wedge^2 \mathbf{U}, \quad H^3(X, \Omega_X^1) = \wedge^3 \mathbf{U} \otimes \mathbf{W}. \quad (5.48)$$

Each has dimension three. The $H^2(\mathcal{O}_X) = \wedge^2 \mathbf{U}$ block of $HT^2(X)$ maps identically to the first summand, using the rank term $f_{n,0} = 1$. The $H^1(T_X) = \mathbf{U} \otimes \mathbf{T}$ block has no contribution there, since $f_{n,1} = 0$, and its projection to the second summand is

$$\eta \otimes \xi \longmapsto -n \eta \wedge j(\xi) \wedge \theta_{\mathbf{H}}. \quad (5.49)$$

For each $k \in \{1, 2, 3\}$, take η and $j(\xi)$ to be the two basis elements u_i, u_j complementary to u_k . Their wedge with $\theta_{\mathbf{H}}$ is a nonzero multiple of $(u_1 \wedge u_2 \wedge u_3) \otimes w_k$. Hence (5.49) spans the second summand. These two source blocks already give rank six; the target in (5.48) has dimension six. This proves surjectivity. The bivector block $H^0(\wedge^2 T_X)$ is not needed for this lower bound. $\square$

Lemma 5.7. *For $\mathcal{I} = I_{\Sigma} \boxtimes I_A$, the map*

$$c_{\mathcal{I}}^2 : HT^2(X^2) \longrightarrow H\Omega_{-2}(X^2)$$

has rank 48.

Proof. Künneth gives the domain decomposition

$$HT^2(X^2) = (HT^2(X) \otimes HT^0(X)) \oplus (HT^1(X) \otimes HT^1(X)) \oplus (HT^0(X) \otimes HT^2(X)).$$

The product character is $f_n \boxtimes f_n$, and $HT^0(X) = \mathbb{C}$ acts by $c_{I_Z}^0(1) = f_n \neq 0$. The written left-action and product formulas (5.3) and (5.4), applied to the Künneth decomposition, give the external-product contraction rule with its total-complex signs. They therefore send the three blocks onto

$$\mathrm{im}(c_{I_{\Sigma}}^2) \otimes \mathbb{C} f_n, \quad \mathrm{im}(c_{I_{\Sigma}}^1) \otimes \mathrm{im}(c_{I_A}^1), \quad \mathbb{C} f_n \otimes \mathrm{im}(c_{I_A}^2).$$

Their dimensions are 6, 36, 6, by Lemma 5.6. The target has the direct Künneth decomposition

$$H\Omega_{-2}(X^2) = \bigoplus_{r+s=2} H\Omega_{-r}(X) \otimes H\Omega_{-s}(X).$$

The first image lies in the factorwise Hodge-imbalance summand $(2, 0)$, the second in $(1, 1)$, and the third in $(0, 2)$. Here the imbalance of $H^q(X, \Omega_X^p)$ is $q - p$; the two-factor Hodge Künneth decomposition is a direct sum for these ordered pairs. Hence the three images cannot cancel, regardless of their Künneth signs. Their ranks add to $6 + 36 + 6 = 48$. $\square$

5.3.4 The exact rank comparison and the fixed transform

The two calculations now have the same dimension. We use the exact local Hochschild results to turn this equality into injectivity on the whole invariant group. Write $o_Q = \text{ob}_Q$ for evaluation and σ_Q for the perfect-trace map of Proposition 5.1. For a perfect object Q on an abelian variety B , their domains and targets are

$$HT^2(B) \xrightarrow{o_Q} \text{Ext}_B^2(Q, Q) \xrightarrow{\sigma_Q} H\Omega_{-2}(B).$$

Proposition 5.8. *For the source $\mathcal{I} = I_\Sigma \boxtimes I_A$ with its canonical $G_{\text{src}} = G_1 \times G_2$ -linearization,*

$$\text{im}(o_{\mathcal{I}}) = \text{Ext}_{X^2}^2(\mathcal{I}, \mathcal{I})^{G_{\text{src}}}, \quad \ker(o_{\mathcal{I}}) = \ker(c_{\mathcal{I}}^2), \quad (5.50)$$

and $\sigma_{\mathcal{I}}$ is injective on this full invariant group.

Let $M = X \times \widehat{X}$ and use the fixed enhanced equivalence

$$\widetilde{\Phi} = \Phi \circ ((H \boxtimes H) \otimes -), \quad \mathcal{G} = \widetilde{\Phi}(\mathcal{I})[-3].$$

Give $\mathcal{G}$ the coherent line-translation action from Proposition 4.8, and denote its underlying translation group by $\bar{G} \simeq G_{\text{src}}$. Then

$$\text{im}(o_{\mathcal{G}}) = \text{Ext}_M^2(\mathcal{G}, \mathcal{G})^{\bar{G}}, \quad \ker(o_{\mathcal{G}}) = \ker(c_{\mathcal{G}}^2), \quad (5.51)$$

and

$$\sigma_{\mathcal{G}}|_{\text{Ext}_M^2(\mathcal{G}, \mathcal{G})^{\bar{G}}} \text{ is injective.}$$

Proof. Proposition 5.1 proves for every perfect Q the exact identity

$$\sigma_Q \circ o_Q = c_Q^2$$

with the definitions just given. In particular it is an identity for the perfect trace on derived Ext, with the same contraction convention as the rank computation. The components with $q > \dim B - 2$ vanish because $H^{q+2}(B, \Omega_B^q) = 0$.

Translations act trivially on $HT^2(X^2)$: they fix the invariant tangent fields and the invariant Dolbeault representatives of $H^a(\mathcal{O}_{X^2})$. The translation-invariant diagonal-kernel HKR construction identifies this with their trivial action on $HH^2(X^2)$. Apply the diagonal-kernel evaluation square (5.26) to each translation equivalence, and identify its translate of $\mathcal{I}$ with $\mathcal{I}$ by the canonical linearization. Naturality of evaluation then gives $g \cdot o_{\mathcal{I}}(\alpha) = o_{\mathcal{I}}(g \cdot \alpha) = o_{\mathcal{I}}(\alpha)$ for $g \in G_{\text{src}}$. Hence

$$\text{im}(o_{\mathcal{I}}) \subset W_{\text{inv}} := \text{Ext}_{X^2}^2(\mathcal{I}, \mathcal{I})^{G_{\text{src}}}.$$

Proposition 5.5 gives $\dim W_{\text{inv}} = 48$, while Lemma 5.7 and the exact identity give

$$48 = \text{rank}(c_{\mathcal{I}}^2) \leq \text{rank}(o_{\mathcal{I}}) \leq \dim W_{\text{inv}} = 48.$$

Thus the inclusion is equality and both maps have rank 48. The identity also gives $\ker(o_{\mathcal{I}}) \subset \ker(c_{\mathcal{I}}^2)$; equality of ranks on their common domain makes the kernels equal. If $w = o_{\mathcal{I}}(\alpha) \in W_{\text{inv}}$ and $\sigma_{\mathcal{I}}(w) = 0$, then $c_{\mathcal{I}}^2(\alpha) = 0$, so the kernel equality gives $w = 0$. This proves (5.50) and source injectivity.

Apply Proposition 5.2 to the fixed enhanced equivalence $Q \mapsto \tilde{\Phi}(Q)[-3]$. It includes tensoring by $H \boxtimes H$, the fixed Fourier–Mukai kernel, and the displayed shift. That proposition gives isomorphisms

$$u : HT^2(X^2) \xrightarrow{\sim} HT^2(M), \quad v : \text{Ext}^2(\mathcal{I}, \mathcal{I}) \xrightarrow{\sim} \text{Ext}^2(\mathcal{G}, \mathcal{G}), \quad \gamma : H\Omega_{-2}(X^2) \xrightarrow{\sim} H\Omega_{-2}(M)$$

such that

$$v o_{\mathcal{I}} = o_{\mathcal{G}} u, \quad \gamma c_{\mathcal{I}}^2 = c_{\mathcal{G}}^2 u. \quad (5.52)$$

The map γ is the HKR expression of the enhanced Hochschild-homology isomorphism: its cap and character naturality, including the shift sign, is exactly what gives the second equality. No identification of γ with a separately normalized Betti integral transform is used.

Proposition 4.8 constructs the target action by coherent conjugation of the canonical source action by this same equivalence. Hence v is an isomorphism of the corresponding representations. The lines in the target line-translation action cancel on derived endomorphisms, and the underlying group is the specified $\bar{G} \simeq G_{\text{src}}$. Therefore

$$v(W_{\text{inv}}) = \text{Ext}_M^2(\mathcal{G}, \mathcal{G})^{\bar{G}}.$$

The first equality of (5.52) transports the source image equality to the image equality in (5.51). Both u and γ are isomorphisms, so its second equality transports the source kernel equality to the target kernel equality. Finally apply $\sigma_{\mathcal{G}} o_{\mathcal{G}} = c_{\mathcal{G}}^2$, the target instance of Proposition 5.1. An invariant z has the form $o_{\mathcal{G}}(\alpha)$; if $\sigma_{\mathcal{G}}(z) = 0$, the target kernel equality gives $z = 0$. This proves injectivity on the full target invariant Ext group. $\square$

The fixed support construction identifies

$$\mathcal{E} = R\mathcal{H}om_M(\mathcal{G}, \mathcal{O}_M)[-1]$$

as the actual coherent perfect sheaf of positive rank, with the dual coherent action on the same underlying translation group. Proposition 7.2 uses the full injection just proved, transfers it by perfect duality, and then performs finite étale twisted descent. The exact dual sign and that derived descent are left to that proposition. The output here is injection on $\text{Ext}^2(\mathcal{G}, \mathcal{G})^{\bar{G}}$ itself, not only on a chosen Hochschild subspace. No identification of that derived Ext group with $H^2(M, \mathcal{H}^0(R\mathcal{H}om_M(\mathcal{G}, \mathcal{G})))$ is used.

6 The spin character and the polarized quadratic seed

The preceding construction gives a coherent perfect sheaf $\mathcal{E}$ of rank $r = 8d$ on $M = X \times \hat{X}$. Its normalized rational Betti character is

$$\kappa_M = \exp\left(-\frac{c_{1,\text{B}}(\mathcal{E})}{r}\right) \text{ch}_{\text{B}}(\mathcal{E}), \quad (6.1)$$

where ch_B and $c_{1,B}$ are the ordinary rational Betti Chern character and first Chern class. We compute κ_M and identify the polarized quadratic structure on this same sixfold. The calculation uses the secant spinors of Markman [25, Sections 2.4 and 6]. We give a finite Clifford proof of the required transform identity, so the identity applies to the algebraic spin group, including its elements that do not preserve the initial Hodge structure.

6.1 The fixed transform and its source character

Put $\ell_X = 6 = 2g_X$, with $g_X = 3$, and set

$$L = H^1(X, \mathbb{Q}), \quad U = L^\vee = H^1(\widehat{X}, \mathbb{Q}), \quad V = L \oplus U.$$

Choose $e_1, \dots, e_{\ell_X}$ with $\int_X e_1 \wedge \dots \wedge e_{\ell_X} = 1$, and let $f_1, \dots, f_{\ell_X}$ be the dual basis of U . Our normalized Poincaré line has class

$$p = c_{1,B}(\mathcal{P}) = \sum_{i=1}^{\ell_X} e_i \wedge f_i.$$

For an ordered subset $P \subset \{1, \dots, \ell_X\}$, write e_P and f_P for the corresponding increasing products. For disjoint ordered subsets I, J , let $\epsilon_{I,J}$ be the sign in $e_I e_J = \epsilon_{I,J} e_{I \cup J}$, and set it to zero if they overlap. Define reversal by

$$\tau(e_P) = (-1)^{|P|(|P|-1)/2} e_P.$$

We use Grothendieck–Riemann–Roch in the following standard form: if $f : B \rightarrow C$ is a proper morphism of smooth projective complex varieties and F is perfect, then

$$\text{ch}_B(Rf_* F) \text{td}(C) = f_*(\text{ch}_B(F) \text{td}(B)).$$

This is the Riemann–Roch theorem of Borel and Serre [3]. In a Fourier transform between abelian varieties the Todd classes are one. This gives the usual cohomological Poincaré transform; compare the calculation in [17, Section 4.2.3]. The second-factor transform in the fixed functor Φ of (4.29) uses the inverse Poincaré line and the shift $[g_X]$. It is therefore

$$F_-(\alpha) = (-1)^{g_X} (\pi_{\widehat{X}})_* (e^{-p} \pi_X^* \alpha). \quad (6.2)$$

The factor $(-1)^{g_X}$ is the Chern-character sign of the shift.

Lemma 6.1 (The fixed Fourier matrix). *For $P \subset \{1, \dots, \ell_X\}$, $k = |P|$, one has*

$$F_-(e_P) = (-1)^{k(k+3)/2} \epsilon_{P, P^c} f_{P^c}.$$

In particular F_- is a rational isomorphism preserving parity.

Proof. In (6.2), only the complementary indices in e^{-p} can contribute to integration on X . Moving their f -factors past the e -factors gives the coefficient

$$(-1)^{g_X + (\ell_X - k)(\ell_X - k + 1)/2} \epsilon_{P, P^c}.$$

The difference of the displayed exponent and $k(k+3)/2$ is $2(g_X + 1)(g_X - k)$, since $\ell_X = 2g_X$, and hence is even. Every basis vector goes to a signed complementary basis vector, proving the remaining assertions. $\square$

For $\mu(x, y) = (x + y, y)$ define the graded Künneth maps

$$\phi = (1 \otimes F_-)\mu^*, \quad \nu = (F_- \otimes 1)\mu^*, \quad D = F_-^{-1} \otimes F_-, \quad \phi' = \phi(1 \otimes \tau) = D\nu(1 \otimes \tau). \quad (6.3)$$

The input of D is written in the order $U \oplus L$, and its output in the order $L \oplus U$; all identifications use the graded shuffle. These are fixed rational isomorphisms, determined only by the oriented lattice, addition, and p . Equivalently,

$$\phi'(s \otimes t) = (-1)^{gx} \int_{X_y} s(x + y) \tau(t)(y) e^{-\sum_i y_i z_i}.$$

This is finite exterior-algebra coefficient extraction in $y_1 \cdots y_{\ell_X}$; the output variables x, z represent L, U . In particular the maps do not vary with a complex period having the given marking.

We now identify the tensor to which this transform applies. Write $\Theta = c_{1,B}(H) \in \Lambda^2 L$ for the principal class of the seed. Both orbit ideals have $n = d + 1$ disjoint Abel components, so Lemma 5.3 gives their common Betti character

$$f_n = 1 - \frac{n}{2}\Theta^2 + \frac{n}{3}\Theta^3.$$

Thus the sheaves $F_1 = I_A(H)$ and $F_2 = I_\Sigma(H)$ used in Φ have common character

$$v = e^\Theta f_{d+1} = 1 + \Theta - \frac{d}{2}\Theta^2 - \frac{d}{6}\Theta^3.$$

Fix $K = \mathbb{Q}(u)$, where $u^2 = -d$ and $d \geq 3$ is the integer of the seed construction, and put

$$\lambda_+ = e^{u\Theta}, \quad \lambda_- = e^{-u\Theta} \quad \text{in } (\Lambda L)_K.$$

Expanding the exponentials in powers of Θ gives

$$v = \frac{1}{2}[(1 + u^{-1})\lambda_+ + (1 - u^{-1})\lambda_-]. \quad (6.4)$$

The two exponential classes are linearly independent: both have constant term 1, and their degree-two terms are opposite and nonzero. Reversal acts by $(-1)^j$ on Θ^j , so it interchanges λ_+ and λ_- . The rational tensor $T = v \otimes \tau v$ therefore lies in the tensor square of their two-dimensional K -span. Grothendieck–Riemann–Roch gives

$$\beta := \text{ch}_B \Phi(F_2 \boxtimes F_1) = \phi(v \otimes v) = \phi'(T). \quad (6.5)$$

To relate this class to the original sheaf, let ι be the ring involution acting by $(-1)^j$ on $H^{2j}(M, \mathbb{Q})$. The shifts in $\mathcal{G} = \Phi(F_2 \boxtimes F_1)[-3]$ and $\mathcal{E} = D_M(\mathcal{G})[-1]$ give

$$\mathcal{E} = D_M(\Phi(F_2 \boxtimes F_1))[2], \quad \text{ch}_B(\mathcal{E}) = \iota\beta, \quad \beta_0 = r = 8d. \quad (6.6)$$

Here the rank is supplied by Proposition 4.7. We next introduce the Clifford action on ΛL and compare its transport through ϕ' with the graded action on $H^*(M)$. The exponential correction will disappear under the normalization defining κ_M .

6.2 The exact Clifford identity

Give V the symmetric split form

$$Q(e_i, f_j) = \delta_{ij}, \quad Q(e_i, e_j) = Q(f_i, f_j) = 0.$$

Use the Clifford algebra with relation $vw + wv = Q(v, w)$. We use the standard Clifford construction of the spin group in this normalization: the norm-one even Clifford normalizer $\text{Spin}(V, Q)$ acts on V by conjugation through $\rho : \text{Spin}(V, Q) \rightarrow SO(V, Q)$, and acts on a Clifford module by multiplication. The norm is Clifford reversion times the element. For this split rational construction, see [17, Sections 3.2 and 3.4–3.5]. Only this construction, not a realization of arbitrary spin elements by derived autoequivalences, is used here.

On $S = \Lambda L$ the Fock action is

$$c(e) = \epsilon_e = e \wedge -, \quad c(f) = \iota_f.$$

Write $m_g = c(g)$. Conjugation by the fixed ϕ' defines the rational algebraic representation

$$\rho'_g = \phi'(m_g \otimes m_g)(\phi')^{-1} \quad \text{on } \Lambda V.$$

The following calculation is the matrix-unit Clifford realization underlying [17, Proposition 3.2.1(e)], with the signs of the fixed transform retained.

Proposition 6.2 (Fixed Clifford identity). *For every algebraic spin element g , one has the exact equality of rational representations*

$$\rho'_g = e^{p/2} \Lambda \rho(g) e^{-p/2} = e^{(p - \Lambda^2 \rho(g)p)/2} \Lambda \rho(g). \quad (6.7)$$

The exponentials denote exterior multiplication. In particular

$$[\rho'_g(1)]_0 = 1, \quad [\rho'_g(1)]_2 = \frac{1}{2}(p - \Lambda^2 \rho(g)p).$$

Proof. Exterior multiplication and contraction give every matrix unit between the monomial basis vectors of S : first project onto a specified monomial by products of the commuting occupation projections $\epsilon_{e_i} \iota_{f_i}$ and $\iota_{f_i} \epsilon_{e_i}$, and then remove and insert the required generators. The ordered Clifford monomials span at most $2^{2\ell_X}$ dimensions, which equals $\dim \text{End}(S)$. Thus

$$C(V, Q) \simeq \text{End}(S).$$

Its even part is the parity-preserving part: the inclusion follows from parity of the generators, and both sides have dimension $2^{2\ell_X - 1}$. Put $F = f_1 \cdots f_{\ell_X}$ in the Clifford algebra and define

$$\varphi : S \otimes S \longrightarrow C(V, Q), \quad \varphi(s \otimes t) = sF\tau(t).$$

On S this is the rank-one operator

$$c(\varphi(s \otimes t))(r) = (-1)^{g_X} [\tau(t) \wedge r]_{\text{top}} s. \quad (6.8)$$

Indeed F contracts the full oriented monomial with sign $(-1)^{\ell_X(\ell_X - 1)/2} = (-1)^{g_X}$. The pairing $\langle t, r \rangle = [\tau(t) \wedge r]_{\text{top}}$ is perfect. Exterior multiplication is self-adjoint for it by reversal of products. Contraction is self-adjoint as well: apply ι_f to a zero product in exterior degree $\ell_X + 1$ and use

$$\tau \iota_f|_{\Lambda^k L} = (-1)^{k-1} \iota_f \tau|_{\Lambda^k L}.$$

Consequently Clifford reversion is adjunction for this pairing. Every spin element has reversion norm one, and therefore preserves the pairing. Formula (6.8) now proves that φ is an isomorphism intertwining the diagonal spin action with conjugation on $C(V, Q)$.

For any bilinear form B on V with $B(v, v) = Q(v, v)/2$, set

$$c_B(v) = \epsilon_v + \iota_{B(v, -)} \quad \text{on } \Lambda V, \quad \psi_B(x) = c_B(x)1.$$

The Clifford relations follow directly from the exterior and contraction relations. On each ordered Clifford monomial ψ_B has that exterior monomial as its leading term, so ψ_B is a triangular isomorphism. Take

$$B_0(e_i, f_j) = \delta_{ij}, \quad B_0(f_i, e_j) = 0, \quad B_s = Q/2, \quad A_0 = B_0 - B_s = \frac{1}{2} \sum_i e^i \wedge f^i,$$

with other entries of B_0 zero. Here e^i, f^i are the covectors dual to e_i, f_i .

The fixed map ν is exactly the nonsymmetric Clifford symbol:

$$\nu(1 \otimes \tau) = \psi_{B_0} \varphi. \quad (6.9)$$

To check every sign, let $I' = P \setminus I$, and let $j = |J|$. Both sides on $e_P \otimes e_J$ equal

$$(-1)^{j(j-1)/2} \sum_{I \subseteq P} \epsilon_{I', I} \epsilon_{I, J} (-1)^{\sum_{i \in I'} i - |I'|} f_{(I')^c} \wedge e_{I \cup J}. \quad (6.10)$$

For the left side, expand $\mu^* e_P$ and apply Lemma 6.1. For the right side use

$$c_{B_0}(e_i) = \epsilon_{e_i} + \iota_{f^i}, \quad c_{B_0}(f_i) = \epsilon_{f_i},$$

and

$$\iota_{f^{i_1}} \cdots \iota_{f^{i_s}} f_{\{1, \dots, \ell_X\}} = (-1)^{\sum_{i \in I'} i - |I'|} f_{(I')^c}.$$

Moving the remaining e_I past the f 's changes $\epsilon_{I, I'}$ into $\epsilon_{I', I}$. This proves (6.9).

For this finite exterior calculation define $\iota_{\alpha \wedge \beta} = \iota_{\beta} \iota_{\alpha}$ for covectors on V . This fixes the contraction convention locally; the Hochschild contraction used earlier was specified independently. The ordinary commutator identity

$$[\iota_{\alpha \wedge \beta}, \epsilon_v] = \alpha(v) \iota_{\beta} - \beta(v) \iota_{\alpha} = \iota_{(\alpha \wedge \beta)(v, -)}$$

and commutation of even contractions with contractions give

$$c_{B_0}(v) = e^{\iota_{A_0}} c_{B_s}(v) e^{-\iota_{A_0}}, \quad \psi_{B_0} = e^{\iota_{A_0}} \psi_{B_s}.$$

The second equality has no scalar, since $e^{-\iota_{A_0}} 1 = 1$. All exponentials are finite. If $R = \Lambda \rho(g)$, preservation of Q gives

$$R c_{B_s}(v) R^{-1} = c_{B_s}(\rho(g)v), \quad R \psi_{B_s}(x) = \psi_{B_s}(g x g^{-1}).$$

Thus the symmetric symbol is spin-equivariant.

It remains to conjugate ι_{A_0} by D . On the input $\Lambda(U \oplus L)$ fix the orientation $\Omega = f_1 \cdots f_{\ell_X} e_1 \cdots e_{\ell_X}$. The exterior determinant pairing of Q defines PD_Q by

$$(PD_Q \eta, \zeta)_Q = [\eta \wedge \zeta]_{\Omega}.$$

For $|P| = k$, $|J| = j$, and $\delta = k + j$, direct evaluation gives

$$PD_Q(f_J e_P) = (-1)^{kj} \epsilon_{J, J^c} \epsilon_{P, P^c} e_{J^c} f_{P^c}.$$

Applying Lemma 6.1 to the two factors of D therefore yields

$$D|_{\Lambda^\delta V} = (-1)^{g_X + \delta(\delta+3)/2} PD_Q. \quad (6.11)$$

For example, before parity reduction the exponent in the coefficient of $D(f_J e_P)$ is

$$(\ell_X - j)(\ell_X - j + 3)/2 + k(k + 3)/2 + j.$$

Its difference from $g_X + \delta(\delta + 3)/2 + kj$ is even; this verifies the factor order as well as the shift.

Raising a covector by Q is denoted by a superscript $\sharp$. The exterior derivation rule, applied twice, gives

$$PD_Q(\iota_{A_0} \eta) = A_0^\sharp \wedge PD_Q(\eta).$$

The signs in (6.11) in degrees $\delta - 2$ and δ have ratio -1 . Moreover Q raises e^i to f_i and f^i to e_i , so

$$A_0^\sharp = \frac{1}{2} \sum_i f_i \wedge e_i = -p/2, \quad D\iota_{A_0} D^{-1} = \epsilon_{p/2}.$$

Thus $De^{\iota_{A_0}} D^{-1} = e^{p/2}$. The map PD_Q , and hence D , commutes with R : R preserves Q , degree, and orientation, since $\det \rho(g) = 1$. It follows that $F_s = D\psi_{B_s} \varphi$ intertwines the diagonal spin action with R . Combining the displayed identities gives

$$\phi' = D\psi_{B_0} \varphi = e^{p/2} F_s,$$

which proves (6.7). Every map used is a rational algebraic map, so the equality is one of algebraic representations. Applying it to 1 proves the two stated components. $\square$

In particular the degree-two term $a(g) = \frac{1}{2}(p - \Lambda^2 \rho(g)p)$ satisfies

$$a(gh) = a(g) + \Lambda^2 \rho(g)a(h).$$

This follows either by direct substitution or by composing (6.7). It fixes the normalization of the character calculation below.

6.3 The quadratic action and its actual polarization

To identify the polarization preserved by the spinor stabilizers, recall the contraction map of the principal class:

$$\vartheta : U \longrightarrow L, \quad \vartheta(y) = \iota_y \Theta, \quad \iota_y(e \wedge e') = y(e)e' - y(e')e.$$

Thus $y'(\vartheta y) = \Theta(y, y')$ and $\vartheta^t = -\vartheta$. The principal polarization gives an isomorphism

$$\lambda : X \longrightarrow \hat{X}, \quad \lambda(z) = t_z^* H \otimes H^{-1}.$$

The isomorphism in the sheaf construction is $a(z) = t_{-z}^* H \otimes H^{-1} = \lambda(-z)$. For the normalized Poincaré class $p = \sum e_i \wedge f_i$, the signs proved in (4.4) are

$$\lambda^* : U \longrightarrow L \text{ is } \vartheta, \quad a^* = -\vartheta.$$

Under the dual identifications on homology, $\lambda_* = -\vartheta$ and $a_* = \vartheta$. No symmetry of H is required.

The two exponential spinors λ_+, λ_- introduced in (6.4) are exchanged by scalar conjugation. Their span is the scalar extension of the rational plane generated by $(\lambda_+ + \lambda_-)/2$ and $(\lambda_+ - \lambda_-)/(2u)$. Let J be the pointwise algebraic spin stabilizer of this rational plane, and let $\tilde{L}_K$ be the ordered stabilizer of the two individual spinor lines over K .

Proposition 6.3 (The polarized quadratic seed). *On the actual sixfold $M = X \times \hat{X}$, define*

$$\eta(u)(x, N) = (a^{-1}(N), [-d]a(x)) = (-\lambda^{-1}(N), [d]\lambda(x)).$$

This gives a unital K -action. The ample line

$$\mathcal{L}_M = \text{pr}_X^* H^{\otimes d} \otimes \text{pr}_{\hat{X}}^* (a^{-1})^* H$$

has a class $\theta_M = c_{1,B}(\mathcal{L}_M)$ fixed by the vector images of J and $\tilde{L}_K$. The action is of Weil type $(3, 3)$, is compatible with this polarization, and its actual homological Hermitian form is hyperbolic.

More precisely, the pullback $f = \eta(u)^$ on $V = L \oplus U$ and its two eigenspaces are*

$$\begin{aligned} f(w, y) &= (d\vartheta y, -\vartheta^{-1}w), & f^2 &= -d, \\ W_+ &= \text{Ann}(\lambda_+) = \{(-u\vartheta y, y) : y \in U_K\}, \\ W_- &= \text{Ann}(\lambda_-) = \{(u\vartheta y, y) : y \in U_K\}. \end{aligned} \tag{6.12}$$

The eigenvalue on W_+ is $+u$, and that on W_- is $-u$.

Proof. The Fock formula $\iota_y e^{u\Theta} = u\vartheta(y) \wedge e^{u\Theta}$ gives the two annihilators in (6.12). They are complementary because ϑ is invertible. Substitution in the displayed f gives the asserted eigenvalues and $f^2 = -d$. It also gives

$$Q(fv, w) = -Q(v, fw).$$

Let $s_a = 1 \times a : X^2 \rightarrow M$ and $s_\lambda = 1 \times \lambda$. Equation (4.4) gives the complete pullback dictionary

	$s^*(w, y)$	s^*W_+	s^*fs^{*-1} on $L \oplus L$
s_a	$(w, -\vartheta y)$	$\{(uz, z)\}$	$f_a = \begin{pmatrix} 0 & -d \\ 1 & 0 \end{pmatrix}$
s_λ	$(w, \vartheta y)$	$\{(-uz, z)\}$	$f_\lambda = \begin{pmatrix} 0 & d \\ -1 & 0 \end{pmatrix}.$

(6.13)

The homology transpose of f_a is

$$T_a = \begin{pmatrix} 0 & 1 \\ -d & 0 \end{pmatrix},$$

which is induced by the algebraic endomorphism $(x, z) \mapsto (z, [-d]x)$ of X^2 . Conjugating by s_a gives the stated $\eta(u)$, so its pullback is f . Since $\eta(u)^2 = [-d]$, the resulting homomorphism $\mathbb{Q}[u]/(u^2 + d) \rightarrow \text{End}^0(M)$ is unital and injective. The map s_a is holomorphic at the original seed. The graph $\{(uz, z)\}$ has three-dimensional intersections with each of $H^{1,0}(X^2)$ and $H^{0,1}(X^2)$; the same is true of the opposite graph. This proves the Weil multiplicities for the actual action.

To identify its invariant cohomology class, first form the covariant alternating form

$$\Xi(v, w) = Q(fv, w).$$

Let $\xi \in \Lambda^2 V$ be obtained by raising both indices of Ξ with Q . If matrices are written on cohomology, its coefficient matrix is

$$C = Q^{-1}(f^t Q)Q^{-1} = Q^{-1}f^t = -fQ^{-1}. \tag{6.14}$$

This raising is necessary: the covariant form Ξ itself is not the degree-two cohomology class.

Here is its exact value on the seed. Put $B_{ij} = \Theta(f_i, f_j)$. The skew invertible matrix B represents the bivector Θ ; the contraction convention means $\vartheta = -B$ in column coordinates, hence $a^* = B$. In s_a -coordinates,

$$Q_a = \begin{pmatrix} 0 & B^{-1} \\ -B^{-1} & 0 \end{pmatrix}, \quad f_a^t Q_a = \begin{pmatrix} -B^{-1} & 0 \\ 0 & -dB^{-1} \end{pmatrix}, \quad Q_a^{-1} = \begin{pmatrix} 0 & -B \\ B & 0 \end{pmatrix}.$$

Multiplication gives

$$Q_a^{-1}(f_a^t Q_a)Q_a^{-1} = \begin{pmatrix} dB & 0 \\ 0 & B \end{pmatrix}, \quad s_a^* \xi = d\Theta_1 + \Theta_2. \quad (6.15)$$

This is exactly the class of $H^{\otimes d} \boxtimes H$. Consequently $\xi = c_{1,B}(\mathcal{L}_M) = \theta_M$, and $\mathcal{L}_M$ is ample.

For a spin element g , $\text{Ann}(m_g s) = \rho(g) \text{Ann}(s)$, as follows by conjugating the Clifford action. Thus the vector image of the ordered line stabilizer preserves Q and both W_+ and W_- . It commutes with f , and hence

$$\Xi(\rho(g)v, \rho(g)w) = Q(f\rho(g)v, \rho(g)w) = Q(\rho(g)fv, \rho(g)w) = \Xi(v, w).$$

Raising by the invariant Q is equivariant, so it fixes θ_M . The same holds for its subgroup J .

It remains to verify compatibility and hyperbolicity on actual homology. Let E be the principal Riemann form of H on $U = H_1(X, \mathbb{Q})$. By (6.15), the actual Riemann form on $U \oplus U$ is

$$e((v, w), (v', w')) = dE(v, v') + E(w, w').$$

It is positive at the original period because $H^{\otimes d} \boxtimes H$ is ample. On the two factor indices put $D_0 = \text{diag}(d, 1)$. Direct multiplication gives

$$D_0^{-1}T_a^t D_0 = -T_a, \quad T_a^t D_0 T_a = dD_0.$$

These scalar matrices commute with E , so $e(T_a x, y) = -e(x, T_a y)$. For $k = b + cu \in K$ it follows that

$$e(\eta(k)x, y) = e(x, \eta(\bar{k})y), \quad \eta(k)^* \theta_M = (b^2 + dc^2)\theta_M = \text{Nm}_{K/\mathbb{Q}}(k)\theta_M.$$

These are precisely the compatibility identities.

Use the actual second-linear form

$$H_{\text{hom}}(x, y) = e(x, T_a y) + ue(x, y).$$

The skew-adjoint identity, $T_a^2 = -d$, and alternation of e give

$$H_{\text{hom}}(T_a x, y) = -uH_{\text{hom}}(x, y), \quad H_{\text{hom}}(x, T_a y) = uH_{\text{hom}}(x, y), \quad H_{\text{hom}}(y, x) = \overline{H_{\text{hom}}(x, y)}.$$

Choose a rational symplectic basis $v_1, \dots, v_6$ for E with $E(v_{2i-1}, v_{2i}) = b_i \in \mathbb{Q}^\times$. The elements $s_i = (v_i, 0)$ form a K -basis because $T_a s_i = (0, -dv_i)$. Directly on this basis,

$$H_{\text{hom}}(s_i, s_j) = du E(v_i, v_j). \quad (6.16)$$

Its Gram matrix is the orthogonal sum of

$$\begin{pmatrix} 0 & dub_i \\ -dub_i & 0 \end{pmatrix} = \begin{pmatrix} 0 & c_i \\ \bar{c}_i & 0 \end{pmatrix}, \quad c_i = dub_i \neq 0.$$

Every block has an isotropic coordinate line and is a hyperbolic Hermitian plane. This proves hyperbolicity of the actual homological form, and its signature is $(3, 3)$.

For comparison, the auxiliary cohomological form $H_V(v, w) = dQ(v, w) + uQ(fv, w)$ is also second-linear: $H_V(fv, w) = -uH_V(v, w)$ and $H_V(v, fw) = uH_V(v, w)$. The homological conclusion above uses (6.16) directly, without a cohomology-to-homology trace-dual convention. $\square$

6.4 The ordered stabilizer and its exterior invariants

The next calculation identifies exactly which invariant part can carry a Weil class. It also distinguishes fixing the two spinor vectors from merely fixing their lines.

Proposition 6.4 (Ordered spinor stabilizer). *Let $W = W_+$. Under $V_K = W \oplus W^*$, the ordered line stabilizer and the pointwise stabilizer are*

$$\tilde{L}_K = \{(g, t) \in GL(W) \times \mathbb{G}_m : t^2 = \det g\}, \quad m(g, t) = t^{-1} \Lambda g, \quad J_K = \{(g, 1) : g \in SL(W)\}.$$

The characters on λ_+ and λ_- are t, t^{-1} . For $0 \leq j \leq 6$,

$$H^{2j}(M, \mathbb{Q})^J = \begin{cases} \mathbb{Q}\theta_M^j, & j \neq 3, \\ \mathbb{Q}\theta_M^3 \oplus \mathcal{W}_K(M), & j = 3, \end{cases} \quad H^{\text{odd}}(M, \mathbb{Q})^J = 0. \quad (6.17)$$

The ordered line stabilizer fixes every θ_M^j and acts on the two scalar Weil lines by t^2, t^{-2} .

Proof. The two annihilators in (6.12) are complementary maximal isotropic spaces, paired perfectly by Q . Identify the second with W^* . The Clifford algebra on $W \oplus W^*$ acts on ΛW by wedge and contraction. As in the matrix-unit argument above it is the full matrix algebra on this module. Hence this module is isomorphic to S_K as a Clifford module. The common annihilator of all wedge operators from W is the top line $K\omega$, while the common annihilator of contractions from W^* is the vacuum line $K1$. The two original spinor lines therefore correspond to $K\omega, K1$, in that order.

An ordered stabilizer must preserve the two annihilators, so its vector action is $\text{diag}(g, g^{-T})$ for $g \in GL(W)$. Conversely a lift of such a block action preserves both one-dimensional annihilator spaces. The operator $P_g = \Lambda g$ on ΛW implements this block action by direct conjugation of wedge and contraction. It is in the even Clifford algebra because it preserves parity. For the top reversal pairing,

$$\langle P_g s, P_g t \rangle = \det(g) \langle s, t \rangle.$$

Thus its Clifford reversion norm is $\tilde{P}_g P_g = \det g$. Any other Clifford lift of the same vector action differs by a scalar: their quotient commutes with all of the full matrix algebra. A norm-one lift is therefore aP_g with $a^2 \det g = 1$. Writing $t = a^{-1}$ proves the asserted algebraic cover and action. The top and vacuum characters are $t^{-1} \det g = t$ and t^{-1} . Fixing both vectors is exactly $t = 1$, hence exactly $SL(W)$. In particular the extra covering element $(1, -1)$, which negates both spinors, is not in J_K . This is a statement of algebraic groups; it does not assert that every K -rational determinant has a square root in K . The map $t \mapsto (\text{diag}(t^2, 1, \dots, 1), t)$ is an algebraic section of the character t ; in particular t^2, t^{-2} are distinct nontrivial characters.

Now

$$\Lambda^k(V_K) = \bigoplus_{a+b=k} \Lambda^a W \otimes \Lambda^b W^*.$$

The invariant vectors in a summand are the intertwiners $\text{Hom}_{SL(W)}(\Lambda^b W, \Lambda^a W)$. Here is the elementary calculation of those intertwiners. In a basis $w_1, \dots, w_6$, the diagonal torus $\prod t_i = 1$ acts on w_I with weight $\prod_{i \in I} t_i$. Two such weights, for $I, J \subset \{1, \dots, 6\}$, agree on this torus exactly when their indicator vectors differ by a constant multiple of $(1, \dots, 1)$. Thus either $I = J$, or $(I, J) = (\{1, \dots, 6\}, \emptyset)$, or the reverse. For $1 \leq a \leq 5$, any invariant subspace of $\Lambda^a W$ is a sum of its one-dimensional torus weight spaces. The elementary unipotents $w_i \mapsto w_i + cw_j$ connect every two a -subsets by successive replacements, so a nonzero invariant subspace contains all the weight spaces. This proves irreducibility. In an endomorphism commuting with the torus the coefficients

on these weight spaces are diagonal, and commuting with the same unipotents makes them all equal. Consequently the only invariant blocks are

$$a = b, \quad (a, b) = (6, 0), (0, 6),$$

and each is one-dimensional. The balanced identity tensor is the degree-two class $\sum_i w_i \wedge w_i^*$; its j -th power is nonzero and spans the balanced block $a = b = j$. The already constructed nonzero invariant θ_M is a scalar multiple of that class. The other two lines are $\det W, \det W^*$, with ordered-group characters $\det g = t^2$ and $(\det g)^{-1} = t^{-2}$.

Invariants commute with extension of scalars: they are the kernel of the coaction difference, and field extension is flat. The group J and θ_M are rational. The two determinant lines are exchanged by scalar conjugation and descend to the rational Weil plane. They occupy the exterior summands $(a, b) = (6, 0), (0, 6)$ in $\Lambda^a W \otimes \Lambda^b W^*$, whereas θ_M^3 occupies the $(a, b) = (3, 3)$ summand. Thus the sum in degree six is direct. Descent gives (6.17), including the vanishing in odd degrees. $\square$

6.5 The normalized character has a nonzero Weil component

For an even class $\alpha = \sum_{j=0}^6 \alpha_j$, with $\alpha_j \in H^{2j}(M, \mathbb{Q})$ and $\alpha_0 \neq 0$, define

$$N(\alpha) = e^{-\alpha_1/\alpha_0} \alpha.$$

If b has degree two, then $N(e^b \alpha) = N(\alpha)$, because the degree-two term changes by $\alpha_0 b$. If R is a graded algebra automorphism, then $N(R\alpha) = RN(\alpha)$.

Proposition 6.5 (The exceptional component of the actual sheaf). *The normalized character κ_M in (6.1) is a rational J -invariant even class, and its degree-six component has a nonzero rational Weil component:*

$$(\kappa_M)_3 = a_M \theta_M^3 + w_M, \quad a_M \in \mathbb{Q}, \quad 0 \neq w_M \in \mathcal{W}_K(M).$$

Proof. We return to the source tensor $T = v \otimes \tau v$ in (6.5). Expanding (6.4) gives

$$\begin{aligned} T &= \frac{d+1}{4d} (\lambda_+ \otimes \lambda_+ + \lambda_- \otimes \lambda_-) \\ &\quad + \frac{d-1}{4d} (\lambda_+ \otimes \lambda_- + \lambda_- \otimes \lambda_+) \\ &\quad + \frac{u}{2d} (\lambda_- \otimes \lambda_+ - \lambda_+ \otimes \lambda_-). \end{aligned} \tag{6.18}$$

It is rational and fixed by J . It is not fixed by $\tilde{L}_K$: the two square components have the distinct nontrivial characters t^2, t^{-2} , with nonzero coefficient $(d+1)/(4d)$, while the mixed components have character one.

Since T is J -fixed, the identity $\beta = \phi'(T)$ makes β J -fixed for ρ' . Proposition 6.2 and the two normalization identities above show that $N(\beta)$ is J -fixed for $\Lambda\rho$.

Suppose its degree-six component had zero Weil projection. Proposition 6.4 would then put every component of $N(\beta)$ in the power ring $K[\theta_M]$, which is fixed by $\tilde{L}_K$. The class $e^{-p/2}\beta$ is J -fixed for $\Lambda\rho$, by (6.7). Its degree-two term

$$b = \beta_1 - rp/2$$

is therefore in $H^2(M, \mathbb{Q})^J = \mathbb{Q}\theta_M$, and is also fixed by $\tilde{L}_K$. But

$$e^{-p/2}\beta = e^{b/r}N(\beta).$$

Under the supposition the right side is fixed by $\tilde{L}_K$. The exact Clifford identity would make β fixed by ρ' on that group, and the invertibility of ϕ' would make T fixed by the diagonal spin action. This contradicts (6.18). Hence

$$N(\beta)_3 = a\theta_M^3 + w, \quad a \in \mathbb{Q}, \quad 0 \neq w \in \mathcal{W}_K(M).$$

The coefficients are rational because the direct sum (6.17) is rational.

The duality identity (6.6) gives $\kappa_M = \iota N(\beta)$. The involution commutes with the graded vector action and preserves the rational decomposition in (6.17). In degree six it multiplies both terms by -1 , so the Weil component remains nonzero. Renaming the two rational terms proves the proposition. $\square$

6.6 The full normalized character is Hodge on the Weil domain

We now relate the vector stabilizer to the actual polarized complex structures. This uses the polarization matrix itself, so no cohomological Hermitian trace-dual convention is needed.

Proposition 6.6 (The actual polarized Hodge circle). *Keep the rational homology of the seed M , the action η , and the Riemann form of $\mathcal{L}_M$ fixed. At every compatible marked complex structure of Weil type $(3, 3)$, every component $(\kappa_M)_j$ has Hodge type (j, j) .*

Proof. Let C be the coefficient matrix of θ_M on cohomology. It is also the matrix of its alternating Riemann form on dual homology. Equation (6.14) gives

$$C = -fQ^{-1}.$$

For a cohomology automorphism g commuting with the invertible f , this implies

$$\begin{aligned} gCg^t = C &\iff -f(gQ^{-1}g^t) = -fQ^{-1} \\ &\iff gQ^{-1}g^t = Q^{-1} \iff g^tQg = Q. \end{aligned} \tag{6.19}$$

Thus among K -linear cohomology operators, fixing the actual polarization bivector is exactly orthogonality for Q .

Let I be a compatible complex structure on homology and let $T = \eta(u)_*$. Put $h_t = \cos t + \sin t I$, and let $g_t = h_t^t$ be its cohomology pullback. Since $IT = TI$ and $T^t = f$, one has $g_tf = fg_t$. Compatibility $e(Ix, Iy) = e(x, y)$ says $I^tCI = C$. Since $I^2 = -1$, it also gives $I^tC + CI = 0$; expansion of $h_t^tCh_t$ then gives $h_t^tCh_t = C$. This is $g_tCg_t^t = C$. Equation (6.19) places g_t in the centralizer of f in $O(Q)$, and its path from the identity places it in $SO(Q)$.

After the usual complex scalar extension of W_+ , the two Hodge-circle weights have multiplicities three and three. Hence

$$\det(g_t|_{W_+}) = e^{3it}e^{-3it} = 1.$$

Using the inverse circle convention only interchanges the factors. By Proposition 6.4, after extending to K the determinant-one vector centralizer is $SL(W_+)$, and its unique lift fixing both spinors is $(g, 1) \in J_K$. This identity of algebraic groups descends from K . The actual polarized Hodge circle is therefore in the vector image of J .

On $H^{p,q}$ the circle has weight $p - q$, up to simultaneous reversal. A degree- $2j$ class fixed by it has only type (j, j) . Proposition 6.5 makes every component of κ_M J -invariant, proving the assertion at every compatible marked period. $\square$

Only a constant rational transport will be needed for the quotient $q : M \rightarrow Y_0$. Transporting homology, the K -action, and the polarization by q_* identifies the marked complex structures; under these identifications q^* is a constant rational cohomology map and respects every corresponding Hodge decomposition. This observation concerns classes on the marked domain, and requires no relative isogeny or relative descent of the sheaf.

7 Descent and deformation

We now pass from the equivariant sheaf on the Jacobian product to an ordinary sheaf whose full obstruction group can be used in deformation. The first step is coherent twisted descent through the finite quotient and a projective-bundle conversion. The second is the restricted smooth-projective-germ semiregularity theorem needed for that ordinary sheaf.

7.1 Twisted descent and conversion to an ordinary sheaf

The deformation argument will use an ordinary coherent sheaf on a projective family. We first construct a twisted sheaf on the quotient of the abelian sixfold and compare its full derived obstruction group with the invariant obstruction group on the covering sixfold. We then cancel the twist on a relative projective bundle. Throughout, the group action on an endomorphism complex is the conjugation action induced by the given line-twisted action; an action on the sheaf by ordinary automorphisms is not required.

7.1.1 Characters and the duality sign

For a perfect complex P on a smooth complex manifold V , write

$$a_P = -\text{At}_{\text{BF}}(P) \in \text{Ext}_V^1(P, P \otimes \Omega_V^1).$$

This is the raw first-jet Atiyah class with positive line-bundle Chern sign. The opposite sign is the convention of Buchweitz–Flenner [4, Definition 3.4 and Definition 4.1]. Use the Atiyah-power and perfect-trace conventions of Proposition 5.1. Define

$$\begin{aligned} \text{ch}_{H,k}(P) &= \frac{\text{tr}(a_P^k)}{k!}, & \text{ch}_H(P) &= \sum_k \text{ch}_{H,k}(P), \\ \sigma_P(v) &= \left(\text{Tr}_{P, \Omega_V^{j[j+2]}} \left(\left(\frac{a_P^j}{j!} \right) [2] \circ v \right) \right)_{j \geq 0}. \end{aligned} \tag{7.1}$$

Here tr is the perfect trace, and

$$\sigma_P : \text{Ext}_V^2(P, P) \longrightarrow \mathcal{H}_{-2}(V) := \bigoplus_{j \geq 0} H^{j+2}(V, \Omega_V^j).$$

The summands with $j > \dim V - 2$ vanish. These definitions also apply to a perfect weight-one complex on a finite-band gerbe: its derived endomorphisms and their trace have weight zero and hence descend to the underlying manifold.

On a smooth projective V , let $\iota_V : H^q(V, \Omega_V^p) \hookrightarrow H^{p+q}(V, \mathbb{C})$ be the Hodge inclusion for the ordinary de Rham differential d . Our normalized Betti convention is

$$\text{ch}_{B,k}(P) = (2\pi i)^{-k} \iota_V(\text{ch}_{H,k}(P)), \quad \text{ch}_B(P) = \sum_k \text{ch}_{B,k}(P). \tag{7.2}$$

For an ordinary perfect complex this is its rational topological Chern character. For a twisted complex the expression initially defines a complex Betti class; rationality in the present construction will follow from finite descent. The inclusion ι_V is multiplicative and commutes with pullback. On the semiregularity summand $H^{j+2}(\Omega_V^j)$, the corresponding normalization is $(2\pi i)^{-j}\iota_V$: its weight is the form degree j , not half the total cohomological degree. Thus, if $\iota_V(\gamma_H) = (2\pi i)\gamma_B$ for $\gamma_H \in H^1(\Omega_V^1)$, multiplication by γ_H on the raw targets corresponds weightwise to multiplication by γ_B . All identities involving σ will remain in raw Hodge cohomology.

Return to the fixed perfect complex $\mathcal{G}$ on $M = X \times \hat{X}$. Write

$$D(\mathcal{G}) = R\mathcal{H}om_M(\mathcal{G}, \mathcal{O}_M), \quad E = D(\mathcal{G})[-1] = \mathcal{E}.$$

By Proposition 4.7, E is a coherent perfect sheaf of rank $r = 8d > 0$. Proposition 4.8 supplies the coherent line-twisted actions on $\mathcal{G}$ and E , both indexed by G_{src} and identified with the translation subgroup $\bar{G}$ through $g \mapsto b_g$. Duality inverts the line factors and preserves the translations and coherent composition law. Proposition 5.8 proves that $\sigma_{\mathcal{G}}$ is injective on the full group $\text{Ext}_M^2(\mathcal{G}, \mathcal{G})^{\bar{G}}$. The next calculation transfers this injection to E .

Lemma 7.1. *Even transpose induces an isomorphism*

$$d_2 : \text{Ext}_M^2(\mathcal{G}, \mathcal{G})^{\bar{G}} \xrightarrow{\sim} \text{Ext}_M^2(E, E)^{\bar{G}}.$$

With the convention (7.1), its components obey

$$\sigma_{E,j}(d_2 v) = (-1)^{j+1} \sigma_{\mathcal{G},j}(v) \quad \text{for every } j \geq 0. \quad (7.3)$$

Consequently, injectivity of $\sigma_{\mathcal{G}}$ on the full $\bar{G}$ -invariant Ext^2 implies injectivity of σ_E on the full $\bar{G}$ -invariant Ext^2 .

Proof. Perfect duality is an anti-equivalence and turns the given line-twisted action into its inverse-line action. The quasi-isomorphism from $D(\mathcal{G})[-1]$ to the degree-zero sheaf E is canonical in the derived category and is equivariant. Transpose therefore gives the asserted invariant isomorphism in even degree.

The sign can be computed in a finite complex of free modules. The dual Atiyah class is the negative transpose of the original class. Transposing a product reverses both its Ext-degree-one factors and its one-form factors. The two reversal Koszul signs cancel: equivalently, an $\Omega^1[1]$ -valued Atiyah operation has even combined parity. The Ext-degree-two operation v is also even, so its movement past these factors gives no further sign. Supertrace is invariant under transpose, while the shift $[-1]$ changes supertrace by -1 . The j negative dual Atiyah factors and this last sign give (7.3). Evaluation, transpose, and the Atiyah construction respect changes of finite free model, so the local calculation glues. The diagonal operator with entries $(-1)^{j+1}$ is invertible, which proves the final assertion. $\square$

7.1.2 Determinant normalization and derived descent

Let

$$q : M \longrightarrow Y_0 := M/\bar{G}$$

be the finite etale translation isogeny of Proposition 4.8. We write the action on E using pullback translations. For $h = -b_g$, set $L_h = N_g^{-1}$. Since $t_{b_g,*} = t_{-b_g}^* = t_h^*$, the action (4.49) becomes

$$L_h \otimes t_h^* E \simeq E \quad (h \in \bar{G}).$$

These isomorphisms retain the unital associative line compositors of that proposition. Reindexing by inversion on $\bar{G}$ preserves the invariant subspace of the derived endomorphisms.

Proposition 7.2. *There are a class $\alpha \in H^2(Y_0, \mu_r)$, a coherent α -twisted sheaf $\mathcal{B}$ of rank r with $\det \mathcal{B} \simeq \mathcal{O}_{Y_0}$, and an invertible $q^*\alpha$ -twisted sheaf $\mathcal{L}$ on M , such that*

$$q^*\mathcal{B} \simeq E \otimes \mathcal{L}. \quad (7.4)$$

Let $\mathfrak{Y}_0$ be the μ_r -gerbe of α . For every integer i , this isomorphism induces

$$\mathrm{Ext}_{\mathfrak{Y}_0}^i(\mathcal{B}, \mathcal{B}) \simeq \mathrm{Ext}_M^i(E, E)^{\bar{G}}, \quad \mathrm{Ext}_0^i(\mathcal{B}, \mathcal{B}) \simeq \mathrm{Ext}_0^i(E, E)^{\bar{G}}, \quad (7.5)$$

where the subscript 0 denotes the kernel of the perfect trace to $H^i(\mathcal{O})$. Put

$$\lambda_H = \frac{a_{\det E}}{r} = \frac{\mathrm{tr}(a_E)}{r}.$$

Under the first isomorphism ϕ in degree two, the raw Hodge square

$$\begin{array}{ccc} \mathrm{Ext}^2(\mathcal{B}, \mathcal{B}) & \xrightarrow{\phi} & \mathrm{Ext}^2(E, E)^{\bar{G}} \\ \downarrow \sigma_{\mathcal{B}} & & \downarrow e^{-\lambda_H} \sigma_E \\ \mathcal{H}_{-2}(Y_0) & \xrightarrow{q^*} & \mathcal{H}_{-2}(M) \end{array} \quad (7.6)$$

commutes, both on full Ext and on its trace-free summand. In particular, full invariant injectivity for E makes $\mathcal{B}$ semiregular. By Lemma 7.1, full invariant injectivity for $\mathcal{G}$ is sufficient for this conclusion.

Proof. We construct the twisted sheaf before comparing its derived endomorphisms. Choose an evenly covered good analytic cover of Y_0 , refined so that the finitely many transition lines are trivial on the required sheet intersections. Take every sheet over every cover member, so the sheets U_i cover M , and retain its image V_i as a labelled member of the cover. Transport $E|_{U_i}$ to a sheaf B_i on V_i . On $V_i \cap V_j$ for $i < j$, the deck transformation comparing the two sheets and a local trivialization of its line L_h give an isomorphism $\rho_{ij} : B_j \rightarrow B_i$. Use the unit and inverse composition data to impose $\rho_{ii} = \mathrm{id}$ and $\rho_{ji} = \rho_{ij}^{-1}$. On a triple intersection, the coherent composition law cancels the translations; the remaining line trivializations differ by an invertible scalar function. Thus $\rho_{ij}\rho_{jk}\rho_{ki} = c_{ijk} \mathrm{id}$.

We use the determinant functor for perfect complexes in its elementary local form: a finite free complex has the alternating tensor product of the determinants of its terms. It commutes with restriction and isomorphism, and the determinant of $c \mathrm{id}$ on a rank- r perfect complex is c^r . Each B_i is perfect, so $\det B_i$ is a line. Trivialize these lines and let η_{ij} be the determinant of ρ_{ij} ; then $\eta_{ii} = 1$ and $\eta_{ji} = \eta_{ij}^{-1}$. After a further good-cover refinement, choose an r -th root $\tilde{\eta}_{ij}$ of this nowhere-zero holomorphic function for $i < j$, and impose $\tilde{\eta}_{ii} = 1$ and $\tilde{\eta}_{ji} = \tilde{\eta}_{ij}^{-1}$. Set $\rho'_{ij} = \tilde{\eta}_{ij}^{-1} \rho_{ij}$. These maps have determinant one and satisfy $\rho'_{ii} = \mathrm{id}$ and $\rho'_{ji} = (\rho'_{ij})^{-1}$. The coherent composition law gives an invertible holomorphic scalar ϵ_{ijk} defined by

$$\rho'_{ij}\rho'_{jk} = \epsilon_{ijk}\rho'_{ik}.$$

Taking determinants gives $\epsilon_{ijk}^r = 1$, so these scalars are locally constant with values in μ_r . Associativity on a fourfold intersection gives the exact Cech cocycle identity

$$\epsilon_{ijk}\epsilon_{ikl} = \epsilon_{jkl}\epsilon_{ijl}.$$

The repeated-index multipliers are one; scalar multiplication is faithful because B_i has positive rank on a smooth manifold. The resulting μ_r -cocycle defines α , and the sheaves B_i with transitions ρ'_{ij}

define $\mathcal{B}$. Their determinant trivializations glue, so $\det \mathcal{B} \simeq \mathcal{O}_{Y_0}$. Rank, coherence, and perfection are local, and hence pass from E to $\mathcal{B}$. This is the determinant-normalized form of the local construction in [25, Lemma 9.3.6 and equation (9.3.3)]; it uses the perfect determinant, not an exterior power of a locally free sheaf.

Let $\theta_i : (q^* B_i)|_{U_i} \xrightarrow{\sim} E|_{U_i}$ be the local identification from transport. On an actual overlap $U_i \cap U_j$, the comparing deck transformation is the identity, so the coherent line action gives an invertible scalar l_{ij} with

$$\theta_i(q^* \rho'_{ij})\theta_j^{-1} = l_{ij} \text{id}_E.$$

The normalized transitions give $l_{ii} = 1$, $l_{ji} = l_{ij}^{-1}$, and $l_{ij}l_{jk} = (q^* \epsilon_{ijk})l_{ik}$. Use l_{ij} as the transition from the j -th trivial line to the i -th. These lines define the weight-one $q^* \alpha$ -twisted line $\mathcal{L}$ with the same multiplier as $q^* \mathcal{B}$, and the local identifications give (7.4). On translated intersections, conjugation by these identifications is the original conjugation action on the derived endomorphisms of E .

For clarity, a twisted sheaf here is a coherent weight-one sheaf on $\mathfrak{Y}_0$, equivalently the just described Čech data. Weight-zero sheaves and complexes descend uniquely to Y_0 . A coherent twisted sheaf on the smooth manifold is absolutely perfect: in a local trivialization of the gerbe it is a finite module over a regular analytic local ring and has a finite free resolution. Write

$$\mathcal{C}_{\mathcal{B}} = R\mathcal{H}om_{\mathfrak{Y}_0}(\mathcal{B}, \mathcal{B})$$

for its weight-zero descent. It is a perfect complex on Y_0 , locally $\mathcal{B}^\vee \otimes^L \mathcal{B}$.

We use the following derived base-change identity. For an absolutely perfect P , any complex Q' , and any morphism $f : T \rightarrow V$,

$$Lf^* R\mathcal{H}om_V(P, Q') \simeq R\mathcal{H}om_T(Lf^* P, Lf^* Q'). \quad (7.7)$$

Indeed, locally the left side is $Lf^*(P^\vee \otimes^L Q')$; derived pullback is symmetric monoidal and carries the dual of a perfect complex to the dual of its pullback. This also proves the identity on gerbe trivializations. The unit and trace, being coevaluation and evaluation, commute with the same derived pullbacks. This is an assertion about derived pullback; ordinary pullback under a nonflat map is not being substituted.

Apply (7.7) to the étale map q , where ordinary and derived pullback agree. Line cancellation in (7.4) gives

$$q^* \mathcal{C}_{\mathcal{B}} \simeq R\mathcal{H}om(q^* \mathcal{B}, q^* \mathcal{B}) \simeq R\mathcal{H}om_M(E, E). \quad (7.8)$$

Both the line factors and their scalar gluing cancel. The descent datum on the right is consequently the honest conjugation action of $\bar{G}$, although the action on E itself is line-twisted.

For a finite covering, q_* on sheaves is exact: over an evenly covered open set it is the direct sum of the finitely many sheet restrictions. Étale descent gives $\mathcal{C}_{\mathcal{B}} \simeq (q_* q^* \mathcal{C}_{\mathcal{B}})^{\bar{G}}$. Over $\mathbb{C}$, invariants are the split summand defined by $|\bar{G}|^{-1} \sum_{h \in \bar{G}} h$. Applying derived global sections to that summand and using exactness of q_* gives

$$R\Gamma(Y_0, \mathcal{C}_{\mathcal{B}}) \simeq R\Gamma(M, q^* \mathcal{C}_{\mathcal{B}})^{\bar{G}}.$$

Taking cohomology and using (7.8) proves the full Ext comparison in (7.5).

The perfect unit and trace are equivariant maps

$$\mathcal{O}_{Y_0} \xrightarrow{u} \mathcal{C}_{\mathcal{B}} \xrightarrow{\text{tr}} \mathcal{O}_{Y_0}, \quad \text{tr } u = r \text{id}. \quad (7.9)$$

The last equality is the alternating-rank calculation in a finite free model. Thus u/r splits the trace. If $\mathcal{C}_{\mathcal{B},0} = \text{fib}(\text{tr})$, then

$$\mathcal{C}_{\mathcal{B}} \simeq \mathcal{O}_{Y_0} \oplus \mathcal{C}_{\mathcal{B},0}, \quad H^i R\Gamma(Y_0, \mathcal{C}_{\mathcal{B},0}) = \ker(\text{Ext}^i(\mathcal{B}, \mathcal{B}) \rightarrow H^i(\mathcal{O}_{Y_0})).$$

The splitting commutes with (7.8), since derived pullback and tensoring by an invertible line preserve evaluation and coevaluation. Descent of this summand proves the trace-free comparison. In particular, the calculation retains the higher sheaf-Ext terms: it does not replace $H^2 R\Gamma(\mathcal{C}_{\mathcal{B},0})$ by $H^2(\mathcal{E}nd_0(\mathcal{B}))$.

It remains to compare the semiregularity maps. The cotangent target on a μ_r -gerbe is that of its underlying manifold. Equivalently, the scalar triple-gluings are locally constant, so their differentials vanish in first jets. The pullback and tensor rules for Atiyah classes, with the differential $q^* \Omega_{Y_0}^1 \simeq \Omega_M^1$, give

$$q^* a_{\mathcal{B}} = a_E + a_{\mathcal{L}} \text{id}_E, \quad a_{\mathcal{L}} = -\frac{a_{\det E}}{r} = -\lambda_H. \quad (7.10)$$

The second equality follows by tracing the first and using $\det \mathcal{B} = \mathcal{O}_{Y_0}$; the trace of a_E is the Atiyah class of its perfect determinant. The pullback compatibility is also [4, Proposition 3.14]; the tensor rule follows from the Leibniz rule for first jets on a local perfect model. The scalar $-\lambda_H$ commutes with the Atiyah operations in the combined Ext/form grading. Expanding the exponential and applying the natural perfect trace yields

$$q^* \sigma_{\mathcal{B}}(v) = e^{-\lambda_H} \sigma_E(\phi v),$$

which is (7.6). Multiplication by $\lambda_H \in H^1(\Omega_M^1)$ raises both Hodge indices and is nilpotent on $\mathcal{H}_{-2}(M)$; $e^{-\lambda_H}$ therefore has inverse e^{λ_H} . Full invariant injectivity on the right implies full injectivity on the left. The trace-free comparison also shows that injectivity restricted to the invariant trace-free group would suffice: the $j = 0$ component of $\sigma_{\mathcal{B}}$ is the perfect trace itself, so its vanishing first puts a class into the trace-free summand. $\square$

The normalized Betti counterpart of λ_H is

$$\lambda_B = \frac{c_1^B(\det E)}{r}, \quad \iota_M(\lambda_H) = (2\pi i)\lambda_B,$$

where c_1^B is the usual integral Betti Chern class. Applying (7.10) to the character and then using (7.2) weight by weight gives

$$\begin{aligned} q^* \text{ch}_H(\mathcal{B}) &= e^{-\lambda_H} \text{ch}_H(E), \\ q^* \text{ch}_B(\mathcal{B}) &= e^{-\lambda_B} \text{ch}_B(E). \end{aligned} \quad (7.11)$$

The right side of the second equality is rational. The Betti trace q_*^B satisfies $q_*^B q^* = (\deg q) \text{id}$, so $(\deg q)^{-1} q_*^B$ proves that $\text{ch}_B(\mathcal{B})$ is rational as well. No integrality of λ_B is required.

7.1.3 A relative projective bundle carrying an ordinary sheaf

Let $a : Y \rightarrow S$ be a holomorphic abelian scheme of relative dimension six over a sufficiently small contractible representative of a smooth analytic germ, with central fiber Y_0 . Assume that the family is polarized. Locally it admits a line $\mathcal{L}_Y$ representing the polarization and ample relative to a . More precisely, if only the polarization homomorphism was specified, the standard rigidified relative

Picard component of representing lines is a torsor under the smooth dual abelian scheme; a local analytic section through a chosen central representative supplies $\mathcal{L}_Y$. Relative ampleness holds after shrinking. This is the only local choice of a representing polarization line that will be used.

We realize the twisting class by a finite projective representation and extend its projective-space bundle over Y . On the central fiber, the inverse-twisted tautological line cancels the twist of the pulled-back sheaf $\mathcal{B}$. The vanishing of higher cohomology of the structure sheaf on projective space retains the full obstruction group in this conversion.

Proposition 7.3. *For the class α and the semiregular twisted coherent sheaf $\mathcal{B}$ in Proposition 7.2, put $m = r^{12}$ and $N = 6 + m - 1$. There exist a holomorphic $\mathbb{P}^{m-1}$ -bundle $p : Z \rightarrow Y$, a determinant-trivial α -twisted vector bundle Q_0 of rank m on Y_0 , and an inverse-twisted tautological line $\ell_0 = \mathcal{O}_{\mathbb{P}(Q_0)}(1)$, with $Z_0 = \mathbb{P}(Q_0)$. The ordinary line*

$$H = \omega_{Z/Y}^{-1} = \det T_{Z/Y} \quad (7.12)$$

satisfies $H|_{Z_0} \simeq \ell_0^m$ and restricts to $\mathcal{O}_{\mathbb{P}^{m-1}}(m)$ on every p -fiber. The normalized Betti class and raw Hodge class

$$\zeta_{B,s} = \frac{c_1^B(H|_{Z_s})}{m}, \quad \zeta_{H,s} = \frac{a_{H|_{Z_s}}}{m}, \quad \iota_{Z_s}(\zeta_{H,s}) = (2\pi i)\zeta_{B,s} \quad (7.13)$$

have the following properties: ζ_B is a horizontal rational class of type $(1,1)$, and its restriction to every p_s -fiber is the normalized hyperplane class. Every Z_s and p_s is projective algebraic, and $a \circ p$ has a relative ample line.

The sheaf

$$F_0 = p_0^* \mathcal{B} \otimes \ell_0$$

is ordinary coherent on Z_0 , and it is semiregular. More precisely, there is a full derived isomorphism

$$\mathrm{Ext}_{Z_0}^2(F_0, F_0) \simeq \mathrm{Ext}_{\mathfrak{Y}_0}^2(\mathcal{B}, \mathcal{B}) \quad (7.14)$$

under which

$$\sigma_{F_0}(p_0^* v) = e^{\zeta_{H,0}} p_0^* \sigma_{\mathcal{B}}(v). \quad (7.15)$$

If a rational horizontal section $\kappa_B = \sum_{k=0}^6 \kappa_{B,k}$ specializes to $\mathrm{ch}_B(\mathcal{B})$ and has type (k,k) in degree $2k$, then

$$A_B = p^* \kappa_B e^{\zeta_B} = \sum_{k=0}^N A_{B,k} \quad (7.16)$$

is rational, horizontal, and of type (k,k) in degree $2k$. Its raw Hodge section in that degree is the edge of $(2\pi i)^k A_{B,k}$, and its central value is $\mathrm{ch}_{H,k}(F_0)$. Finally, on every fiber,

$$p_{s*}^B(\zeta_{B,s}^{m-1} e^{-\zeta_{B,s}} A_B(s)) = \kappa_B(s), \quad (7.17)$$

where p_{s}^B is normalized Betti fiber integration.*

Proof. We first realize the exact finite-band class α . Set $\Lambda = H_1(Y_0, \mathbb{Z}) \simeq \mathbb{Z}^{12}$. The universal coefficient theorem for the torus gives

$$H^2(Y_0, \mu_r) = \mathrm{Hom}\left(\bigwedge^2 \Lambda, \mu_r\right);$$

there is no Ext term because $H_1(Y_0, \mathbb{Z})$ is free. Let b be the alternating pairing corresponding to α , and put $G_r = \Lambda/r\Lambda$. For an ordered basis $e_1, \dots, e_{12}$ of Λ , define

$$c(x, y) = \prod_{i < j} b(e_i, e_j)^{x_i y_j} \quad (x, y \in G_r).$$

This is a well-defined bimultiplicative cocycle with $c(x, y)c(y, x)^{-1} = b(x, y)$. On the $m = |G_r| = r^{12}$ dimensional vector space with basis $\{v_y : y \in G_r\}$, set

$$T_x v_y = c(x, y)v_{x+y}.$$

Then $T_x T_y = c(x, y)T_{x+y}$, so $x \mapsto [T_x]$ is a projective representation. For $r = 1$ this is the trivial one-dimensional representation.

The multiplication map $[r] : Y \rightarrow Y$ is a finite etale torsor under $Y[r]$. Its differential on the relative tangent bundle is multiplication by r ; it is proper, and every abelian fiber has r^{12} inverse images of a point. Thus it is a proper local biholomorphism of that degree. Contractibility of S trivializes the finite local system $Y[r]$ as G_r , with its central identification. Form Z as the quotient of $Y \times \mathbb{P}^{m-1}$ by the free diagonal action covering translation by G_r , taking the inverse action on the projective coordinate. If local sections of the torsor satisfy $s_j = s_i + x_{ij}$, this convention gives $[s_j, z] = [s_i, [T_{x_{ij}}]z]$. Local sections of the finite covering show that $p : Z \rightarrow Y$ is a holomorphic projective-space bundle with these projective transitions.

We must check the class in $H^2(Y_0, \mu_r)$, not only its image in the analytic Brauer group. Rescale each T_x to have determinant one. The multiplier of these special-linear lifts is μ_m -valued: it is scalar with determinant one. It differs from c by a scalar coboundary and still has commutator b . The monodromy of the central $[r]$ -torsor is $\Lambda \rightarrow \Lambda/r\Lambda$. For any constant coefficient group C' ,

$$H^2(Y_0, C') = \text{Hom}(\bigwedge^2 \Lambda, C').$$

In the bar complex the cycle $[e_i|e_j] - [e_j|e_i]$ represents $e_i \wedge e_j$, and an inflated multiplier evaluates on it as its alternating commutator. Thus the special-linear obstruction class has pairing b . The inclusion $\mu_m \hookrightarrow \mathbb{Q}/\mathbb{Z}$ induces an injection on this H^2 , so that class is the image of α under $\mu_r \hookrightarrow \mu_m$. On a common good cover, its μ_m -cocycle therefore differs from a chosen μ_r -cocycle for α by the coboundary of a μ_m -valued one-cochain. Multiplying the special-linear lifts by that cochain gives transitions with exactly the chosen μ_r -multiplier and still determinant one. They define Q_0 , and their projectivizations give Z_0 .

Use the convention that $\mathbb{P}(Q_0)$ parametrizes lines in Q_0 . The tautological $\ell_0 = \mathcal{O}(1)$ then has the inverse twist. The twisted factors cancel in the Euler sequence

$$0 \longrightarrow \mathcal{O}_{Z_0} \longrightarrow p_0^* Q_0 \otimes \ell_0 \longrightarrow T_{Z_0/Y_0} \longrightarrow 0.$$

Since $\det Q_0 = \mathcal{O}_{Y_0}$, its determinant is $H|_{Z_0} = \ell_0^m$. On any projective-space fiber the same anticanonical line is $\mathcal{O}(m)$.

On each s , the quotient construction is the algebraic quotient of the projective variety $Y_s \times \mathbb{P}^{m-1}$ by a free finite algebraic action. It is projective: tensor the translates of an ample line on the product, use the permutation linearization on that tensor product, descend it along the free finite quotient, and use descent of ampleness under a finite surjection. The map p_s is algebraic and projective, for instance by its closed graph in a projective product. The global line H is p -ample. The composition rule for relative ampleness gives positive u, v for which $H^u \otimes p^* \mathcal{L}_Y^v$ is ample for $a \circ p$. The class $\zeta_{B,s}$ is the restriction of the global class $c_1^B(H)/m$. Restrictions of a total class along a proper smooth submersion are horizontal, and a holomorphic line gives type $(1, 1)$. Its fiber degree is one by the calculation of H . The raw equality in (7.13) is the degree-one comparison (7.2); centrally it gives $a_{\ell_0} = \zeta_{H,0}$.

We now verify ordinary semiregularity. The twists of $p_0^* \mathcal{B}$ and ℓ_0 cancel, so F_0 is ordinary coherent. It need not be locally free. Since p_0 is flat, the derived identity (7.7) and line cancellation give

$$R\mathcal{H}om_{Z_0}(F_0, F_0) \simeq p_0^* \mathcal{C}_B.$$

Moreover $Rp_{0*}\mathcal{O}_{Z_0} = \mathcal{O}_{Y_0}$. To check this equality, pull back along the finite etale cover $[r] : Y_0 \rightarrow Y_0$; the bundle becomes $Y_0 \times \mathbb{P}^{m-1}$, where the assertion is the vanishing of the higher cohomology of $\mathcal{O}_{\mathbb{P}^{m-1}}$. Flat base change for this fixed projective fiber and faithful flatness give the equality on Y_0 . The perfect projection formula now yields

$$\begin{aligned} \mathrm{Ext}_{Z_0}^2(F_0, F_0) &\simeq H^2 R\Gamma(Z_0, p_0^* \mathcal{C}_B) \\ &\simeq H^2 R\Gamma(Y_0, \mathcal{C}_B) = \mathrm{Ext}_{\mathfrak{Y}_0}^2(\mathcal{B}, \mathcal{B}). \end{aligned}$$

This is (7.14); it uses the full derived Hom complex. Pullback and line tensoring preserve evaluation and coevaluation, so this identification commutes with the unit and perfect trace under $p_0^* : H^2(Y_0, \mathcal{O}_{Y_0}) \xrightarrow{\sim} H^2(Z_0, \mathcal{O}_{Z_0})$ and identifies their trace-free kernels. The tensor rule and $a_{\ell_0} = \zeta_{H,0}$ give (7.15).

The raw Hodge pullback in that equation is injective. Indeed, for $\eta_H \in H^{j+2}(Y_0, \Omega_{Y_0}^j)$, put $\eta_B = (2\pi i)^{-j} \iota_{Y_0}(\eta_H)$, a complex Betti class. Comparison commutes with pullback. With $f = m - 1$, normalized Betti integration gives

$$p_{0*}^B(\zeta_{B,0}^f p_0^* \eta_B) = \eta_B, \quad \int_{\mathbb{P}^f} c_1^B(\mathcal{O}(1))^f = 1.$$

Thus $p_0^* \eta_H = 0$ implies $\eta_B = 0$, and then $\eta_H = 0$. This avoids any choice of normalization for Hodge Gysin. The exponential $e^{\zeta_{H,0}}$ is invertible on the finite direct sum $\mathcal{H}_{-2}(Z_0)$. Equation (7.15) and semiregularity of $\mathcal{B}$ prove full semiregularity of F_0 .

Finally, pullback is a morphism of cohomology local systems and ζ_B is horizontal, so (7.16) is horizontal. Each degree- $2k$ part is rational of type (k, k) . Its flat lift $(2\pi i)^k A_{B,k}$ lies in F^k , and its edge in $R^k(a \circ p)_* \Omega_{Z/S}^k$ is the raw Hodge class to be used in deformation theory. Multiplicativity of the comparison and the one Tate factor contributed by each ζ_H give, centrally,

$$\begin{aligned} A_H(0) &= p_0^* \mathrm{ch}_H(\mathcal{B}) e^{\zeta_{H,0}} = \mathrm{ch}_H(F_0), \\ A_B(0) &= p_0^* \mathrm{ch}_B(\mathcal{B}) e^{\zeta_{B,0}} = \mathrm{ch}_B(F_0). \end{aligned} \tag{7.18}$$

Here $A_{H,k}$ denotes the edge of $(2\pi i)^k A_{B,k}$, and $A_H = \sum_k A_{H,k}$. The normalized projection formula and the hyperplane degree give

$$\begin{aligned} p_{s*}^B(\zeta_{B,s}^{m-1} e^{-\zeta_{B,s}} A_B(s)) &= p_{s*}^B(\zeta_{B,s}^{m-1} p_s^* \kappa_B(s)) \\ &= \kappa_B(s) p_{s*}^B(\zeta_{B,s}^{m-1}) = \kappa_B(s). \end{aligned}$$

All exponentials are finite polynomials in cohomology. This proves (7.17) and the proposition. $\square$

7.2 Deformation over a smooth projective analytic germ

The ordinary coherent-sheaf deformation statement below is the projective case of the argument in Buchweitz–Flenner [4, Theorem 5.1 and its proof, pp. 174–179], whose proof constructs a flat analytic sheaf with the prescribed central fiber. We reconstruct that argument for the projective bundle in Proposition 7.3, keeping the full coefficient obstruction group explicit. Projectivity permits a comparison of the Chern characters of all infinitesimal lifts and an effectivity argument through compatible quotients.

Theorem 7.4. *Let $S = (S, 0)$ be a smooth complex analytic germ, and let $\pi : \mathcal{X} \rightarrow S$ be a proper holomorphic submersion of relative dimension n , equipped with a holomorphic π -ample line $\mathcal{L}_{\mathcal{X}}$. Let*

F_0 be an ordinary coherent sheaf on X_0 . For $I \subseteq \{1, \dots, n\}$, suppose the following raw semiregularity map is injective:

$$\begin{aligned} \sigma_{I,0} : \text{Ext}_{X_0}^2(F_0, F_0) &\longrightarrow \bigoplus_{k \in I} H^{k+1}(X_0, \Omega_{X_0}^{k-1}), \\ v &\longmapsto \left(\text{Tr}_{F_0, \Omega_{X_0}^{k-1}[k+1]} \left(\left(\frac{a_{F_0}^{k-1}}{(k-1)!} \right) [2] \circ v \right) \right)_{k \in I} \end{aligned} \quad (7.19)$$

Suppose there are rational flat sections $\beta_{B,k}$ of $R^{2k}\pi_*\mathbb{Q}$, for $k \in I$, such that

$$\beta_{B,k}(0) = \text{ch}_{B,k}(F_0), \quad \beta_{B,k} \in F^k(R^{2k}\pi_*\mathbb{C} \otimes \mathcal{O}_S).$$

The second condition means that the flat de Rham class itself lies in the indicated Hodge filtration. Then, after shrinking S , there is a coherent S -flat sheaf F on $\mathcal{X}$, with $F|_{X_0} \simeq F_0$, for which

$$\text{ch}_{B,k}(F|_{X_s}) = \beta_{B,k}(s) \quad (s \text{ near } 0, k \in I). \quad (7.20)$$

These classes are rational algebraic cycle classes. No rank, simplicity, stability, local-freeness, reflexivity, or fixed-support hypothesis is imposed on F_0 .

We use the raw convention $a = -\text{At}_{\text{BF}}$ from (7.1). With ordinary de Rham differential d , the raw degree- k Chern character is the Hodge edge of $(2\pi i)^k \text{ch}_{B,k}$. Thus the prescribed raw Hodge section in the theorem is the edge of $(2\pi i)^k \beta_{B,k}$. The factor does not change horizontality or membership in F^k .

7.2.1 Absolute perfectness and the coefficient obstruction group

Write

$$R = \mathcal{O}_{S,0} = \mathbb{C}\{t_1, \dots, t_e\}, \quad A_\nu = R/\mathfrak{m}^{\nu+1}, \quad X_\nu = \mathcal{X} \times_S \text{Specan } A_\nu.$$

Here $\mathfrak{m}$ is the maximal ideal of R . The integer $\nu \geq 0$ records the infinitesimal order.

We construct compatible A_ν -flat sheaves E_ν with central fiber F_0 . If E_ν has been constructed, the obstruction to the next lift has coefficients in $J_\nu = \ker(A_{\nu+1} \rightarrow A_\nu)$. The first lemma identifies its group with $\text{Ext}_{X_0}^2(F_0, F_0) \otimes_{\mathbb{C}} J_\nu$, on which central semiregularity remains injective. For these extensions, differentiation gives an injection $J_\nu \rightarrow \Omega_{A_{\nu+1}/\mathbb{C}}^1 \otimes_{A_{\nu+1}} A_\nu$. The comparison with Gauss–Manin then forces the obstruction to vanish if each selected component of the relative Chern character of E_ν is the edge of its prescribed flat filtered class. We prove this character identity for every infinitesimal lift in Lemma 7.7; thus any lift produced at one stage satisfies the hypothesis needed at the next. Finally, we express the sheaves as quotients of one vector bundle by lifting the quotient-defining sections compatibly. The resulting formal section of the relative Quot space admits an analytic approximation preserving the central quotient. Pulling back the universal quotient gives a flat sheaf with central fiber F_0 , whose fiber characters are determined by flat transport.

Lemma 7.5. *Let $A = A_\nu$, $X = X_\nu$, and let E be an A -flat coherent sheaf on X whose central fiber is F_0 . Then E is absolutely perfect as an $\mathcal{O}_X$ -complex, with local projective dimension at most n . If W is a finite A -module killed by the maximal ideal and $W_X = \mathcal{O}_X \otimes_A W$, there are natural identifications*

$$\begin{aligned} \text{Ext}_X^2(E, E \otimes_A W) &\simeq \text{Ext}_{X_0}^2(F_0, F_0) \otimes_{\mathbb{C}} W, \\ H^q(X, \Omega_{X/A}^j \otimes_A W) &\simeq H^q(X_0, \Omega_{X_0}^j) \otimes_{\mathbb{C}} W. \end{aligned} \quad (7.21)$$

Under them the coefficient semiregularity map

$$\sigma_{I,W}(v) = \left(\text{Tr}_{E, (\Omega_{X/A}^{k-1} \otimes_A W)[k+1]} \left(\left(\frac{a_{X/A}(E)^{k-1}}{(k-1)!} \otimes_A^{\mathbf{L}} 1_W \right) [2] \circ v \right) \right)_{k \in I}$$

is exactly $\sigma_{I,0} \otimes 1_W$, and is therefore injective.

Proof. At $x \in X_0$, put $B = \mathcal{O}_{X,x}$ and $M_E = E_x$. Both are A -flat, while $B_0 = B \otimes_A \mathbb{C}$ is regular local of dimension n . Choose a minimal finite-rank free B -resolution of M_E , initially allowed to be infinite. Tensoring it over A with $\mathbb{C}$ remains exact because its terms and M_E are A -flat. It is still minimal over B_0 , since maximal-ideal entries remain in the maximal ideal after reduction. The regular-local projective-dimension theorem gives projective dimension at most n over B_0 , so this minimal reduced resolution has no terms in higher degree. Its ranks are the ranks before reduction, and the original resolution therefore stops as well. This proves absolute $\mathcal{O}_X$ -perfectness. It is separate from relative Tor-amplitude $[0, 0]$, which is precisely the assumed A -flatness; the nonreduced total space has not been declared regular. For the general fiberwise perfectness criterion, see [4, Proposition 4.5 and Lemma 4.6, p. 169].

Let $i : X_0 \hookrightarrow X$. Flatness of $\mathcal{O}_X$ and E over A , together with derived associativity, gives

$$Li^* E \simeq E \otimes_A^{\mathbf{L}} \mathbb{C} = F_0, \quad E \otimes_{\mathcal{O}_X}^{\mathbf{L}} W_X \simeq E \otimes_A^{\mathbf{L}} W = E \otimes_A W = i_*(F_0 \otimes_{\mathbb{C}} W). \quad (7.22)$$

In particular the last derived tensor is an ordinary sheaf. We do not use the generally false identity $Li^* W_X = \mathcal{O}_{X_0} \otimes_{\mathbb{C}} W$. Perfect duality and the projection formula for the closed immersion give

$$R\mathcal{H}om_X(E, E \otimes_A W) \simeq E^\vee \otimes^{\mathbf{L}} i_*(F_0 \otimes W) \simeq i_*(F_0^\vee \otimes^{\mathbf{L}} F_0 \otimes W).$$

Derived global sections prove the first identity in (7.21). The coefficient sheaf $\Omega_{X/A}^j \otimes_A W$ is the closed pushforward of $\Omega_{X_0}^j \otimes_{\mathbb{C}} W$, proving the second. These are finite direct-sum identities, not higher-direct-image base change for an endomorphism complex.

The relative first-jet Atiyah class pulls back to that of F_0 , and evaluation $E^\vee \otimes E \rightarrow \mathcal{O}_X$ pulls back to central evaluation. Apply these identities to the displayed perfect-Hom formula. Each coefficient semiregularity component becomes the corresponding central component tensored with W . This proves the asserted equality and injectivity. $\square$

7.2.2 A square-zero lift and its semiregularity obstruction

Fix ν , put $A' = A_{\nu+1}$, $A = A_\nu$, and write

$$J = \mathfrak{m}^{\nu+1}/\mathfrak{m}^{\nu+2}, \quad K = \Omega_{A'/\mathbb{C}}^1 \otimes_{A'} A, \quad j : X = X_\nu \hookrightarrow X' = X_{\nu+1}.$$

The ideal of X is $J_X = \mathcal{O}_X \otimes_A J$. We have $J^2 = 0$ and $\mathfrak{m}_A J = 0$. Moreover

$$K = \Omega_{R/\mathbb{C}}^1 / \mathfrak{m}^{\nu+1} \Omega_{R/\mathbb{C}}^1 \simeq A^{\oplus e}, \quad d : J \hookrightarrow K. \quad (7.23)$$

Indeed $d(\mathfrak{m}^{\nu+2}) \subset \mathfrak{m}^{\nu+1} \Omega_R^1$. On homogeneous terms the last map is

$$\text{Sym}^{\nu+1}(\mathfrak{m}/\mathfrak{m}^2) \longrightarrow \text{Sym}^\nu(\mathfrak{m}/\mathfrak{m}^2) \otimes (\mathfrak{m}/\mathfrak{m}^2),$$

given by differentiation. Euler's formula $\sum_i t_i \partial_i f = (\nu+1)f$ proves injectivity in characteristic zero.

Lemma 7.6. *Let E be an A -flat coherent sheaf on X with central fiber F_0 . Suppose, for every $k \in I$, that its raw relative character $\text{ch}_{H,k}^{\text{rel}}(E) = \text{tr}(a_{X/A}(E)^k)/k!$ is the restriction of the Hodge edge of a flat F^k de Rham class on X'/A' . Then E has a coherent A' -flat sheaf lift on X' , with its prescribed restriction to X .*

Proof. We first record the precise coefficient exactness needed for (7.23). The relative ample line algebraizes every X_ν , and every coherent sheaf on it, over the finite complex algebra A_ν . We use projective analytic GAGA in the form of an exact equivalence on coherent sheaves and comparison on their cohomology; local analytification is faithfully flat [37]. The algebraic model is projective and smooth over A_ν : its analytification is locally a smooth product, and smoothness for these finite-type models is reflected by analytification. Deligne's proper-smooth theorem [12, Theorem 5.5] applies in characteristic zero, including to nonreduced Artinian bases. It states that $H^q(X_\nu, \Omega_{X_\nu/A_\nu}^j)$ is finite locally free, commutes with arbitrary algebra base change, and that the Hodge-to-de-Rham spectral sequence degenerates.

Here is the finite-module consequence. For a finite A -module W , apply algebra base change to the split square-zero algebra $A \oplus W$. On the same underlying space the relative differential sheaf is $\Omega_{X/A}^j \oplus (\Omega_{X/A}^j \otimes_A W)$. The base-change isomorphism respects the inclusion and augmentation of $A \oplus W$. Taking the W -summand gives, naturally in W ,

$$H^q(X, \Omega_{X/A}^j \otimes_A W) \simeq H^q(X, \Omega_{X/A}^j) \otimes_A W. \quad (7.24)$$

The latter Hodge module is free, so this functor of W is exact. In particular $d : J \hookrightarrow K$ stays injective after (7.24). GAGA transfers the same assertion to analytic sheaves.

For the remaining derived calculation, algebraize X' , reduce it to obtain X , and algebraize E ; use the same symbols for this projective algebraic pair. Faithfully flat local analytification reflects A -flatness, and the proof of Lemma 7.5 gives absolute perfectness on the algebraic model too. The derived foundations used here are the usual tensor and adjunction formalism for modules, cotangent transitivity, the first conormal homology $H^{-1}(L_{X/X'}) = J_X/J_X^2$, and the functorial derived principal-parts construction of the Atiyah class. For a perfect complex the trace is evaluation; its cyclicity and its compatibility with pullback and coefficient maps follow in a finite free model. We also use the characteristic-zero shifted exterior convention, for a cotangent complex L , $L \text{Sym}^k(L[1]) \simeq (L \wedge^k L)[k]$. These are the standard algebraic cotangent and perfect-duality constructions; no obstruction theorem for a general perfect complex is imported.

Only ordinary relative objects have to be compared back to analysis. For the smooth morphism X/A , its cotangent complex is $\Omega_{X/A}^1[0]$ and its derived self-product is underived by flatness. The first-principal-parts sequence is split for the input $\mathcal{O}_X$ -action, so tensoring it with any coherent E is exact. Thus the relative derived Atiyah triangle is the ordinary first-jet extension. Ordinary differentials, conormal sequences, and first jets commute with analytification, as follows from finite presentations; perfect trace does too by local free evaluation. For $Q = E \otimes_A J$, GAGA for the bounded coherent complex $E^\vee \otimes^L Q$ compares $\text{Ext}_X^2(E, Q)$ with the analytic group, and ordinary GAGA compares the Hodge targets. Hence the coefficient map of Lemma 7.5 is the same one on the model. We do not compare an absolute nonreduced analytic cotangent complex or an Ext^1 group on X' ; the extension below will be constructed algebraically and then analytified.

Define the cotangent extension class

$$\delta_X : L_{X/\mathbb{C}} \longrightarrow L_{X/X'} \longrightarrow H^{-1}(L_{X/X'})[1] = J_X[1].$$

The whole complex $L_{X/X'}$ need not be $J_X[1]$. Smoothness of X'/A' , restricted to X , gives the exact sequence

$$0 \longrightarrow \mathcal{O}_X \otimes_A K \longrightarrow \Omega_{X'/\mathbb{C}}^1 \otimes \mathcal{O}_X \longrightarrow \Omega_{X/A}^1 \longrightarrow 0. \quad (7.25)$$

The conormal map $J_X \rightarrow \Omega_{X'/\mathbb{C}}^1 \otimes \mathcal{O}_X$ factors through its first arrow by $1 \otimes d$. It is injective by (7.23) and A -flatness. Consequently the ordinary conormal sequence ending in $\Omega_{X/\mathbb{C}}^1$ is short exact as well.

Map the transitivity triangle

$$Lj^* L_{X'/\mathbb{C}} \longrightarrow L_{X/\mathbb{C}} \longrightarrow L_{X/X'}$$

to this conormal triangle by the degree-zero truncations. The induced cone map $L_{X/X'} \rightarrow J_X[1]$ is the canonical truncation: on $H^{-1} = J_X/J_X^2$ it is the identity, and, since $L_{X/X'} \in D^{\leq -1}$, truncation adjunction determines a map to $J_X[1]$ by that map on H^{-1} . Now map the conormal sequence to (7.25), using $1 \otimes d$ on the left and the identity in the middle. Naturality of boundaries and exterior comultiplication give the derived square

$$\begin{array}{ccc} L \bigwedge^k L_{X/\mathbb{C}} & \xrightarrow{\rho_k} & \Omega_{X/A}^k \\ \downarrow \delta_{X,k} & & \downarrow \nabla'_k \\ (\Omega_{X/A}^{k-1} \otimes_A J)[1] & \xrightarrow{1 \otimes d} & (\Omega_{X/A}^{k-1} \otimes_A K)[1]. \end{array} \quad (7.26)$$

Here ρ_k projects each factor to relative forms, $\delta_{X,k}$ contracts one factor with δ_X and projects the others, and ∇'_k is the exterior boundary of (7.25). The exterior signs are those of the shifted symmetric convention just specified. This argument does not require a regular immersion.

We next prove the required sheaf obstruction statement. By (7.22),

$$Q = E \otimes_A J \simeq E \otimes_{\mathcal{O}_X}^L J_X$$

is an ordinary coherent sheaf. Put $\mathcal{K} = Lj^* j_* E$. The Tor sequence for $0 \rightarrow J_X \rightarrow \mathcal{O}_{X'} \rightarrow \mathcal{O}_X \rightarrow 0$ gives $H^0(\mathcal{K}) = E$ and $H^{-1}(\mathcal{K}) = Q$; lower cohomology is allowed. Its truncation triangle is

$$Q[1] \longrightarrow \tau_{\geq -1} \mathcal{K} \longrightarrow E \xrightarrow{b_E} Q[2]. \quad (7.27)$$

Derived adjunction identifies $\text{Ext}_{X'}^1(j_* E, j_* Q)$ with $\text{Hom}_{D(X)}(\mathcal{K}, Q[1])$. Because Q is a sheaf, this Hom factors through $\tau_{\geq -1} \mathcal{K}$. Applying $\text{Hom}(-, Q[1])$ to (7.27) gives the exact segment

$$\text{Ext}_{X'}^1(j_* E, j_* Q) \longrightarrow \text{Hom}_X(Q, Q) \longrightarrow \text{Ext}_X^2(E, Q). \quad (7.28)$$

Use the cochain convention in which a connecting map is the differential of a lift. The last map sends id_Q to b_E . If the first class is represented by $0 \rightarrow j_* Q \rightarrow E' \rightarrow j_* E \rightarrow 0$, the Tor boundary after restriction sends $j_0 \otimes e$ to $j_0 \tilde{e}$, where $\tilde{e}$ lifts e in E' . This follows directly from the two-term presentation $J_X \rightarrow \mathcal{O}_{X'}$ of $\mathcal{O}_X$, with the same connecting-map convention. It is the map $Q \rightarrow Q$ in (7.28). Thus $b_E = 0$ precisely when there is an extension

$$0 \longrightarrow E \otimes_A J \longrightarrow E' \longrightarrow E \longrightarrow 0 \quad (7.29)$$

whose kernel identification is the multiplication map $J_X \otimes_{\mathcal{O}_X} E \rightarrow E'$. Yoneda Ext supplies an actual extension; it is coherent because coherent modules on the noetherian model are closed under extensions.

We identify b_E with the Atiyah contraction, including its orientation. Keep the two actions on

$$\mathcal{C} = \mathcal{O}_X^{(1)} \otimes_{\mathcal{O}_{X'}}^L \mathcal{O}_X^{(2)}, \quad \mathcal{I} = \text{fib}(\mathcal{C} \rightarrow \mathcal{O}_X).$$

Tensor over action (1), using action (2) as output. Then $\mathcal{C} \otimes_{(1)}^L E = \mathcal{K}$, $\mathcal{I} \in D^{\leq -1}$, and $H^{-1}(\mathcal{I}) = J_X$. Universal derivation on action (2) gives the diagonal linearization $\ell : \mathcal{I} \rightarrow L_{X/X'}$, oriented by

$1 \otimes f - f \otimes 1 \mapsto df$. To check its first homology, resolve factor (2) by a local free DG algebra over $\mathcal{O}_{X'}$. A degree- -1 generator u with $d_{\text{DG}} u = j_0 \in J_X$ represents j_0 in $H^{-1}(\mathcal{I})$; its universal differential represents the same class in $H^{-1}(L_{X/X'}) = J_X/J_X^2$. Thus ℓ induces the identity under the canonical first-conormal-homology identifications. Universal derivation makes this check compatible on overlaps.

The homotopy pushout of $\mathcal{I} \otimes_{(1)}^L E \rightarrow \mathcal{K}$ along $\ell \otimes E$ is the derived first-principal-parts triangle

$$E \otimes^L L_{X/X'} \longrightarrow P_{X/X'}^1(E) \longrightarrow E \xrightarrow{\text{At}_{X/X'}(E)} E \otimes^L L_{X/X'}[1].$$

The boundary is the principal-parts definition of Atiyah. Its sign relative to Buchweitz–Flenner can be read from first jets. For ordinary degree-zero jets put $j(p) = (1 \otimes 1) \otimes p$, with p on action (1) and the module action on action (2). With $df = 1 \otimes f - f \otimes 1$,

$$j(pf) - j(p)f = -p df.$$

A connection satisfies $\nabla(pf) = \nabla(p)f + p df$, so $j + \nabla$ is the module splitting. In a free DG model its failure to commute with the differential is $[d_{\text{DG}}, \nabla]$. This is precisely At_{BF} in [4, Definition 3.4], not its negative.

Let $t : L_{X/X'} \rightarrow J_X[1]$ be truncation. Since $t\ell$ is the identity on H^{-1} , truncation adjunction identifies it with $\mathcal{I} \rightarrow J_X[1]$. After tensoring with E , the analogous assertion has target $Q[1]$: the tensor cohomology spectral sequence has in degree -1 only $H^{-1}(\mathcal{I}) \otimes E = Q$, and all higher Tor terms have lower degree. The derived target is $Q[1]$ by (7.22). Naturality of the fiber and truncation triangles therefore identifies b_E with the principal-parts Atiyah class followed by $t[1]$. Naturality under $L_{X/\mathbb{C}} \rightarrow L_{X/X'}$ gives

$$b_E = (1_E \otimes \delta_X)[1] \text{At}_{\text{BF}, X/\mathbb{C}}(E).$$

Define $o_E = -b_E$. With $a = -\text{At}_{\text{BF}}$, we have proved

$$o_E = \langle \delta_X, a_{X/\mathbb{C}}(E) \rangle. \quad (7.30)$$

It vanishes if and only if (7.29) exists with its multiplication identification. The use of the two highest cohomology degrees of $\mathcal{K}$ depends on Q being a sheaf. For a general perfect complex, the Hom in (7.28) need not factor through this truncation.

Absolute perfectness supplies the trace in the next identity. Put $L = L_{X/\mathbb{C}}$. Under $L \text{Sym}^k(L[1]) \simeq (L \wedge^k L)[k]$, the Atiyah morphism $a : E \rightarrow E \otimes L[1]$ is a degree-zero coefficient-valued endomorphism. The map $\delta_X[1] : L[1] \rightarrow J_X[2]$ extends as the derivation contracting each symmetric generator. Its output coefficient has even shift two. Expanding contraction of a^k gives k terms with the same supertrace by graded cyclicity: the symmetric algebra accounts for the internal Koszul signs, and rotation of the degree-zero Atiyah operations and the even coefficient leaves no residual sign. Trace commutes with coefficient maps. Dividing by $k!$ proves

$$\sigma_{k,J}(o_E) = \delta_{X,k} \left(\frac{\text{tr}(a_{X/\mathbb{C}}(E)^k)}{k!} \right). \quad (7.31)$$

This supplies the particular obstruction and trace identities corresponding to [4, Propositions 4.4 and 4.2, pp. 166–169].

The multiplication identification also gives A' -flatness. For completeness, the square-zero flatness criterion says that an A' -module M' is flat if M'/JM' is A -flat and $J \otimes_A (M'/JM') \rightarrow M'$ is injective. The latter condition is $\text{Tor}_1^{A'}(A, M') = 0$. The change-of-rings Tor sequence then gives

$\mathrm{Tor}_1^{A'}(W, M') = 0$ for every A -module W . Filtering an arbitrary A' -module by $JW \subset W$ gives the same vanishing for every W , which is flatness. For (7.29), the kernel identification gives $E'/JE' = E$ and the required injection. Apply the criterion to the algebraic stalks. Flat local analytification preserves this A' -flatness, coherence, and the extension.

We now relate the obstruction to the assumed horizontal class. The exterior boundary ∇'_k in (7.26) is the restricted Hodge-graded Gauss–Manin map. Indeed, quotient the absolute de Rham complex of $X'/\mathbb{C}$ by forms with at least two base differentials. The connecting map of the resulting sequence is base differentiation in product coordinates. Filtering by relative form degree gives on the graded term exactly the exterior boundary of (7.25). Hodge degeneration identifies the degree- $2k$ graded term with $H^k(\Omega^k)$, and (7.24) identifies its restriction to A . Therefore the restriction of the edge of a flat F^k class on X'/A' is killed by ∇'_k .

The image under ρ_k of the absolute character in (7.31) is the relative character of E , by Atiyah naturality. Evaluate (7.26) on that absolute character and use (7.31). The result is

$$(1 \otimes d)_* \sigma_{k,J}(o_E) = \nabla'_k(\mathrm{ch}_{H,k}^{\mathrm{rel}}(E)). \quad (7.32)$$

This is the restricted Gauss–Manin square used in [4, Lemma 5.8 and Proposition 5.9]. The hypothesis of the lemma kills its right side for $k \in I$. The left coefficient map is injective by (7.24), and $\sigma_{I,J}$ is injective by Lemma 7.5, since $\mathfrak{m}_A J = 0$. Thus $o_E = 0$. Equations (7.30) and (7.29), followed by the flatness criterion, give the required analytic flat sheaf lift. $\square$

7.2.3 The Chern character of every infinitesimal lift

The one-step lemma can be iterated only after checking that a chosen lift has the prescribed relative character, rather than merely the prescribed central restriction. The projectivity assumption supplies a direct proof for every stage.

Lemma 7.7. *For every A_ν -flat coherent sheaf E_ν on X_ν with central fiber F_0 , and every $0 \leq k \leq n$, the ordinary flat comparison*

$$H_{\mathrm{dR}}^l(X_\nu/A_\nu) \simeq H^l(X_0, \mathbb{C}) \otimes_{\mathbb{C}} A_\nu$$

satisfies

$$\begin{aligned} \mathrm{ch}_{H,k}^{\mathrm{rel}}(E_\nu) &= \mathrm{edge}_k((2\pi i)^k \mathrm{ch}_{B,k}(F_0) \otimes 1), \\ (2\pi i)^k \mathrm{ch}_{B,k}(F_0) \otimes 1 &\in F^k H_{\mathrm{dR}}^{2k}(X_\nu/A_\nu). \end{aligned} \quad (7.33)$$

The assertion holds independently of the choice of the lift.

Proof. For $k = 0$ the assertion is constancy of rank on each connected component. Assume $k \geq 1$. The projective model of X_ν has an ample line. Repeated Serre global-generation surjections from finite sums of negative powers of that line resolve E_ν . The local projective-dimension bound in Lemma 7.5 makes the n -th syzygy locally free. GAGA gives a global finite holomorphic vector-bundle resolution $V^\bullet \rightarrow E_\nu$. No compatibility of these resolutions as ν varies is asserted or needed.

Choose a differentiable trivialization of the proper submersion $\mathcal{X} \rightarrow S$. Holomorphic t -jets through order ν of relative smooth forms define a morphism

$$\Omega_{X_\nu/A_\nu}^\bullet \longrightarrow \mathcal{A}_{X_0}^\bullet \otimes_{\mathbb{C}} A_\nu,$$

where $\mathcal{A}_{X_0}^\bullet$ is the smooth de Rham complex. Explicitly, after the trivialization take

$$\sum_{|\mathbf{a}| \leq \nu} \frac{(\partial_t^{\mathbf{a}} \omega)|_{t=\bar{t}=0}}{\mathbf{a}!} t^{\mathbf{a}}.$$

Leibniz's rule makes this compatible with products, and jets commute with the fiber differential. The jets of the fiberwise type projections give a bigrading on the target; their integrability identities persist under taking jets. In a local holomorphic product chart the resulting Dolbeault complex is the ordinary one with coefficients A_ν , up to an invertible formal smooth coordinate change. The coefficientwise Dolbeault lemma therefore makes it a fine resolution of each Ω_{X_ν/A_ν}^p , respecting the form filtration.

Both total complexes resolve the constant sheaf A_ν . For the holomorphic complex, filter by powers of $\mathfrak{m}$; its associated graded complex is $\Omega_{X_0}^\bullet \otimes_{\mathbb{C}} \text{gr } A_\nu$, to which the holomorphic Poincaré lemma applies. The jet map is the identity on constants. It therefore realizes the ordinary flat de Rham comparison in the statement.

A holomorphic vector bundle V on X_ν becomes a smooth A_ν -bundle under the same jet map, and is smoothly isomorphic to $V_0^\infty \otimes A_\nu$. Indeed, modulo successive powers of $\mathfrak{m}$, the discrepancy between its transition matrices and the central ones is a Čech one-cocycle with values in smooth sections of $\text{End}(V_0^\infty) \otimes \mathfrak{m}^a/\mathfrak{m}^{a+1}$. This sheaf is fine, so a smooth gauge change removes that discrepancy. Induction through the finitely many powers of $\mathfrak{m}$ gives the isomorphism.

Choose a smooth right-module connection $D = \bar{\partial}_V + \nabla^{1,0}$ on this bundle, and put $F_D = D^2$. Its curvature has no $(0,2)$ -part. In the Dolbeault resolution the BF commutator $[\bar{\partial}, \nabla]$, with its form coefficient written after the $(0,q)$ -form, maps under exterior multiplication to $F_D^{1,1}$. The sign also follows from the first-jet calculation above: for a line with frames $e_j = e_i g_{ij}$, the BF cocycle is $-d \log g_{ij}$, while the connection matrices differ by $d \log g_{ij}$; the Čech–Dolbeault boundary carries this cocycle to $+\bar{\partial} A_i$. Thus $a = -\text{At}_{\text{BF}}$ is represented by $-F_D^{1,1}$. It follows that the closed form

$$\frac{(-1)^k}{k!} \text{tr}(F_D^k)$$

belongs to F^k and that its edge is the raw degree- k Atiyah character.

This de Rham class is the constant central class. Use a constant connection on $V_0^\infty \otimes A_\nu$ and the smooth isomorphism just constructed. For $D_u = D_0 + u(D_1 - D_0)$, the Chern–Simons identity is

$$\partial_u \frac{\text{tr}(F_{D_u}^k)}{k!} = d \frac{\text{tr}((D_1 - D_0)F_{D_u}^{k-1})}{(k-1)!}.$$

Integration in u proves independence of the connection; the polynomial identity is valid over the finite complex algebra A_ν . For this right-connection convention, the normalized topological character is $\text{tr} \exp(-F_D/(2\pi i))$. Hence the signed unscaled form above represents $(2\pi i)^k \text{ch}_{B,k}(V_0)$ in the flat comparison.

Apply the calculation termwise to $V^\bullet$. A grading-preserving connection represents At_{BF} by its commutator with $d_V^\bullet + \bar{\partial}$, and a by the negative of that commutator. In a power, every term involving $d_V^\bullet$ has positive internal degree and zero supertrace. The remaining terms are the alternating sum of the signed bundle curvature traces. Their edge is therefore $\text{ch}_{H,k}^{\text{rel}}(E_\nu)$. By (7.22), the central restriction of the resolution resolves F_0 . Its alternating central character is $\text{ch}_{B,k}(F_0)$, which proves (7.33). $\square$

7.2.4 Compatible quotients and analytic effectivity

Proof of Theorem 7.4. Let $\beta_{H,k,\nu}$ be the edge of the restriction of $(2\pi i)^k \beta_{B,k}$ to X_ν/A_ν . A flat section is uniquely determined near the origin by its central value. Thus Lemma 7.7 identifies the relative character of every E_ν with $\beta_{H,k,\nu}$. Starting with $E_0 = F_0$, the assumption that $\beta_{B,k}$ lies in F^k provides the flat lift to order $\nu + 1$. Lemma 7.6 then gives $E_{\nu+1}$ with its identification on X_ν .

This constructs a compatible tower of flat coherent sheaves. Only the extensions $A_{\nu+1} \rightarrow A_\nu$ and their modules J_ν were used, not arbitrary Artinian extensions.

We make the classifying maps compatible before applying analytic approximation. Choose c large enough that $F_0(c) = F_0 \otimes \mathcal{L}_{\mathcal{X},0}^c$ is generated by b global sections and has $H^1(X_0, F_0(c)) = 0$. Let $\mathcal{V} = (\mathcal{L}_{\mathcal{X}}^c)^{\oplus b}$ and fix the central quotient $\mathcal{V}_0 \twoheadrightarrow F_0$. Given a quotient $\mathcal{V}_\nu \twoheadrightarrow E_\nu$, tensor the extension for $E_{\nu+1}$ by $\mathcal{L}_{\mathcal{X}}^c$. Its kernel is $i_{0*}(F_0(c) \otimes_{\mathbb{C}} J_\nu)$, whose first cohomology vanishes. The global-section map to order ν is therefore surjective. Lift the b sections defining the quotient. Their map to $E_{\nu+1}$ is surjective by Nakayama's lemma, because its central restriction is the fixed central quotient. We obtain compatible quotients $\mathcal{V}_\nu \twoheadrightarrow E_\nu$ for every ν .

We use the relative analytic Quot theorem of Pourcin [32, Theorem 2, pp. 451–452]: proper flat quotients of a coherent sheaf on a separated finite-dimensional complex space over a separated finite-dimensional complex base, which may be nonreduced, are represented by a complex space with a universal flat quotient. These hypotheses hold here; in particular quotient supports are proper because $\mathcal{X} \rightarrow S$ is proper. Let T be the germ of this Quot space at the central quotient. The compatible quotients give unique compatible S -maps $\text{Specan } A_\nu \rightarrow T$. Passing to the inverse limit gives a local R -algebra map

$$\mathcal{O}_{T,0} \longrightarrow \hat{R} = \mathbb{C}[[t_1, \dots, t_e]],$$

that is, a formal section of $T \rightarrow S$.

A graph embedding presents this analytic S -germ as

$$\mathcal{O}_{T,0} = R\{z_1, \dots, z_h\}/(g_1, \dots, g_v).$$

The formal section gives zero-constant formal series $z_j(t)$ solving these finitely many convergent equations. Artin's analytic approximation theorem [1, Theorem 1.2, p. 277] says that such a formal solution over $\mathbb{C}$ has an exact convergent solution with any prescribed finite-order congruence. Take congruence modulo $\mathfrak{m}$. The solution still maps the origin to the central quotient, and, with t fixed as the parameters, defines an analytic section $S \rightarrow T$. Pullback of the universal quotient gives a coherent S -flat sheaf F with central fiber F_0 . The convergent solution need not realize the entire chosen formal tower; retaining the central fiber is sufficient.

It remains to identify characters on actual fibers. The total space $\mathcal{X}$ is smooth, so F is absolutely perfect. Separately, S -flatness and derived associativity give, for every s ,

$$Li_s^* F \simeq F \otimes_{\mathcal{O}_{\mathcal{X}}}^L (\mathcal{O}_{\mathcal{X}} \otimes_{\mathcal{O}_S}^L k(s)) \simeq F \otimes_{\mathcal{O}_S}^L k(s) = F|_{X_s}[0]. \quad (7.34)$$

Near X_0 , F has a finite resolution by total holomorphic vector bundles: use relative Serre generation repeatedly and the regular-local projective-dimension bound on the smooth total space, then shrink by properness at the final locally free syzygy. The alternating normalized topological Chern character of these bundles restricts by (7.34) to $\text{ch}_B(F|_{X_s})$. Restrictions of a total cohomology class along a proper smooth submersion are flat. Their central value is $\text{ch}_B(F_0)$, so uniqueness of flat transport proves (7.20).

Every X_s is projective algebraic, and projective GAGA algebraizes $F|_{X_s}$. On the smooth projective fiber a finite locally free resolution identifies its normalized topological character with the cycle class of its rational algebraic Chern character. Its raw degree- k Atiyah character is the weightwise $(2\pi i)^k$ image of that class. This proves algebraicity and completes the theorem. $\square$

For $p : Z \rightarrow Y$ in Proposition 7.3, the morphism $a \circ p$ satisfies the theorem's projectivity hypothesis, F_0 is ordinary coherent and fully semiregular, and its relative dimension is $N = 6 + m - 1$. Taking $I = \{1, \dots, N\}$, the raw input in degree k is exactly the edge of $(2\pi i)^k A_{B,k}$. The theorem uses neither a relative version of the finite isogeny q nor a deformation of the twisted sheaf $\mathcal{B}$. Its

proof also does not assert arbitrary classical base change for global Ext groups or higher direct images. The smooth projective germ and the ordinary coherent-sheaf condition are the stated limits of the deformation result.

8 Algebraicity at every compatible period

We apply the descent and deformation results to the normalized character computed for the polarized seed. Projective-bundle operations first give local algebraicity. A marked period family and proper Hilbert parameter spaces then extend the Weil plane to every compatible period, and rational isogenies remove the dependence on the initial lattice.

8.1 The descended character on the marked Weil domain

Fix $K = \mathbb{Q}(u)$, with $u^2 = -d$ and $d \geq 3$, and use the seed $M = X \times \widehat{X}$ constructed above. Its coherent perfect sheaf $\mathcal{E}$ has rank $r = 8d$. We write cl_B for the usual rational Betti cycle-class map, normalized so that the first Chern class $c_1^B(L)$ of a line bundle is integral. The normalized character on the seed is

$$\kappa_{M,B} = \exp\left(-\frac{c_1^B(\mathcal{E})}{r}\right) \text{ch}_B(\mathcal{E}) = \sum_{j=0}^6 \kappa_{M,B,j}. \quad (8.1)$$

Here $c_1^B(\mathcal{E}) = c_1^B(\det \mathcal{E})$, so this is the class κ_M of Section 6 in normalized Betti notation. Proposition 6.3 supplies the unital K -action on M and its ample class θ_M , and Proposition 6.5 gives

$$\kappa_{M,B,3} = c_\theta \theta_M^3 + w_M, \quad c_\theta \in \mathbb{Q}, \quad 0 \neq w_M \in \mathcal{W}_K(M). \quad (8.2)$$

Proposition 6.6 proves that every $\kappa_{M,B,j}$ has type (j, j) at every marked complex structure of Weil type $(3, 3)$ compatible with this action and polarization. We will transport these properties to the quotient on which the descended sheaf is defined.

8.1.1 The actual homological Hermitian space

Let e_M be the Riemann form of the ample class θ_M . The actual homological Hermitian form is

$$h_M(x, y) = e_M(x, \eta_M(u)_* y) + u e_M(x, y).$$

In the second-linear convention of (3.1), Proposition 6.3 proves that this form is hyperbolic of signature $(3, 3)$; its three hyperbolic blocks are computed in (6.16).

Let $q : M \rightarrow Y_0$ be the isogeny in Proposition 7.2. Put

$$V_0 = H_1(Y_0, \mathbb{Q}), \quad \Lambda_0 = H_1(Y_0, \mathbb{Z}), \quad \theta_0 = (q^*)^{-1} \theta_M.$$

Transport the rational action through q in the category up to isogeny, and write $T_0 = \eta_0(u)_*$ for the transported action on V_0 . The alternating form e_0 associated to θ_0 is characterized by

$$e_0(q_* x, q_* y) = e_M(x, y).$$

It is rational and is a positive Riemann form at Y_0 ; a positive integer multiple is integral on Λ_0 and is an ample polarization. The second-linear Hermitian form

$$h_0(x, y) = e_0(x, T_0 y) + u e_0(x, y) \quad (8.3)$$

is carried to h_M by q_* , so it too is hyperbolic of signature $(3, 3)$.

8.1.2 The full compatible domain

Let $\mathcal{D}$ be the set of real complex structures I on $V_{0,\mathbb{R}}$ satisfying

$$I^2 = -1, \quad IT_0 = T_0I, \quad e_0(Ix, Iy) = e_0(x, y), \quad e_0(x, Ix) > 0 \quad (x \neq 0). \quad (8.4)$$

This is the full marked compatible domain for the fixed rational space, form, and lattice. We verify both its connectedness and the complex structure used below.

Choose the embedding $\sigma : K \hookrightarrow \mathbb{C}$ for which $\sigma(u) = i\sqrt{d}$, and view $V_{0,\sigma} = V_0 \otimes_{K,\sigma} \mathbb{C}$ with its Hermitian form $h_{0,\sigma}$. A real I commuting with K is complex-linear on this space. Commutation with T_0 together with preservation of e_0 implies that I is unitary for $h_{0,\sigma}$. Its i - and $-i$ -eigenspaces are therefore orthogonal. On the first of them $T_0 = \sqrt{d}I$, and on the second $T_0 = -\sqrt{d}I$. Since $h_0(x, x) = e_0(x, T_0x)$, Riemann positivity makes the first eigenspace positive and the second negative. Each has dimension three because h_0 has signature $(3, 3)$.

Conversely, if $P \subset V_{0,\sigma}$ is a positive three-plane, then $P^\perp$ is negative of dimension three. Set $I = i$ on P and $I = -i$ on $P^\perp$. This is a real complex structure commuting with K and preserving h_0 , hence also e_0 , which is the coefficient of u in (8.3). For $x = x_+ + x_-$ in this orthogonal decomposition,

$$e_0(x, Ix) = d^{-1/2}(h_{0,\sigma}(x_+, x_+) - h_{0,\sigma}(x_-, x_-)) > 0 \quad (x \neq 0).$$

Thus positive three-planes parameterize exactly $\mathcal{D}$. Fix a positive/negative orthogonal splitting $V_{0,\sigma} = V_+ \oplus V_-$. Projection of a positive three-plane to V_+ is an isomorphism, so that plane is the graph of a unique map $Z : V_+ \rightarrow V_-$. Positivity is precisely

$$1 - Z^*Z > 0, \quad \text{equivalently} \quad \|Z\|_{\text{op}} < 1. \quad (8.5)$$

The operator-norm unit ball in $\text{Hom}_{\mathbb{C}}(V_+, V_-)$ is convex. In particular the underlying domain is connected and contractible and has complex dimension nine.

For the holomorphic convention, use cohomological periods. Write $W_\sigma \oplus W_{\bar{\sigma}} = H^1(Y_0, \mathbb{C})$ for the two K -eigenspaces. The cohomological polarization induced by e_0 pairs these two spaces perfectly and vanishes on each separately. If F^1 is a compatible Lagrangian Hodge subspace, then $P^\vee = F^1 \cap W_\sigma$ is a three-plane and

$$F^1 = P^\vee \oplus \text{Ann}(P^\vee) \subset W_\sigma \oplus W_{\bar{\sigma}}, \quad (8.6)$$

where the annihilator is taken using that perfect pairing. Conversely this formula, restricted by the open Riemann positivity condition, gives precisely the compatible periods. The annihilator map between the two Grassmannians is algebraic. We give $\mathcal{D}$ the complex structure in the coordinate $P^\vee$; then (8.6) is holomorphic into the Siegel period domain. This coordinate can be the conjugate of the homological graph coordinate in (8.5); the connected underlying space and its complex dimension are unchanged.

8.1.3 A fine principal-level family and the seed lattice

Choose a rational symplectic basis $a_1, \dots, a_6, b_1, \dots, b_6$ for e_0 , normalized by $e_0(a_i, b_j) = \delta_{ij}$, and let Λ_{pp} be its integral span. Such a basis follows by repeatedly choosing a pair of pairing one and taking its symplectic orthogonal complement. The restriction of e_0 to Λ_{pp} is integral and unimodular. The Riemann criterion for polarized tori therefore makes

$$A_I^{\text{pp}} = (V_{0,\mathbb{R}}, I) / \Lambda_{\text{pp}}, \quad I \in \mathcal{D}, \quad (8.7)$$

principally polarized abelian sixfolds. The rational K -action persists in $\text{End}^0(A_I^{\text{pp}})$: every rational K -operator has an integer multiple preserving Λ_{pp} , and it commutes with I .

We use the following precise fine-level theorem. Over $\mathbb{C}$, fix a primitive fourth root for the symplectic Weil pairing. The functor of principally polarized abelian schemes of relative dimension six with a full symplectic level-4 isomorphism to the standard $(\mathbb{Z}/4)^6 \times \mu_4^6$ has a smooth quasi-projective fine moduli scheme $\mathcal{A}_6(4)$ and a universal abelian scheme $\pi : \mathcal{U} \rightarrow \mathcal{A}_6(4)$. On the chosen analytic component it is the quotient of the universal marked torus family on $\mathfrak{H}_6$ by the principal congruence subgroup $\Gamma_6(4)$; both this component and the universal total space are quasi-projective algebraic varieties. These are Chai's symplectic-level and uniformization statements [9, Chapter I, Theorems 1.4 and 1.7.2, and Sections 2.3–2.5, pp. 5–13]. The hypotheses here are explicit: e_0 is perfect on Λ_{pp} , the chosen basis supplies the level, $4 \geq 3$ removes stabilizers, and the ground field has characteristic zero.

The universal π is smooth and projective. Smoothness and properness are part of being the universal abelian scheme. To see projectivity from the stated theorem without choosing a universal theta line, embed the quasi-projective $\mathcal{U}$ locally closed in some $\mathbb{P}_{\mathbb{C}}^b$. Its graph over the separated scheme $\mathcal{A}_6(4)$ is a locally closed immersion

$$\mathcal{U} \longrightarrow \mathbb{P}_{\mathcal{A}_6(4)}^b.$$

This morphism is proper: for any morphism from an $\mathcal{A}_6(4)$ -proper scheme to an $\mathcal{A}_6(4)$ -separated scheme, the graph is closed and the second projection is a base change of a proper morphism. A proper immersion is a closed immersion. The displayed map is therefore a projective embedding over $\mathcal{A}_6(4)$.

The symplectic basis identifies the space of all positive Lagrangians with $\mathfrak{H}_6$. The compatible Lagrangians in (8.6) give a holomorphic map $\mathcal{D} \rightarrow \mathfrak{H}_6$. Composition with the level quotient gives

$$\nu : \mathcal{D} \longrightarrow \mathcal{A}_6(4)^{\text{an}}. \quad (8.8)$$

The quoted marked uniformization identifies $\nu^* \mathcal{U}^{\text{an}}$ with the family (8.7). This construction uses the ordinary principally polarized level family. It asserts neither that $\mathcal{D}$ or its image is algebraic nor that the ambient universal family has a K -action.

We also obtain the precise family with central fiber Y_0 . Choose an integer $c_{\text{lat}} > 0$ such that $c_{\text{lat}} \Lambda_{\text{pp}} \subset \Lambda_0$. The finite group

$$G_{\text{lat}} = c_{\text{lat}}^{-1} \Lambda_0 / \Lambda_{\text{pp}} \subset V_0 / \Lambda_{\text{pp}}$$

consists of torsion points of uniformly bounded order. In the marked family on the contractible $\mathcal{D}$ these are holomorphic torsion sections. Indeed, the kernel of multiplication by their common order is a finite étale holomorphic cover of $\mathcal{D}$, and the marking identifies it with the constant torsion lattice. The chosen torsion classes therefore select sections, which are holomorphic because the cover is locally biholomorphic. Translation by G_{lat} is free. Taking the quotient, locally by disjoint translated coordinate neighborhoods, gives a smooth holomorphic group family with zero section. It is proper because it is the finite quotient of a proper family. Its fiber is $(V_{0,\mathbb{R}}, I) / (c_{\text{lat}}^{-1} \Lambda_0)$. Multiplication by c_{lat} identifies it with

$$Y_I = (V_{0,\mathbb{R}}, I) / \Lambda_0, \quad (8.9)$$

and identifies the seed fiber with Y_0 .

This family is polarized in the relative sense required in Proposition 7.3. Indeed, choose an integer $c_{\text{pol}} > 0$ so that $c_{\text{pol}} e_0$ is integral on Λ_0 . Let $\mathcal{P}_I$ be the normalized Poincaré line on $Y_I \times \hat{Y}_I$. Mark the dual homology by

$$H_1(\hat{Y}_I, \mathbb{Z}) \xrightarrow{\sim} \Lambda_0^{\vee}, \quad \xi \longmapsto [y \longmapsto c_{1,B}(\mathcal{P}_I)((y, 0), (0, \xi))].$$

In this marking the dual complex structure is $-I^t$. The constant lattice map

$$F : \Lambda_0 \longrightarrow \Lambda_0^\vee, \quad x \longmapsto [y \longmapsto c_{\text{pol}}e_0(y, x)]$$

extends to the marked real vector spaces. It is complex-linear from I to the dual complex structure $-I^t$, since

$$F(Ix)(y) = c_{\text{pol}}e_0(y, Ix) = -c_{\text{pol}}e_0(Iy, x) = (-I^tF(x))(y).$$

Equivalently, its constant complexification preserves the holomorphic Hodge subbundles in the marked uniformizations, so it induces a holomorphic homomorphism from the family Y to its dual family. On each fiber the Riemann criterion supplies an ample line L_I with alternating form $c_{\text{pol}}e_0$. The normalized mixed-biextension identity (4.3), evaluated on $(y, 0), (0, x)$, gives

$$((\phi_{L_I})_*x)(y) = c_{\text{pol}}e_0(y, x), \quad \phi_{L_I}(z) = t_z^*L_I \otimes L_I^{-1}.$$

Homomorphisms of complex tori are determined by their maps on the universal covers, so the family homomorphism is fiberwise this polarization. Its ample class is the marked transport of $c_{\text{pol}}\theta_0$. In particular all fibers are projective. The local representing line, and its relative ampleness after shrinking, are supplied by the relative Picard argument immediately preceding Proposition 7.3. Restricting (8.9) to a small contractible open neighborhood S of the seed gives the polarized holomorphic abelian scheme $a : Y \rightarrow S$ used there; S is a smooth nine-dimensional analytic germ.

The quotient also gives, on each fiber, an isogeny

$$j_I : A_I^{\text{pp}} \longrightarrow Y_I \tag{8.10}$$

induced by $c_{\text{lat}} \text{id}_{V_0}$. It is K -linear in the category up to isogeny. It is algebraic: both tori are projective, and Serre's GAGA Proposition 15 says that a holomorphic map from a compact algebraic variety to an algebraic variety is regular [37, Section 4, no. 19, Proposition 15, pp. 29–30]. Only this fiberwise isogeny is needed; the central isogeny q has not been extended as a relative algebraic isogeny.

8.1.4 The flat descended character

Proposition 7.2 supplies the determinant-trivial twisted coherent sheaf $\mathcal{B}$ of rank r on Y_0 , with full twisted-derived semiregularity. Its character comparison (7.11) is the equality

$$\kappa_B := \text{ch}_B(\mathcal{B}) \in \bigoplus_{j=0}^6 H^{2j}(Y_0, \mathbb{Q}), \quad q^* \kappa_B = \kappa_{M, B}. \tag{8.11}$$

Rationality here is part of that comparison, or follows by applying the rational trace $(\deg q)^{-1}q_*^B$ to its right side. Because q_* is an isomorphism of rational first homology, q^* is an isomorphism of rational cohomology rings. It is K -linear on first cohomology for the transported action and therefore identifies the two Weil planes. Consequently (8.2) becomes

$$\kappa_{B,3} = c_\theta \theta_0^3 + w_0, \quad 0 \neq w_0 \in \mathcal{W}_K(Y_0). \tag{8.12}$$

Use the lattice marking in (8.9) to transport κ_B , θ_0 , and w_0 , denoting their values by $\kappa_B(I)$, θ_I , and w_I . They are rational flat sections. For each $I \in \mathcal{D}$, transporting I through q_* gives a compatible polarized complex structure on the rational homology of M . The constant map q^* is a

Hodge map for these corresponding structures. Proposition 6.6, applied to the invariant character in (8.1), thus proves

$$\kappa_{B,j}(I) \in H^{j,j}(Y_I) \cap H^{2j}(Y_I, \mathbb{Q}) \quad (0 \leq j \leq 6, I \in \mathcal{D}). \quad (8.13)$$

Also θ_I is a rational polarization class at every period, and (8.12) remains the flat equality

$$\kappa_{B,3}(I) = c_\theta \theta_I^3 + w_I, \quad 0 \neq w_I \in \mathcal{W}_K(Y_I).$$

This is an assertion on the marked domain. No section on an arithmetic quotient and no relative twisted sheaf is used to define these flat classes.

8.2 Local algebraicity and projective extraction

Proposition 8.1 (Local algebraicity of the descended character). *There is a nonempty analytic neighborhood of the seed in $\mathcal{D}$ on which every component $\kappa_{B,j}(I)$ is a rational algebraic cycle class on Y_I .*

Proof. Take the smooth contractible nine-dimensional germ S and its polarized abelian scheme $a : Y \rightarrow S$ constructed above. Apply Proposition 7.3 to the descended sheaf $\mathcal{B}$, and retain its notation

$$m = r^{12}, \quad N = 6 + m - 1, \quad p : Z \rightarrow Y, \quad F_0 = p_0^* \mathcal{B} \otimes \ell_0.$$

Thus $a \circ p$ is a proper holomorphic submersion of relative dimension N with a relative ample line. The sheaf F_0 is ordinary coherent, and its raw semiregularity map is injective on the entire group $\text{Ext}_{Z_0}^2(F_0, F_0)$. The proposition also supplies the relative anticanonical line and its normalized Betti class

$$H = \omega_{Z/Y}^{-1}, \quad \zeta_{B,s} = \frac{c_1^B(H|_{Z_s})}{m}. \quad (8.14)$$

The class ζ_B is rational, flat, and of type $(1,1)$, and restricts to the hyperplane class on every p_s -fiber. Each Z_s is projective algebraic, p_s is a projective algebraic morphism, and $H|_{Z_s}$ is an algebraic line bundle.

Now put

$$A_B(s) = p_s^* \kappa_B(s) e^{\zeta_{B,s}} = \sum_{k=0}^N A_{B,k}(s). \quad (8.15)$$

Equations (8.13) and (8.14) show that $A_{B,k}$ is a rational flat section of $R^{2k}(a \circ p)_* \mathbb{Q}$ of type (k,k) . In particular its flat de Rham lift lies in F^k . The central character identity (7.18) gives

$$A_B(0) = p_0^* \text{ch}_B(\mathcal{B}) e^{\zeta_{B,0}} = \text{ch}_B(F_0). \quad (8.16)$$

In the raw $d \log$ convention, the prescribed Hodge section in degree k is the edge of $(2\pi i)^k A_{B,k}$. Its central value is $\text{ch}_{H,k}(F_0)$ by (7.2) and (8.16). Apply Theorem 7.4 with $\beta_{B,k} = A_{B,k}$ and $I_{\text{BF}} = \{1, \dots, N\}$. This index set retains the full semiregularity injection: every potentially nonzero target $H^{j+2}(Z_0, \Omega_{Z_0}^j)$ has $j \leq N - 2$ and occurs at $k = j + 1$; the target for $k = N$ is zero. All the theorem's hypotheses now hold, so after one common shrinking of S every $A_{B,k}(s)$, $1 \leq k \leq N$, is rational algebraic. The degree-zero term $A_{B,0} = r$ is already algebraic.

On a fiber of this smaller germ, choose rational Chow classes representing all $A_{B,k}(s)$, and represent $\zeta_{B,s}$ by $c_1^{\text{CH}}(H|_{Z_s})/m$. Products and the finite exponential polynomial give a rational Chow class whose Betti image is $\zeta_{B,s}^{m-1} e^{-\zeta_{B,s}} A_B(s)$. Proper pushforward along the algebraic p_s has Betti class $\kappa_B(s)$ by (7.17). Every exponential is truncated in the cohomology or Chow ring at the dimension N . Taking homogeneous components of this proper pushforward proves that every $\kappa_{B,j}(s)$ is rational algebraic. $\square$

8.2.1 Two algebraic generators of the rational plane

Lemma 8.2. *Let A be an abelian sixfold with a unital action of the imaginary quadratic field K . If a nonzero vector of $\mathcal{W}_K(A)$ is a rational algebraic class, then every vector of $\mathcal{W}_K(A)$ is a rational algebraic class.*

Proof. Choose $k = 1 + tu \in K^\times$ with $t \in \mathbb{Q}$ such that $\sigma(k)^6 \neq \bar{\sigma}(k)^6$. Such t exists because

$$(1 + tu)^6 - (1 - tu)^6$$

is a nonzero polynomial in t , with linear coefficient $12u$; only finitely many rational t are excluded. Over K (or $\mathbb{Q}$) the plane is the direct sum of the two conjugate sixth exterior eigenspaces from the definition of $\mathcal{W}_K(A)$. A nonzero rational vector w has nonzero coordinates in both lines: scalar conjugation exchanges them, so one zero coordinate would force the other to be zero. On the two lines, $k^* = \eta(k)^*$ has respective eigenvalues $\sigma(k)^6$ and $\bar{\sigma}(k)^6$. The determinant formed by w and k^*w in those coordinates is therefore nonzero. They are linearly independent over $\mathbb{Q}$, and form a basis of $\mathcal{W}_K(A)$.

Choose an integer $n_k > 0$ clearing the denominator of $\eta(k) \in \text{End}^0(A)$. The resulting $h = n_k \eta(k)$ is an algebraic endomorphism, and on H^6 one has $k^* = n_k^{-6} h^*$. Pullback by h preserves rational algebraic cycle classes, so k^*w is algebraic when w is. Both basis vectors, hence the whole rational plane, are algebraic. $\square$

On the open supplied by Proposition 8.1, the class

$$w_I = \kappa_{B,3}(I) - c_\theta \theta_I^3$$

is rational algebraic: an integer multiple of θ_I is an ample polarization class. It is nonzero everywhere because it is the transport of the single nonzero marked vector w_0 . Lemma 8.2 makes $\mathcal{W}_K(Y_I)$ algebraic on that open. The isogeny (8.10) pulls this plane isomorphically to $\mathcal{W}_K(A_I^{\text{pp}})$: its pullback on H^1 is a K -linear rational isomorphism and thus identifies both sixth exterior eigenspaces. Algebraic pullback preserves their cycle classes.

Let

$$\mathbb{W}_{\text{pp}} \subset \nu^* R^6 \pi_*^{\text{an}} \mathbb{Q}$$

be the constant rank-two rational local subsystem defined by the fixed marked K -eigenspaces; its fiber is $\mathcal{W}_K(A_I^{\text{pp}})$. We have proved that all its fibers are algebraic on a nonempty analytic open of $\mathcal{D}$. Only this subsystem, not the individual character κ_B , will be used in the global step.

8.3 Spreading the plane in the fine family

Proposition 8.3 (Analytic Hilbert spread). *Let $\pi : \mathcal{X} \rightarrow S$ be a smooth projective morphism of relative dimension n over a smooth quasi-projective complex variety. Let $\nu : T \rightarrow S^{\text{an}}$ be holomorphic, where T is a connected complex manifold. Fix $0 \leq c \leq n$ and a finite-rank rational local subsystem*

$$\mathbb{V} \subset \nu^* R^{2c} \pi_*^{\text{an}} \mathbb{Q}.$$

If every vector of $\mathbb{V}_t$ is a rational algebraic codimension- c class on $\mathcal{X}_{\nu(t)}$ for all t in a nonempty analytic open subset of T , then the same is true for every $t \in T$.

Proof. The assertion is immediate if the rank b of $\mathbb{V}$ is zero, so assume $b > 0$. Fix a relatively very ample line for $\mathcal{X}/S$. For each of b candidate generators choose a finite list of rational coefficients and, for each term in that list, a Hilbert polynomial of degree $n - c$. There are only countably

many such finite choices. For one choice let $\mathcal{H}$ be the product over S of the corresponding relative Hilbert schemes, with structure map $\rho : \mathcal{H} \rightarrow S$.

The exact representability result used here is Grothendieck's fixed-polynomial Hilbert theorem: for a projective scheme over a noetherian base, a fixed relatively very ample line and Hilbert polynomial P give a projective relative Hilbert scheme representing closed subschemes flat over a locally noetherian test scheme with fiber polynomial P , with its universal flat subscheme. It is the case $F = \mathcal{O}_{\mathcal{X}}$ of the fixed-polynomial Quot theorem [19, Section 3, Theorem 3.2, pp. 259–260, and the Hilbert specialization on p. 265]. Here S is noetherian and π is projective, so the theorem applies. In particular $\mathcal{H} \rightarrow S$ is projective. The space $\mathcal{H}$ may be singular or nonreduced; we do not discard any of its fibers.

After analytic base change to T , the parameters whose b rational combinations span $\mathbb{V}$ will form an open and closed locus. Properness will make its image a closed analytic subset of T . The countably many choices cover the assumed open algebraicity locus, so Baire's theorem and the identity principle will force one image to be all of T . The first step is to prove the cycle-class constancy that makes the spanning condition locally constant, including at singular parameter points.

Put $X_{\mathcal{H}} = \mathcal{X} \times_S \mathcal{H}$. For each factor of $\mathcal{H}$, let $\mathcal{F}$ be the structure sheaf of its pulled-back universal subscheme on $X_{\mathcal{H}}$. It is coherent and flat over $\mathcal{H}$. It is also perfect on $X_{\mathcal{H}}$, with a uniform local projective-dimension bound n . To see this, at $x \mapsto h$ write

$$R = \mathcal{O}_{\mathcal{H},h}, \quad B = \mathcal{O}_{X_{\mathcal{H}},x}, \quad M = \mathcal{F}_x, \quad \overline{B} = B \otimes_R k(h).$$

Both B and M are R -flat. The local fiber ring $\overline{B}$ is regular of dimension at most n , since $X_{\mathcal{H}} \rightarrow \mathcal{H}$ is smooth of relative dimension n . Derived associativity and this flatness give

$$M \otimes_B^{\mathbf{L}} k(x) \simeq (M \otimes_R k(h)) \otimes_{\overline{B}}^{\mathbf{L}} k(x). \quad (8.17)$$

A regular local ring of dimension e has global dimension at most e , as in the regular-local theorem [39, Tag 00O7, Proposition 10.110.1]. The right side of (8.17) has no Tor homology above n . Over the noetherian local ring B , build a minimal resolution of the finite module M by choosing minimal finite generators of each successive kernel. All its differentials become zero after tensoring with $k(x)$. The vanishing of $\mathrm{Tor}_{n+1}^B(M, k(x))$ therefore forces its $(n+1)$ -st free term to vanish. This proves $\mathrm{pd}_B M \leq n$.

The scheme $X_{\mathcal{H}}$ is quasi-projective over $\mathbb{C}$: both $\mathcal{H} \rightarrow S$ and $X_{\mathcal{H}} \rightarrow \mathcal{H}$ are projective and S is quasi-projective. Choose an ample line on $X_{\mathcal{H}}$. The ample-line resolution property says that on a quasi-compact scheme with such a line every finite-type quasi-coherent sheaf is a quotient of a finite direct sum of negative powers of the line [39, Tag 01Q3, Proposition 28.27.13(8)]. Successively take these vector-bundle surjections onto $\mathcal{F}$ and its coherent kernels. The n -th syzygy is locally free by the local projective-dimension bound just proved (and if $n = 0$, $\mathcal{F}$ itself is locally free). Thus $\mathcal{F}$ has a finite global vector-bundle resolution on $X_{\mathcal{H}}$. Derived restriction of that resolution to h is an ordinary resolution of $\mathcal{F}_h$: its terms are $\mathcal{H}$ -flat because $X_{\mathcal{H}}$ is $\mathcal{H}$ -flat, and $\mathcal{F}$ is $\mathcal{H}$ -flat, so derived tensoring with $k(h)$ introduces no higher Tor.

Here is a topological justification of local constancy that does not require a smooth Hilbert space. The original $\pi^{\mathrm{an}} : \mathcal{X}^{\mathrm{an}} \rightarrow S^{\mathrm{an}}$ is a proper smooth submersion between manifolds, because S and π are smooth. Ehresmann's fibration theorem gives a differentiable product over a sufficiently small coordinate ball B around s_0 : $\mathcal{X}_{s_0}^{\mathrm{an}} \times B \simeq \pi^{-1}(B)$. Pull back this trivialization of the original smooth family to $\mathcal{H}^{\mathrm{an}}$; smoothness of the Hilbert space is not required.

Under this product, restrict one vector bundle in the global resolution to $F \times V$, where $F = \mathcal{X}_{s_0}^{\mathrm{an}}$ is compact and V is a neighborhood of a chosen $h_0 \in \mathcal{H}^{\mathrm{an}}$ over B . Shrinking V , finitely many

product neighborhoods $U_i \times V$ trivialize the bundle and cover $F \times V$. A partition of unity on F subordinate to the U_i , multiplied into these trivializations, embeds the bundle continuously in a fixed trivial complex vector bundle on $F \times V$. Let $P(x, h)$ be the orthogonal projection onto its image; it is continuous. Compactness of F permits a further shrinking such that

$$\|P(x, h) - P(x, h_0)\| < 1 \quad (x \in F, h \in V).$$

The map $P(x, h)$ from $\text{im } P(x, h_0)$ to $\text{im } P(x, h)$ is then injective, since a nonzero vector in its kernel would contradict this inequality; the ranks agree, so it is an isomorphism. These maps are continuous bundle isomorphisms. Thus the topological Chern character of the bundle on each fiber is locally constant in h . Apply this to all the finitely many terms of the resolution. Their alternating Chern character is $\text{ch}_B(\mathcal{F}_h)$ by the exact fiber restriction above. It defines a locally constant section of the cohomology local system pulled back to $\mathcal{H}^{\text{an}}$. This argument is valid at singular and nonreduced parameter points, since their underlying topology and the restricted topological bundles are all that was used.

To identify the cycle class including nonreduced multiplicities, use Baum–Fulton–MacPherson’s map $\tau : K_0 \rightarrow \text{CH}_*(-)_{\mathbb{Q}}$ for quasi-projective schemes over a field. It preserves the support-dimension filtration and sends $[\mathcal{O}_V]$ on the associated graded to $[V]$; on smooth X , it sends a vector bundle E to $\text{ch}_{\text{CH}}(E) \text{td}(T_X) \cap [X]$ [2, Chapter II, Theorem, pp. 119–120; Chapter III, Section 1, Theorem and Lemma, pp. 128–129]. Put $s = \rho(h)$. For $\mathcal{F}_h$ supported in dimension at most $n - c$, let V_i be its dimension- $(n - c)$ components with generic points η_i , and put $\ell_i = \text{length}_{\mathcal{O}_{\mathcal{X}_s, \eta_i}}(\mathcal{F}_{h, \eta_i})$. Devissage gives

$$[\mathcal{F}_h] \equiv \sum_i \ell_i [\mathcal{O}_{V_i}] \pmod{\text{classes supported in dimension } < n - c}.$$

Indeed, generic composition series spread to dense opens; their discrepancies are lower-dimensional. Thus $\tau(\mathcal{F}_h)$ has no term above dimension $n - c$ and has top term $\sum_i \ell_i [V_i]$. Additivity on a vector-bundle resolution gives

$$\tau(\mathcal{F}_h) = \text{ch}_{\text{CH}}(\mathcal{F}_h) \text{td}(T_{\mathcal{X}_s}) \cap [\mathcal{X}_s].$$

Since $\text{td}_0 = 1$, comparison in successive codimensions gives $\text{ch}_{\text{CH}, j} = 0$ for $j < c$ and $\text{ch}_{\text{CH}, c}(\mathcal{F}_h) \cap [\mathcal{X}_s] = \sum_i \ell_i [V_i]$. Compatibility of the cycle map with normalized Betti Chern classes [2, Section 0.8, p. 106] therefore gives, for the universal fiber subscheme $\mathcal{Z}_h$,

$$\text{ch}_{B, c}(\mathcal{O}_{\mathcal{Z}_h}) = \text{cl}_B([\mathcal{Z}_h]_{n-c}). \quad (8.18)$$

Here $[\mathcal{Z}_h]_{n-c}$ is the fundamental cycle with its generic scheme lengths. Lower-dimensional and embedded contributions affect only higher character terms. The preceding local constancy of the character therefore proves precisely the local constancy of these rational cycle classes on $\mathcal{H}^{\text{an}}$.

Now form the analytic base change

$$\mathcal{H}_T = \mathcal{H}^{\text{an}} \times_{S^{\text{an}}} T \longrightarrow T.$$

It is proper and holomorphic: a projective embedding of $\mathcal{H}$ over S realizes this base change as a closed analytic subspace of $T \times \mathbb{P}^a$, whose projection is proper. The fixed rational combinations of the universal classes in (8.18) give locally constant sections of the pulled-back cohomology local system. In local trivializations the condition that these b sections lie in and span $\mathbb{V}$ is a condition on fixed rational vectors. It is locally constant. Its good locus is open and closed in $\mathcal{H}_T$, hence is a closed analytic subspace and remains proper over T .

We use Remmert's proper-image theorem in its reduced-space form: the image of a proper holomorphic map between reduced complex spaces is an analytic subset. Grauert gives this precise statement and its derivation from the coherent direct-image theorem in Section 7, Satz 1, p. 60 [18]. Reduce the good locus before applying it. Reduction does not change its underlying proper map or its image, while T is already reduced. The theorem thus applies even though the original Hilbert space could be nonreduced or singular. The image of the good locus is a closed analytic subset of T , closed also by properness.

The countably many choices of coefficients and polynomials give countably many such good images. Their union is exactly the locus where $\mathbb{V}_t$ is algebraic. In one direction, a good point supplies b rational algebraic classes spanning the fiber. Conversely, if that fiber is algebraic, choose a rational basis and express each basis vector as a finite rational combination of integral codimension- c subvarieties. Their reduced subschemes have Hilbert polynomials of degree $n - c$ and give a good point for one of the choices.

A closed analytic subset unequal to the connected manifold T is nowhere dense by the analytic identity principle. The assumed nonempty open subset of T is itself a Baire space, so its countable covering by the good images cannot consist entirely of such nowhere dense subsets. Hence one good image equals T , which proves the assertion. $\square$

Apply Proposition 8.3 with

$$S = \mathcal{A}_6(4), \quad \mathcal{X} = \mathcal{U}, \quad T = \mathcal{D}, \quad n = 6, \quad c = 3, \quad \mathbb{V} = \mathbb{W}_{\text{pp}}.$$

The family is smooth projective, $\mathcal{D}$ is a connected complex manifold, and the local algebraicity hypothesis was proved above. Therefore

$$\mathcal{W}_K(A_I^{\text{pp}}) \subset \text{im}(\text{CH}^3(A_I^{\text{pp}})_{\mathbb{Q}} \xrightarrow{\text{cl}_B} H^6(A_I^{\text{pp}}, \mathbb{Q})) \quad (I \in \mathcal{D}). \quad (8.19)$$

The proof applies to the analytically pulled-back rational subsystem. It does not require an algebraic Weil locus or an algebraic K -action on the ambient fine family.

8.4 Every hyperbolic rational lattice

Proposition 8.4. *For the fixed $K = \mathbb{Q}(u)$ with $u^2 = -d$, $d \geq 3$, let A be any complex abelian sixfold with a unital K -action and a compatible polarization. If its actual homological Hermitian form $h_A(x, y) = e_A(x, uy) + ue_A(x, y)$ is hyperbolic, then $\mathcal{W}_K(A)$ is a rational algebraic plane.*

Proof. The hyperbolic-basis assertion of Lemma 3.2 identifies every nondegenerate rank-six hyperbolic Hermitian space over K , in the second-linear convention, with three planes of Gram matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. That elementary assertion was proved by splitting isotropic pairs, independently of the Weil-plane theorem. Since both h_A and the actual seed form h_0 are hyperbolic, it gives a K -isometry

$$f_A : (H_1(A, \mathbb{Q}), h_A) \longrightarrow (V_0, h_0). \quad (8.20)$$

Let $\Lambda_A = H_1(A, \mathbb{Z})$ and let I_A be the complex structure of A . Transport it to $I = f_A I_A f_A^{-1}$ on $V_{0, \mathbb{R}}$. The isometry is K -linear, so I commutes with T_0 . The Riemann form is recovered without any dual convention: it is exactly the coefficient of u when $h(x, y) = e(x, uy) + ue(x, y)$ is expanded in the $\mathbb{Q}$ -basis $1, u$ of K . Thus (8.20) implies

$$e_0(f_A x, f_A y) = e_A(x, y).$$

It follows that I preserves e_0 and satisfies $e_0(x, Ix) > 0$ for $x \neq 0$. Hence I is a point of the full connected domain $\mathcal{D}$. Equation (8.19) algebraizes the Weil plane of A_I^{pp} at this particular period.

The full rational lattices $f_A(\Lambda_A)$ and Λ_{pp} in V_0 are commensurable. Choose an integer $n_A > 0$ such that $n_A f_A(\Lambda_A) \subset \Lambda_{\text{pp}}$. The invertible complex-linear map $n_A f_A$ then induces a surjective holomorphic homomorphism with finite kernel

$$\varphi : A = (H_1(A, \mathbb{R}), I_A) / \Lambda_A \longrightarrow A_I^{\text{pp}} = (V_{0, \mathbb{R}}, I) / \Lambda_{\text{pp}}. \quad (8.21)$$

Both tori are projective, so this homomorphism is algebraic by Serre's Proposition 15 cited above. It is a K -linear isogeny in the category up to isogeny. Its pullback on first rational cohomology is therefore a K -linear isomorphism. It maps each of the two embedding eigenspaces isomorphically, and taking their sixth exterior powers gives

$$\varphi^* \mathcal{W}_K(A_I^{\text{pp}}) = \mathcal{W}_K(A).$$

Pullback along the algebraic isogeny preserves rational codimension-three cycle classes. The left side is algebraic by (8.19), proving the proposition.

The isogeny need not identify the integral polarizations: its pullback scales the rational Riemann form by n_A^2 . Nor was a lift of f_A to a rational spin group used. Only the elementary hyperbolic-basis argument and a lattice denominator were required, so extra endomorphisms or a product decomposition of A impose no restriction. $\square$

Proof of Theorem 3.1. Every imaginary quadratic field can be written $K = \mathbb{Q}(u_0)$, with $u_0^2 = -d_0$ for a positive squarefree integer d_0 . Set $u = 2u_0$ and $d = 4d_0 \geq 4$. This is the same field and is a parameter in the range of the seed construction. For any compatible Riemann form,

$$h_u(x, y) = e(x, uy) + ue(x, y) = 2h_{u_0}(x, y).$$

More generally, changing a purely imaginary generator by any nonzero rational factor scales this Hermitian form by that factor. Such scaling does not change its isotropic subspaces; in signature (3, 3) it does not change the signature either. Thus the hyperbolic hypothesis of the theorem is the hyperbolic hypothesis for the generator u used by this seed. Proposition 8.4 applies and gives the stated algebraicity of $\mathcal{W}_K(A)$. The full domain in (8.4) and the lattice-clearing isogeny include every compatible period, with no simplicity or endomorphism-generality assumption. $\square$

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